UNIVERSITY OF SANTO TOMAS
COLLEGE OF INFORMATION AND COMPUTING SCIENCES
Department of Information Systems

SmartFlow NLEX: A Multisource Corridor Analytics and Decision-Intelligence System for Expressway Traffic Management

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CHAPTER 1

INTRODUCTION

The North Luzon Expressway (NLEX) is the main expressway connecting Metro Manila to Central and Northern Luzon. Its main line runs from Balintawak in Quezon City to Sta. Ines in Mabalacat City, Pampanga (Metro Pacific Tollways Corporation, n.d.-b), about 76 kilometres between kilometre posts 12 and 88.25. Together with the Subic–Clark–Tarlac Expressway (SCTEX) and the NLEX Connector, it shortens travel between the capital and the provinces of Central and Northern Luzon.

NLEX links the growing regions of Central and Northern Luzon with Metro Manila and helps spread economic activity beyond the capital. It carries manufactured goods, raw materials and agricultural products between Metro Manila markets and provincial producers, so smooth traffic on NLEX means faster deliveries and lower transport costs for logistics companies. It is also a daily route for many employees and students who commute between nearby provinces and Metro Manila, so congestion or disruption on the expressway quickly delays both freight and commuters.

Traffic congestion has been a significant challenge along NLEX. Congestion occurs when demand for road use exceeds the capacity of the road, resulting in slower traffic and longer travel times (Afrin & Yodo, 2020). Motorists and commuters frequently experience delays during peak hours, and congestion intensifies during long weekends and major holidays. For Holy Week 2025, traffic on NLEX, SCTEX and the NLEX Connector was projected to peak at 493,000 vehicles per day, against regular NLEX traffic of about 350,000 vehicles per day (Reyes-Estrope, 2025). Navigation applications and social media give drivers information on current conditions, but they offer little guidance on how congestion is likely to develop in the hours or days ahead.

To address congestion and the limited availability of forward-looking information, this capstone project proposes SmartFlow NLEX, an analytics and decision-intelligence system for the NLEX corridor. It integrates NLEX toll and incident records with crowdsourced Waze data, weather, holiday and event calendars, and applies descriptive, predictive and prescriptive analytics to explain past traffic patterns, forecast volume and congestion, and recommend operational responses.

The project also includes a mobile application that gives drivers and commuters live corridor status, congestion forecasts by exit and operator advisories. Through these, SmartFlow NLEX aims to support proactive traffic management by NLEX operators while helping road users make more informed travel decisions.

1.1 Definition of Terms

The following terms are defined according to their use in this study:

a.) Agent-Based Simulation Sandbox

A computational traffic simulation in which individual vehicles are modelled as agents that follow car-following and lane-changing rules, so that congestion emerges from their interactions (Bastarianto et al., 2023). In SmartFlow NLEX it is the Scenario Sandbox, where operators compare an intervention such as a lane closure with a no-intervention baseline in a risk-free virtual environment before acting on the road.

b.) Carbon Dioxide (CO_2) Emissions

The release of greenhouse gases into the atmosphere is primarily caused by the combustion of fossil fuels from motorized vehicles. In the context of traffic systems, these emissions significantly escalate during recurring and nonrecurring congestion due to prolonged vehicle delays and excessive fuel consumption (Afrin & Yodo, 2020).

c.) Champion Model

The candidate forecasting model that achieves the lowest error on chronologically held-out data while also outperforming naive baseline forecasts. The champion is the model whose forecasts are published to the dashboard. It is re-selected whenever the models are retrained.

d.) Corridor

A continuous stretch of road managed as a single unit for traffic operations. In this study, the corridor is the NLEX main line from Balintawak (Km 12) to Sta. Ines (Km 87), covering the toll plazas recorded in the data, which is about 76 kilometres of measured centreline.

e.) Decision Intelligence

The use of data, forecasts and decision models to recommend and evaluate operational actions, combining descriptive, predictive and prescriptive analytics with human judgement. In this study, the system advises and the operators decide.

f.) Medallion Architecture

A layered data-warehouse design in which raw data are stored in a bronze layer, cleaned and standardized data in a silver layer, and analysis-ready aggregates and model outputs in a gold layer (Armbrust et al., 2021).

g.) NLEX Commuters and Drivers

The primary end users of the mobile application, who use NLEX for personal or professional travel and use corridor status and forecasts to plan their trips.

h.) NLEX Data Analysts

Technical staff who interpret historical data to identify traffic trends and system performance. In SmartFlow NLEX they also manage data uploads, review model accuracy and audit system activity.

i.) NLEX Incident Operators

Specialized personnel focused on the documentation, verification, and management of road crashes, vehicle breakdowns, and other non-recurring traffic events.

j.) NLEX Traffic Control Center (TCC) Operators

The personnel responsible for monitoring expressway operations, resource deployment, and emergency response coordination.

k.) Reactive and Proactive

Reactive refers to responding to a problem after it has already occurred, while proactive refers to anticipating and addressing a problem before it takes place.

l.) Rush Hour

The busy part of the day when roads are crowded, typically in the morning when people travel to work and in the evening when they travel home (Cambridge University Press, n.d.). In this study it is used interchangeably with peak hours.

m.) Traffic Congestion

A condition in which vehicle movement on the road is slower due to various factors, including reaching maximum road capacity, occurring incidents, or extreme weather conditions, which may result in increased travel time, accident rates, and carbon emissions (Shaygan et al., 2022).

n.) Waze for Cities (Partner Hub) Feed

A data-sharing programme through which partner agencies receive Waze users' crowdsourced jam and alert reports. SmartFlow NLEX uses its historical archive and its live feed as indicators of congestion and incidents (Hoseinzadeh et al., 2020; Liu et al., 2024).

1.2 Company Background

NLEX Corporation, formerly the Manila North Tollways Corporation (MNTC), is the builder-concessionaire and operator of the North Luzon Expressway, the Subic–Clark–Tarlac Expressway and the NLEX Connector (Metro Pacific Tollways Corporation, n.d.-b). Its work covers the development, design, construction, financing, operation and maintenance of toll roads.

NLEX Corporation belongs to the Metro Pacific Tollways Corporation (MPTC) group. MPTC is a subsidiary of Metro Pacific Investments Corporation and describes itself as the Philippines' leading toll-road developer and operator, with the largest expressway network in the country by traffic volume, revenue and asset base, and as the first Filipino private toll-road company with major stakes in Indonesia and Vietnam (Metro Pacific Tollways Corporation, n.d.-a). Its other Philippine concessions include the Manila–Cavite Expressway (CAVITEX), the Cavite–Laguna Expressway (CALAX) and the Cebu–Cordova Link Expressway (CCLEX).

The operator also supports electronic toll collection through Easytrip RFID accounts, which motorists use and reload at the toll plazas (Reyes-Estrope, 2025). SmartFlow NLEX is developed independently by the proponents as an academic project and is not an NLEX Corporation system.

1.3 Project Context (Business Analytics)

NLEX Corporation already plans ahead for predictable, calendar-based demand, positioning resources before Friday rush periods, long weekends and declared holidays. This proactivity is limited in scope. Traffic, incident and toll data are collected continuously, but they are presented mainly as summaries of current or past conditions. When surges arise from factors outside the standard calendar, such as sudden weather, uncleared incidents or large events at the Philippine Arena, operators have little data-driven forewarning and no structured way to test an intervention before deploying it. Traffic engineering that relies mainly on past observations inevitably responds with some lag after conditions change (GoodVision, 2023). Congestion also raises vehicle emissions through idling and stop-and-go driving (Afrin & Yodo, 2020). The data needed to anticipate such compound surges already exist in toll records, incident logs, crowdsourced reports, weather records and event schedules, but they are held separately and are not combined in a single analytical pipeline.

To address this, the proponents develop SmartFlow NLEX, a decision-intelligence system that integrates hourly toll records, incident records, Waze jam and alert data, weather, holidays and event schedules. Descriptive analytics identify recurring congestion and incident patterns. Predictive analytics forecast traffic volume, congestion state, incidents and emissions, and every forecast is evaluated against naive baselines. Prescriptive analytics recommend operational responses such as toll-booth staffing, patrol positioning and incident response.

The system also includes an agent-based simulation sandbox in which operators can test interventions in a virtual corridor before executing them on the road. This approach follows the use of digital-twin traffic simulation to evaluate congestion measures before implementation (Fujishima et al., 2023). Through these capabilities, the proponents seek to extend NLEX's existing proactive management beyond scheduled peak-day planning toward a data-driven approach that prepares operators for both predictable and unpredictable conditions.

1.4. Purpose and Objectives

The proponents aim to develop SmartFlow NLEX, a centralized decision-intelligence system that addresses two specific pain points identified within NLEX Corporation's current traffic management practice. The first pain point is the absence of in-depth predictive and prescriptive intelligence. The current system lacks a machine learning forecasting layer, so operators cannot anticipate volume surges from weather, holidays, or regional events before they materialize into congestion. The second pain point is siloed and underutilized data. Toll records, weather data, incident logs, and crowdsourced reports exist independently and are never fused into a unified analytical pipeline, leaving critical cross-source insights untapped.

1.4.1. General Objective

The general objective of this capstone project is to develop and evaluate SmartFlow NLEX, a centralized decision-intelligence platform that leverages a 4-year historical NLEX dataset alongside real-time crowdsourced and external data to optimize proactive traffic management and incident response. By reducing traffic congestion, the capstone project proposal will support SDG 9 (Industry, Innovation, and Infrastructure) through lower vehicular carbon emissions.

1.4.2. Specific Objectives

To achieve the general objective, the project aims to fulfill the following specific objectives:

1. To evaluate response strategies within an Agent-Based Simulation Sandbox in order to minimize expressway downtime and maximize vehicle throughput during incidents or demand surges, providing NLEX management with a risk-free, data-driven environment to test operational decisions before live deployment using a custom HTML5 Canvas-based simulation interface.
2. To develop the "SmartFlow NLEX" decision-intelligence system for NLEX operators using a custom-built Next.js and TypeScript web dashboard. This system aims to merge and visualize real-time, multi-source data streams across web and conversational channels. By doing so, we will provide operators with a highly usable, accessible platform that simplifies complex data, streamlines operational decision-making, and ensures widespread user adoption.
3. To analyze a 4-year historical NLEX dataset to map and identify recurring systemic inefficiencies and traffic congestion patterns for strategic capacity planning using Python and RStudio.
4. To deploy existing multi-source predictive models within the SmartFlow NLEX dashboard that forecast expressway conditions using integrating historical toll records, crowdsourced data, weather inputs, and event schedules using existing machine learning algorithms.

1.4.3. Alignment of Objectives with System Features and Evaluation

Table 1.4.3.1 links each objective to the system functions that address it, the expected output and the criterion by which it is evaluated. Prescriptive analytics, described in Section 1.3, support the general objective and Specific Objectives 1 and 2.

Table 1.4.3.1 Objectives, System Features, Outputs and Evaluation Criteria

Objective	System functionality	Expected output	Evaluation criterion
General Objective	Integrated warehouse of toll, incident, Waze, weather, holiday and event data; descriptive, predictive and prescriptive modules; emission estimation	Operator dashboard and mobile application that present validated forecasts, recommendations and emission estimates	KPIs K1–K16 (Table 3.6.5); FR-01 to FR-22 (Table 3.5.1.1); UAT acceptance
Specific Objective 1 (agent-based simulation sandbox)	Scenario Sandbox (IDM/MOBIL microsimulation on HTML5 Canvas); natural-language scenario commands; prescriptive models on the Traffic and Incident pages	Comparison of an intervention with a no-intervention baseline (speed, throughput, queue, travel time, CO_2) with confidence intervals	Sandbox intervention effect (K10), run time (K11), operator acceptance (K12); engine verification checks
Specific Objective 2 (Next.js and TypeScript dashboard)	Next.js/TypeScript dashboard with role-based access, live map, analytics pages, maintenance, data management, audit log, model narratives; mobile application	A single interface that combines the integrated data, forecasts and recommendations	System Usability Scale (K14), API latency (K13), functional and security tests (Section 3.13)
Specific Objective 3 (historical analysis)	Descriptive analytics on the January 2022 – December 2025 record: temporal profiles, holiday and event impact, incident hotspots, wet-versus-dry	Baselines of recurring congestion, incident and emission patterns	Coverage and reproducibility of the descriptive measures; MTTC baseline (K9); emission baselines

	crash rates, clearance times, emissions		(K7, K8)
Specific Objective 4 (multi-source predictive models)	Candidate models for volume, congestion state, event uplift, incidents, severity, clearance time, CO_2 and fleet mix, with weekly retraining and an accuracy gate	Published champion forecasts with their held-out accuracy	Forecast error against naive baselines (K1–K5)

1.5 Scope and Limitations

This study covers the development of a decision-intelligence system for the NLEX corridor. The system is delivered as an operator dashboard and a commuter mobile application. It integrates historical and live data, analyses traffic, incidents and emissions, and provides forecasts, recommendations and simulation for operators and commuters. Its scope and limitations are defined below.

1.5.1 Scope

The functional boundaries and core features of the system are defined by the following areas:

a) Corridor Coverage

The geographic scope is the NLEX main line from the Balintawak toll plaza (Km 12) to the Sta. Ines exit (Km 87) in Mabalacat City, Pampanga, covering its 20 exits, including the SCTEX interchange at Km 85. All 20 exits are covered by the live map and the congestion forecasts. Toll-volume analytics and emission estimates are reported for 19 of them, because trips paid at the SCTEX interchange are not part of the NLEX toll record. The kilometres that such trips drive on NLEX are still included in the corridor emission estimate. The Volume by Plaza chart shows SCTEX separately as the vehicles that entered NLEX through the Mabiga toll plaza in Mabalacat; these entries are not included in corridor totals or forecasts. The SCTEX expressway itself, the NLEX Connector, Harbor Link spurs not represented in the data, and connecting local roads are outside the scope.

b) Multi-Source Data Integration

The system integrates hourly toll transactions by plaza, direction and vehicle class (January 2022 to December 2025); NLEX accident, breakdown and crash records; the Waze for Cities historical archive and live feed; historical hourly weather and air quality; the Philippine holiday calendar; and a Philippine Arena event calendar. The data are stored in a layered PostgreSQL and PostGIS warehouse.

c) Real-Time Data Monitoring and Visualization

The live Waze feed is refreshed about every two minutes and is shown on the Live Map and the Overview page. Toll and incident records are added in batches as NLEX exports are uploaded. The operator dashboard is delivered as a desktop application, and the commuter interface as a mobile application.

d) External Factor Integration

The analytics account for factors other than traffic volume that influence expressway behaviour: holidays, Philippine Arena events and weather. These are used to measure past effects, as inputs to selected forecasting models, and as scenario settings in the sandbox.

e) Predictive Forecasting

The system forecasts the congestion state of each exit from 1 to 168 hours ahead (refreshed hourly), hourly toll volume by plaza, direction and vehicle class for the 48 hours after the latest upload, daily corridor volume (validated up to 14 days ahead; longer horizons are shown as unvalidated), event-day uplift at the Philippine Arena exit, daily incident counts, incident clearance time, daily corridor CO_2 (validated up to 7 days ahead) and the vehicle-class mix. Every model is compared with naive baseline forecasts on held-out data, its accuracy is displayed with the forecast, and the weekly retraining is designed to publish a retrained model only when it passes this accuracy check (Section 3.11.3.6).

f) Carbon Emission Monitoring

The system estimates CO_2, NO_x and PM2.5 emissions by multiplying the kilometres each toll trip drove on NLEX, taken from its entry and exit plazas, by fixed per-class emission factors, and reports the CO_2 total with a low–high range. It also estimates the additional CO_2 emitted by vehicles idling in the queues behind lane-blocking incidents.

g) Agent-Based Sandbox Simulation

The system includes a virtual simulation environment in which vehicle agents follow car-following and lane-changing rules on a chosen stretch of the corridor. Operators can run what-if scenarios, such as lane closures, speed limits, counterflow and incidents, and compare them with a no-intervention baseline before applying them on the road.

h) User-Facing Decision-Intelligence Interfaces

The platform delivers its outputs through two interfaces. The Operator Dashboard is a desktop console for NLEX personnel that provides analytics, forecasts, recommendations, a live map, maintenance scheduling, data management and the simulation sandbox. The Commuter Mobile Application provides live corridor status, congestion forecasts by exit up to seven days ahead, maintenance notices, community reports and a conversational assistant that answers questions about current corridor conditions.

1.5.2 Limitations

The operational execution and technical performance of the platform are governed by the following constraints:

a) Data Integrity and Availability

The accuracy of all analyses and forecasts depends on the quality and completeness of the data supplied by NLEX Corporation and third parties. The toll and incident record used in this study covers 1 January 2022 to 31 December 2025. Some incident records lack clearance timestamps, and the Waze history has gaps, notably from 19 April to 3 August 2026. Incident records dated 2020–2021 could not be verified and are excluded.

b) Connectivity and Network Dependence

The interfaces require a continuous internet connection. Commuters in low-coverage areas may experience delays, and the demonstration API runs on a free hosting tier that may take several seconds to respond after a period of inactivity.

c) Event Scraping Restrictions

Event data are limited to a calendar of 159 events at three venues near the corridor (the Philippine Arena, the Philippine Sports Stadium and Clark Global City). The calendar was collected once from the venues' published schedules in August 2026; later events have been added by hand, because no automated event feed or scraper runs. Capacity is recorded for every event and a stated attendance for 63 of them, as free text, but the system has no ticket-sales data and its models do not use attendance: event effects are estimated from past traffic on event days.

d) Granularity of Emission Calculations

The carbon monitoring feature provides corridor-level estimates. Each toll trip's distance is taken from its entry and exit plazas, but trips paid on entry in the open section have no recorded exit, so their distance is assumed from the exits ahead and reported as a range. Each trip's emissions are assigned to the plaza where it paid, so the estimates cannot isolate short road sections, and the fixed per-class factors do not reflect changes in driving speed. The factors are European average values from the EMEP/EEA emission inventory guidebook, and the idling rates are United States values; neither has been calibrated to the Philippine fleet. They are estimates and are not validated against roadside measurements.

e) Third-Party API Uptime

Live features depend on external services: the Waze feed, the identity provider, the language-model providers and the map-tile provider. Outages or changes at these providers can interrupt the live map, the assistant, scenario commands or model narratives.

f) Lack of Operational Autonomy

The system is purely advisory. It does not control toll barriers, change roadside signs, or dispatch patrol, ambulance or towing units. All recommendations require an operator's decision.

g) AI Knowledge Domain Restriction

The mobile assistant answers only questions about NLEX corridor conditions, using live corridor data from the system. It declines general-knowledge questions and does not give travel-time estimates or departure recommendations.

h) Geographic Boundaries

The system's monitoring and forecasting do not extend beyond the NLEX main line. Connecting local roads, SCTEX and the NLEX Connector are not analysed.

i) Deployment and Evaluation Scope

SmartFlow NLEX is evaluated as a capstone demonstration and is not deployed in live NLEX operations. Reductions in congestion, clearance time or emissions are therefore estimated from historical data and simulation, not observed in operation.

j) Model Skill

Several models improve only modestly on naive baselines, and some candidate models (for example, the daily LSTM volume model) did not beat them. The congestion-state forecast is only modestly more accurate than the historical exit-by-hour profile. The 48-hour volume forecast starts from the latest uploaded day, so it describes the two days after the end of the record until new data are uploaded, and its uncertainty band is least reliable on public holidays. Forecasts beyond the validated horizons are shown as unvalidated.

1.6 Conceptual Framework of Data Integration

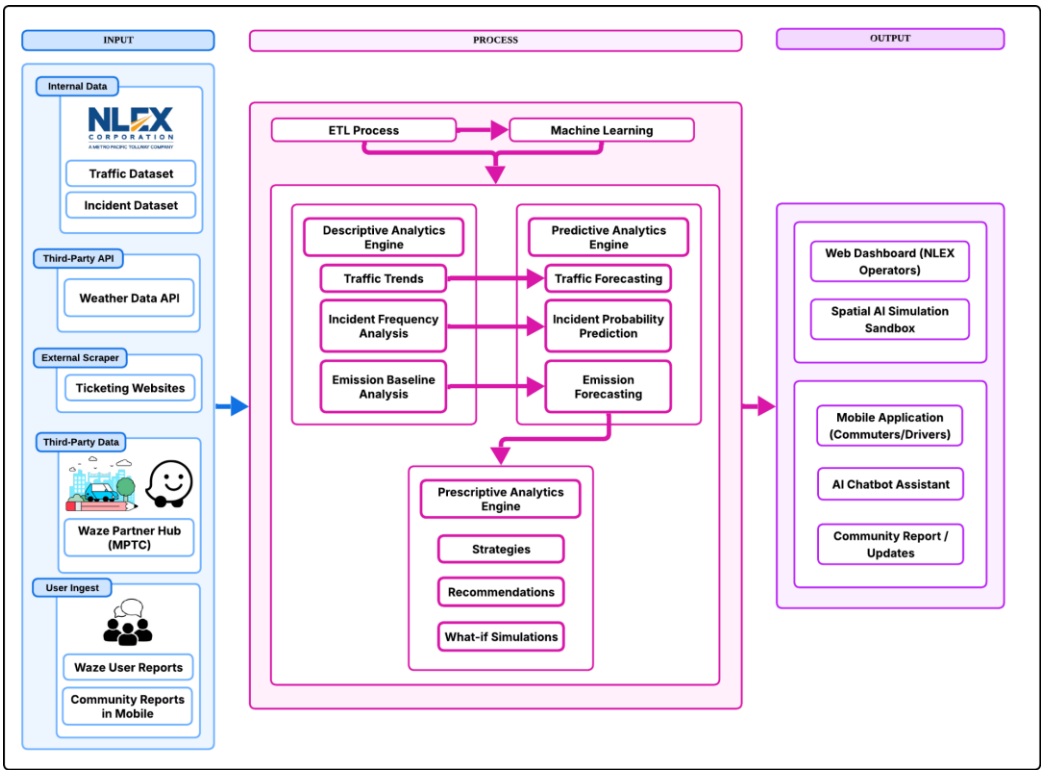


Figure 1.6.1 IPO Conceptual Framework of the SmartFlow NLEX

The Input phase (Figure 1.6.1) collects the internal NLEX data (hourly toll transactions and incident records), the Waze for Cities historical archive and live feed, historical weather and air quality, the holiday calendar and a Philippine Arena event calendar. The figure labels the event source “External Scraper”. As implemented, the calendar was collected once from the venue's published schedule. Community reports are submitted through the mobile application and shown in its feed.

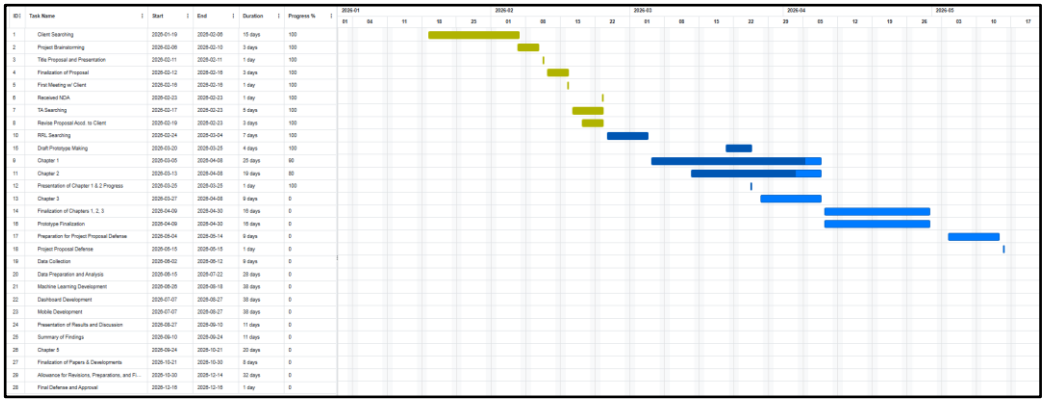
In the Process phase, the ETL pipeline validates, cleans and integrates the data into a layered warehouse. Descriptive analytics establish baselines for traffic, incidents and emissions. Predictive models, each compared with naive baselines on held-out data,

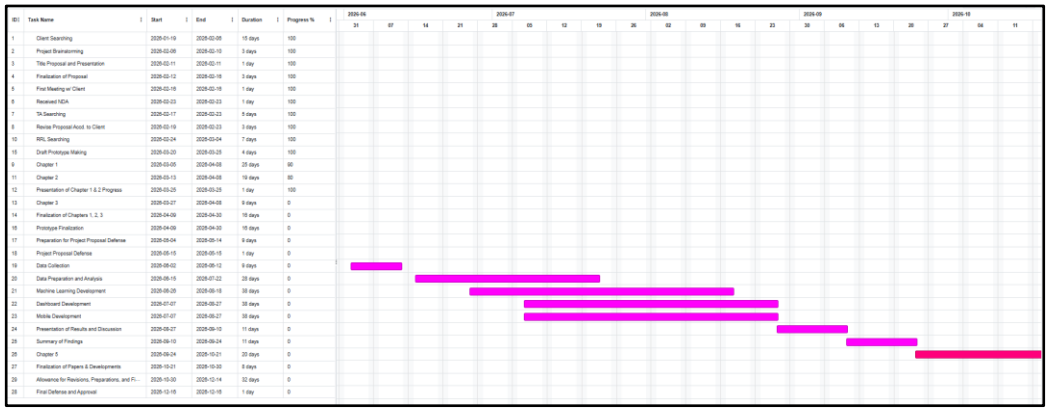
forecast volume, congestion, incidents and emissions. Prescriptive models then recommend staffing, patrol positions, incident responses and emission strategies, and operators can test interventions in the simulation sandbox.

The Output phase delivers the results to two audiences. NLEX operators use the dashboard, which includes the simulation sandbox, and commuters use the mobile application, which includes a conversational assistant and community reports. The framework is iterative: new NLEX data are loaded through the dashboard, the models are retrained weekly and republished only if they pass the accuracy gate, and operator feedback from user-acceptance testing informs later refinements.

1.7 Project Milestones/Timeline

To ensure timely completion, this timeline tracks each project phase from start to finish, highlighting key tasks and deadlines.





ID	Task Name	Start	End	Duration	Progress %
1	Client Searching	2026-01-19	2026-02-06	15 days	100
2	Project Brainstorming	2026-02-06	2026-02-10	3 days	100
3	Title Proposal and Presentation	2026-02-11	2026-02-11	1 day	100
4	Finalization of Proposal	2026-02-12	2026-02-16	3 days	100
5	First Meeting w/ Client	2026-02-16	2026-02-16	1 day	100
6	Received NDA	2026-02-23	2026-02-23	1 day	100
7	TA Searching	2026-02-17	2026-02-23	5 days	100
8	Revise Proposal Accd. to Client	2026-02-19	2026-02-23	3 days	100
10	RRL Searching	2026-02-24	2026-03-04	7 days	100
15	Draft Prototype Making	2026-03-20	2026-03-25	4 days	100
9	Chapter 1	2026-03-05	2026-04-08	25 days	90
11	Chapter 2	2026-03-13	2026-04-08	19 days	80
12	Presentation of Chapter 1 & 2 Progress	2026-03-25	2026-03-25	1 day	100
13	Chapter 3	2026-03-27	2026-04-08	9 days	0
14	Finalization of Chapters 1, 2, 3	2026-04-09	2026-04-30	16 days	0
16	Prototype Finalization	2026-04-09	2026-04-30	16 days	0
17	Preparation for Project Proposal Defense	2026-05-04	2026-05-14	9 days	0
18	Project Proposal Defense	2026-05-15	2026-05-15	1 day	0
19	Data Collection	2026-06-02	2026-06-12	9 days	0
20	Data Preparation and Analysis	2026-06-15	2026-07-22	28 days	0
21	Machine Learning Development	2026-06-26	2026-08-18	38 days	0
22	Dashboard Development	2026-07-07	2026-08-27	38 days	0
23	Mobile Development	2026-07-07	2026-08-27	38 days	0
24	Presentation of Results and Discussion	2026-08-27	2026-09-10	11 days	0
25	Summary of Findings	2026-09-10	2026-09-24	11 days	0
26	Chapter 5	2026-09-24	2026-10-21	20 days	0
27	Finalization of Papers & Developments	2026-10-21	2026-10-30	8 days	0
29	Allowance for Revisions, Preparations, and Fi...	2026-10-30	2026-12-14	32 days	0
28	Final Defense and Approval	2026-12-16	2026-12-16	1 day	0

Figure 1.7.1 Project Timeline

CHAPTER 2

REVIEW OF RELATED LITERATURE, STUDIES, SYSTEMS, AND TECHNOLOGIES

This chapter reviews the literature, studies, systems and technologies relevant to the development of SmartFlow NLEX: A Multisource Corridor Analytics and Decision-Intelligence System for Expressway Traffic Management. It examines prior findings and methods, explains how each relates to SmartFlow NLEX, and identifies the gap that the project addresses.

2.1 Review of Related Literature and Studies

This section reviews the literature and studies relevant to SmartFlow NLEX: A Multisource Corridor Analytics and Decision-Intelligence System for Expressway Traffic Management.

2.1.1 Intelligent Transportation Systems and Traffic Management

2.1.1.1 Traffic Analysis and Management

Afrin and Yodo (2020) emphasize that traffic congestion is mostly influenced by the increase in vehicle numbers and define it as a condition where travel demand exceeds road capacity. The authors divide this problem into two subcategories: "recurring" traffic congestion influenced by predictable peak hour volumes, and "non-recurring" congestion influenced by unpredictable events such as accidents or adverse weather. Because both categories lead to significant socio-economic losses, the authors argue that modern traffic management requires proactive and continuous monitoring.

Rainfall also changes traffic-stream behaviour on expressways. The Highway Capacity Manual applies larger reductions in free-flow speed and capacity as rain intensity increases (National Academies of Sciences, Engineering, and Medicine, 2022). The operating limits of a corridor therefore shift with weather, so analyses and simulations of NLEX should account for rainfall rather than assume fixed road characteristics.

Non-recurring disruptions such as crashes and stalled vehicles are a major source of delay, and their duration depends on incident type, response and conditions. Korkmaz and Erturk (2024) review the statistical and machine-learning methods used to analyse and predict incident duration, and Grigorev et al. (2022) note that accurate duration predictions help operators manage non-recurring congestion and help drivers choose routes.

2.1.1.2 Decision-Intelligence Systems

Decision-support systems extend monitoring by evaluating candidate responses. Chung et al. (2020) developed a decision-support system for integrated active traffic management that uses the METANET macroscopic model to predict near-future conditions on connected freeways and arterials and evaluates combinations of strategies, such as variable speed limits, queue warning and ramp metering, by their effect on travel-time reliability.

In a Federal Highway Administration report, Miller et al. (2021) classify decision-support tools (DSTs) for traffic management as knowledge-driven, data-driven or model-driven. Knowledge-driven tools apply stored rules and procedures, data-driven tools query historical and current data, and model-driven tools provide “what-if” analysis that predicts the possible outcomes of response strategies on measures such as travel time, throughput and environmental impact before they are deployed.

Model-driven decision support is increasingly delivered through digital twins, virtual replicas of the road network kept in step with observed data. Rudskoy et al. (2021) describe how digital twins can modernize traffic control centres within intelligent transportation systems, and Fujishima et al. (2023) show how a digital-twin traffic simulation can be used to evaluate congestion-mitigation measures before they are implemented.

These studies support integrating fragmented data into a central interface that both describes current conditions and lets operators evaluate responses before acting. SmartFlow NLEX applies this idea by combining a knowledge- and model-driven dashboard with a simulation sandbox, while leaving every decision to the operators.

2.1.2 Traffic Data Sources and Multisource Integration

This section reviews why multisource data are needed for corridor analytics. Single sources such as fixed sensors are costly and cover limited areas (Hoseinzadeh et al., 2020), while crowdsourced reports and external factors such as weather and events add information that toll counts alone do not capture (Liang et al., 2024; Nigam & Srivastava, 2023). Combining these sources is expected to improve the understanding and forecasting of expressway traffic (Kashinath et al., 2021).

2.1.2.1 Multisource Data Integration in Traffic Systems

For dealing with the specific problem of missing data and anomalies, robust data fusion techniques are required. P. Wang et al. (2021) demonstrate an effective data fusion methodology by combining conventional hardware sensor data with toll collection data. Because raw sensor data frequently suffers from missing values due to faulty equipment, spatiotemporal dependency modeling restores the missing data while utilizing accurate toll logs to validate overall traffic volumes. Leinonen et al. (2025) further point out that relying on single-source data is inadequate for representing complex traffic behaviors; integrating diverse data sources significantly improves the accuracy and reliability of predictive models compared to using isolated datasets.

For high-dimensional tabular data, tree-based ensembles are widely used. Lam (2024) compared random forest, XGBoost and LSTM models on junction vehicle counts and found XGBoost the most accurate and efficient, and Sun et al. (2021) showed that an improved XGBoost model captures spatio-temporal dependencies in segmented traffic-flow prediction.

These studies support fusing heterogeneous sources such as toll counts, crowdsourced reports and weather, and evaluating tree-based ensembles among the candidate models. In SmartFlow NLEX, XGBoost proved most useful for classifying congestion state, while a decomposable time-series model forecasts daily volume best (Section 3.11.2).

2.1.2.2 Crowdsourcing and Real-Time Traffic Data Collection

Crowdsourced reports vary in reliability. Yu et al. (2025) proposed a consortium-blockchain reputation scheme that scores each reporting vehicle so that low-quality reports can be filtered before they affect traffic applications.

Evaluations of Waze data support its use as a supplementary source. Hoseinzadeh et al. (2020) found that Waze speeds closely tracked Bluetooth ground truth, with lower accuracy at night and on low-volume segments, and Liu et al. (2024) compared Waze crash and stopped-vehicle reports from 2018 to 2021 with official state incident records, finding that Waze covered a varying share of the recorded incidents, that more than half of the matched crashes appeared in Waze first and that location differences averaged less than half a mile. These findings justify SmartFlow NLEX's use of Waze jams and alerts as a live congestion signal, while recognizing that Waze is not a complete or error-free record.

2.1.2.3 External Factors Affecting Traffic Patterns

Rainfall degrades traffic-stream characteristics on freeways, reducing free-flow speed and capacity as rain becomes heavier (National Academies of Sciences, Engineering, and Medicine, 2022). U.S. guidance on active traffic management likewise describes managing recurrent and non-recurrent congestion dynamically, including through real-time monitoring of traffic combined with weather conditions (National Academies of Sciences, Engineering, and Medicine, 2024).

Planned events are another external factor. Giuliano and Lu (2021) analysed traffic around a large stadium by comparing event days with comparable control days and found measurable increases in traffic near the venue, showing that event effects can be isolated when the comparison days are chosen independently of the outcome.

Following this approach, SmartFlow NLEX measures the effect of Philippine Arena events and holidays against local control days and estimates per-exit event uplift from a calendar of events, rather than from days selected because traffic was high.

2.1.3 Descriptive Analytics for Traffic Systems

In transportation research, descriptive analytics summarizes historical data into structured insights. Bachechi et al. (2022) describe big-data analytics and visual dashboards that aggregate traffic sensor data over space and time, and Attioui and Lahby (2025) note that understanding historical patterns is a foundation for forecasting.

These establish the importance of the project's initial exploratory phase. By using data summarization and clustering, the study can objectively identify historical traffic trends, creating the necessary baseline for more advanced predictive and prescriptive features.

2.1.3.1 Traffic Volume Trend Analysis (Time Series Aggregation)

Traffic volume trend analysis fundamentally begins with the descriptive analytics and visual aggregation of highly variable spatio-temporal data. To uncover structural patterns within continuous traffic flows, researchers aggregate raw observations across specific temporal dimensions, such as by day of the week, month, or year. This systematic approach allows analysts to isolate recurring routines from irregular fluctuations and effectively map the evolution of traffic over both time and space (Bachechi et al., 2022).

Beyond visual mapping, recurring traffic patterns can be analysed with formal time-series methods. Hyndman and Athanasopoulos (2021) show how a series can be decomposed into trend, seasonal and remainder components, and how autocorrelation (ACF) and partial autocorrelation (PACF) functions help identify candidate model orders.

These methods provide a framework for separating NLEX traffic into trend and seasonal cycles, so that recurring routines can be distinguished from irregular spikes.

2.1.3.2 Peak Hour and Congestion Pattern Analysis

Peak-hour analysis summarizes when congestion recurs. Aggregating volume by hour of day and day of week reveals distinct morning and evening peaks and differences between weekdays, weekends and holidays, which traffic managers use to time resources (Bachechi et al., 2022).

SmartFlow NLEX applies this through hour-by-weekday profiles of toll entries and incidents, which identify when staffing and patrol coverage are most needed.

2.1.3.3 Event-Based and Holiday Traffic Analysis

Planned events and holidays cause non-recurrent changes in demand. Forecasting texts treat such calendar effects as explicit components or regressors, so that they are not mistaken for trend or noise (Hyndman & Athanasopoulos, 2021).

To isolate the effect of a specific event, Giuliano and Lu (2021) compared event days with comparable control days at the same location, which measures the net change attributable to the event.

SmartFlow NLEX implements this as a local control-day comparison: each holiday or event day is compared with the same weekday within 45 days, excluding other holidays and events (Section 3.11.1.3).

2.1.3.4 Location-Based Hotspot Analysis

Geographic information systems and spatial statistics can identify crash hotspots. Alam and Tabassum (2023) used hot-spot analysis to locate statistically significant clusters of severe crashes in Ohio and to prioritize safety interventions.

2.1.3.4.1 Spatial Density Estimation and Hotspot Mapping

Kernel density estimation converts crash points into a continuous density surface (Berhanu et al., 2023), and the Getis-Ord Gi∗ statistic tests whether high values cluster beyond chance (Alam & Tabassum, 2023). SmartFlow NLEX currently ranks five-kilometre segments by incident counts. Kernel density and Gi∗ testing are proposed extensions.

2.1.3.5 Incident Frequency and Severity Analysis

A baseline of incident frequency, severity and clearance time is needed to evaluate incident management. Grigorev et al. (2024) review methods for predicting incident duration, and Islam et al. (2021) model crash clearance time on freeways, showing that clearance time varies with incident and roadway characteristics.

Mapping these clearances into a visual response time distribution, analysts can identify temporal bottlenecks in emergency operations that often lead to further network degradation. Consequently, tracking the secondary incident rate, the percentage of crashes caused by the queue of an existing primary incident, is critical for comprehensive safety evaluation.

SmartFlow NLEX reports incident frequency by type and cause, severity counts, and the mean, median and 90th-percentile clearance time by incident type with the share of records that have valid timestamps.

2.1.3.6 Weather-Condition Traffic Impact Analysis

Adverse weather affects traffic flow. Romanowska and Budzyński (2022) used expressway data to quantify how weather and time of day change speed, headway and capacity, and derived weather adjustment factors. Comparing traffic and crash outcomes in wet and dry conditions is a direct way to measure such effects.

Weather affects both flow and safety by changing driver behaviour, which supports including weather data in the analysis of NLEX traffic and incidents.

2.1.3.6.1 Mejia and Sigua Speed-Reduction Model

For NLEX specifically, Mejia and Sigua (2018) provide rain-dependent speed parameters. SmartFlow NLEX scales its 108 km/h clear-weather sandbox speed by the ratio of their rain to clear-weather free-flow speeds, giving speed caps of 100, 100 and 96 km/h in light, moderate and heavy rain, and its descriptive analysis compares crash rates in wet and dry hours (Section 3.11.1.5).

2.1.3.7 Traffic-Driven Carbon Emission Estimation

Estimating the environmental footprint of the corridor requires emission factors by vehicle type. Road-transport emission models such as COPERT provide speed-dependent emission factors by vehicle category (Ntziachristos & Samaras, 2023), while simpler inventories apply fixed per-kilometre factors to vehicle-kilometres travelled. Congestion adds emissions through idling and stop-and-go driving (Afrin & Yodo, 2020).

SmartFlow NLEX applies fixed per-class factors to the vehicle-kilometres of each toll trip, estimates the additional CO_2 from vehicles idling behind lane-blocking incidents, and reports daily, hourly and monthly totals (Section 3.9.3).

2.1.3.8 Vehicle Classification and Fleet Mix Analysis

The composition of the vehicle fleet shapes both traffic flow and emissions. Heavy vehicles take up more of a road's capacity than passenger cars: the Highway Capacity Manual counts each truck or bus as the equivalent of two passenger cars on level freeway segments and three on rolling terrain (National Academies of Sciences, Engineering, and Medicine, 2022; Washburn et al., 2024). They also emit several times more CO_2 per kilometre, because heavy-duty vehicles burn several times more fuel per kilometre than passenger cars (Ntziachristos & Samaras, 2023).

SmartFlow NLEX therefore tracks the shares of Class 1, 2 and 3 vehicles in traffic and in emissions, and forecasts the class mix (Section 3.11.2.10).

2.1.3.9 Descriptive Geospatial Analysis using Heatmaps

Descriptive geospatial analysis relies heavily on advanced visualization techniques to interpret massive volumes of spatio-temporal data for continuous traffic management. By integrating traffic sensor observations into interactive decision-support dashboards, researchers and public authorities can visually track the geographic evolution of traffic conditions over time. These geospatial visualizations, often utilizing color gradients or heatmaps to represent data intensity, enable analysts to efficiently pinpoint congested expressway sections, monitor dynamic traffic flows, and translate complex spatial data into actionable insights for infrastructure monitoring (Bachechi et al., 2022).

In the realm of transportation safety and incident management, geospatial analysis is frequently applied to visualize the spatial distribution of adverse traffic events. By employing point density techniques and geographic mapping, researchers can generate heatmaps that clearly highlight high-risk accident hotspots across a regional road network. This method of visually clustering event data not only simplifies the interpretation of complex geographic patterns but also provides the critical, location-based evidence necessary to prioritize specific corridors for targeted safety interventions and infrastructural improvements (Berhanu et al., 2023).

SmartFlow NLEX uses hour-by-weekday heatmaps for traffic, incidents and emissions, and a live corridor map that shows Waze jams and alerts by exit.

2.1.4 Predictive Analytics for Traffic Forecasting

The ability to accurately predict traffic conditions represents a core element of contemporary Intelligent Transportation Systems (ITS), shifting urban traffic management from reactive observation to proactive mitigation (Shaygan et al., 2022). While predictive algorithms efficiently handle short-term route planning using continuous streaming data, long-term forecasting projects network conditions weeks into the future. This extended horizon requires robust analytical models capable of addressing highly complex, non-linear spatiotemporal dependencies and sudden, non-recurrent anomalies (Attioui & Lahby, 2025).

Deep-learning architectures such as long short-term memory (LSTM) networks can learn sequential dependencies in large traffic datasets and are widely used for traffic and travel-time prediction (Leinonen et al., 2025; Attioui & Lahby, 2025).

Benchmark studies also caution that complex models do not automatically outperform simpler ones. In the M4 competition, statistical methods and combinations remained highly competitive (Makridakis et al., 2020), and Manibardo et al. (2022) found in a benchmark of short-term road-traffic forecasting that deep learning was not the best choice in every case. SmartFlow NLEX therefore evaluates statistical, machine-learning and deep-learning candidates on held-out data and publishes only those that beat naive baselines.

2.1.4.1 Time-Series Forecasting Models for Traffic, Incident, and Sustainability Data

2.1.4.1.1 Prophet Forecasting Model

Prophet models a time series as the sum of a piecewise-linear trend with changepoints, Fourier seasonal components and holiday effects, and is designed to produce forecasts with little manual tuning (Hyndman & Athanasopoulos, 2021).

When this architecture is fused with deep learning approaches, the Prophet forecasting model operates as an advanced filtering mechanism. Liu et al. (2023) demonstrate that Prophet efficiently isolates macro-level seasonal volume trends, leaving only the complex, non-linear residual errors for complex neural networks to process.

In SmartFlow NLEX, Prophet with weather regressors is the current champion for daily corridor volume, selected over Holt-Winters, SARIMAX and LSTM by held-out accuracy. Holiday and event effects are not yet included as regressors in the volume model and are estimated separately.

2.1.4.1.2 Seasonal Autoregressive Integrated Moving Average (SARIMA)

Forecasting the frequency of accidents and general highway volumes requires addressing pronounced seasonal unevenness and temporal volatility. Deretić et al. (2022) establish that the Seasonal Autoregressive Integrated Moving Average (SARIMA) model serves as an ideal analytical tool for this task by systematically modeling non-stationarity both within distinct seasonal periods and between them.

By applying mathematical differencing to smooth out erratic data points, the model extracts pure linear relationships and recurring cyclical trends from highly volatile traffic datasets. Hyndman and Athanasopoulos (2021) corroborate this, demonstrating that autoregressive techniques provide the foundational statistics necessary to separate underlying trend signals from random, non-recurrent traffic noise.

In SmartFlow NLEX, a seasonal-naive forecast (the value one week earlier) serves as the baseline that every volume model must beat, and SARIMA-family models are candidates.

2.1.4.1.3 SARIMAX Model

While standard time-series models handle internal historical data effectively, they are completely blind to outside environmental forces. Hyndman and Athanasopoulos (2021) establish that extending traditional frameworks into SARIMAX architectures addresses this limitation by allowing for the explicit integration of exogenous variables, or external regressors, directly into the forecasting equation.

Romanowska and Budzyński (2022) derived adjustment factors that quantify how weather and time of day change traffic-flow characteristics, which supports treating weather as an exogenous input.

SmartFlow NLEX tests SARIMAX$(1,1,1)(1,1,1)_7$ with weather regressors. Once future weather is replaced by climatology to avoid leakage, weather adds little accuracy, and SARIMAX is currently not accepted for daily volume.

2.1.4.1.4 Holt's Linear Trend and Holt-Winters Exponential Smoothing

Exponential smoothing methods forecast by weighting recent observations more heavily. Holt's method adds a trend component, and the Holt-Winters method adds seasonality (Hyndman & Athanasopoulos, 2021). Hyndman and Athanasopoulos (2021) also present these methods as innovations state-space (ETS) models, which give the same point forecasts, produce prediction intervals and support automatic model selection by minimising the AICc.

Because they are fast and robust, exponential smoothing methods are standard benchmarks against which more complex models must be justified (Makridakis et al., 2020).

SmartFlow NLEX evaluates Holt's linear trend and additive Holt-Winters with a weekly season as volume candidates and benchmarks. Holt-Winters is currently accepted as a benchmark that beats the seasonal-naive baseline.

2.1.4.2 Machine Learning Models for Traffic, Incident, and Sustainability Prediction

2.1.4.2.1 Random Forest Regression

Processing high-dimensional traffic and emissions data requires ensemble learning algorithms capable of capturing complex non-linear relationships while aggressively avoiding data overfitting. Smelyakov et al. (2023) emphasize that Random Forest (RF) algorithms achieve this by applying an averaging mechanism across multiple, independent decision trees that are fitted on random subsets of the overall dataset.

By averaging many trees, random forests reduce variance and are robust to noise. Lam (2024) included random forests among the models compared for traffic prediction.

SmartFlow NLEX uses random forests as comparators for congestion-state classification and incident counts. They have not been selected as champions.

2.1.4.2.2 Extreme Gradient Boosting (XGBoost)

Extreme Gradient Boosting (XGBoost) builds trees sequentially, each correcting the residual errors of the previous ones, with regularization and efficient handling of sparse data (Bentéjac et al., 2021). Sun et al. (2021) applied an improved XGBoost to spatio-temporal traffic-flow prediction.

Lam (2024) found XGBoost the most accurate and efficient of the models compared for junction traffic prediction.

In SmartFlow NLEX, XGBoost is one of the candidate congestion-state classifiers that forecast low, medium and high states by exit for 1 to 168 hours ahead. The served classifier is chosen again at every hourly refresh, and recurrent networks have often been chosen over XGBoost (Section 3.11.2.3). XGBoost is also a comparator for incident severity and fleet mix and the residual-correction stage of a hybrid hourly volume model.

2.1.4.2.3 Gradient Boosting Classifier and Regressor (GBR)

Gradient boosting builds an additive model in stages, fitting each new regression tree to the negative gradient of a loss function, which for squared error is the residual of the current model (Bentéjac et al., 2021). Changing the loss adapts the same procedure to regression or classification, and XGBoost is a regularized, scalable implementation of it (Bentéjac et al., 2021).

The method has been applied to transport emissions. Li et al. (2022) compared ordinary least squares, support vector machines and gradient boosting regression for forecasting transportation CO_2 emissions in the 30 highest-emitting countries and found gradient boosting the most accurate, and Wen et al. (2021) used gradient boosting regression to predict the on-road CO_2 emissions of light diesel vehicles.

In SmartFlow NLEX, gradient boosting regression is the champion daily CO_2 forecaster, while XGBoost classifies congestion state.

2.1.4.2.4 Long Short-Term Memory (LSTM)

Modern Intelligent Transportation Systems increasingly rely on deep learning architectures to process massive, multi-source datasets spanning long horizons. Leinonen et al. (2025) establish that Long Short-Term Memory (LSTM) networks are specifically engineered to retain complex time-series patterns over extended periods without suffering from severe prediction degradation.

Within this domain, LSTM networks specifically address the vanishing gradient problem found in older neural networks. Attioui and Lahby (2025) demonstrate that by utilizing internal memory cells, LSTMs successfully learn the complex, long-term sequential dependencies that dictate severe traffic disruptions and non-recurrent congestion.

In SmartFlow NLEX, LSTM networks trained for daily volume and CO_2 did not beat the baselines, so they are retained as comparators there. This is consistent with benchmark evidence that deep networks do not always outperform simpler methods on traffic series (Manibardo et al., 2022). On hourly data the result differs. Stacked LSTMs, in which the output sequence of one LSTM layer is the input of the next, are widely used for short-term traffic forecasting (Cui et al., 2020; Lim et al., 2021). In SmartFlow NLEX, a stacked LSTM is the production model of the 48-hour hourly volume forecast, and single-layer, stacked and dual-path LSTMs are candidates for the congestion-state forecast.

2.1.4.2.5 Spatial and Multivariate LSTM

Traffic at one location is influenced by neighbouring locations. Bogaerts et al. (2020) combined graph convolution and LSTM in a single network that captures spatial and temporal patterns for short- and long-term traffic forecasting.

Spatially aware models can therefore represent how conditions at one point propagate to others.

SmartFlow NLEX includes a spatial LSTM for exit-level incident counts that adds the lagged counts of the two nearest exits. Its congestion classifier likewise uses lagged states of neighbouring exits.

2.1.4.2.6 Gated Recurrent Unit (GRU)

While LSTMs are highly capable, handling the computational demands of real-time forecasting requires optimized network architectures. Hewamalage et al. (2021) establish that the Gated Recurrent Unit (GRU) functions as a streamlined recurrent neural network that utilizes an efficient gating mechanism.

This simpler design reduces training and inference time. Hewamalage et al. (2021) report that GRUs can achieve accuracy comparable to LSTMs with fewer parameters.

In SmartFlow NLEX, a GRU network is one of the candidate congestion-state classifiers and a comparator for incident counts. Separate GRU models for wet and dry conditions are a proposed extension.

2.1.4.2.7 Geographically Weighted Regression (GWR)

Global regression assumes that the relationship between predictors and crash frequency is the same everywhere. Geographically weighted regression relaxes this by estimating local coefficients that vary across space, using a distance-weighting kernel (Al-Hasani et al., 2021; Comber et al., 2023).

Soroori et al. (2021) compared geographically weighted Poisson and negative binomial regressions with global negative binomial models of injury crashes across 253 traffic analysis zones and found that the geographically weighted models captured spatially varying effects of road, socioeconomic, demographic and land-use factors and outperformed the global models.

SmartFlow NLEX fits a Poisson geographically weighted regression across its 20 exits. With so few locations, the local coefficients are interpreted descriptively.

2.1.4.2.8 Negative Binomial Regression

Crash counts are typically overdispersed, with variance larger than the mean. Lord et al. (2021) review crash-frequency models and identify the negative binomial model as a standard choice for such data. Deretić et al. (2022) show that accident counts also have strong seasonality.

Count models with overdispersion therefore give more reliable uncertainty than ordinary regression for incident frequencies.

SmartFlow NLEX uses a negative binomial model as the champion daily incident forecaster, compared with Poisson, tree-based and recurrent models. Its improvement over naive forecasts is modest.

2.1.4.2.9 Ordinal and Logistic Regression

Incident severity is an ordered outcome, from property damage only to injury and fatal, so it is commonly modelled with ordered-response models that use the ordering of the categories without assigning them numeric scores (Washington et al., 2020). Mo et al. (2024) analysed crashes at mainline toll plazas, which act as bottlenecks, and found that crash likelihood rose with upstream occupancy and with speed and occupancy differences between the upstream and downstream sides and fell as the share of electronic toll collection lanes increased, which is relevant to a tolled expressway.

Logistic regression is likewise used for binary incident outcomes. Salek et al. (2021) used logistic regression to estimate the likelihood of secondary crashes on freeways and found that traffic conditions (peak periods and AADT), lighting and characteristics of the primary crash, such as a rear-end collision, affected that likelihood. More broadly, Grigorev et al. (2024) review the techniques used to predict incident duration, and Shang et al. (2022) showed that incident outcomes can be estimated from incomplete multi-source data available at report time.

SmartFlow NLEX applies ordinal logistic regression and XGBoost to classify accident severity, and a ridge-penalized logistic regression to estimate secondary-incident risk. The severity models currently perform about as well as predicting the most common class, which is reported as a limitation.

2.1.4.2.10 Cox Proportional Hazard (Cox PH)

Survival analysis is the standard framework for incident duration. Li et al. (2020) applied hazard-based (survival) duration models to highway incidents and found that predictions improve as incident information accumulates, and the Cox proportional hazards model estimates how covariates scale the hazard of clearance without assuming a baseline distribution (Collett, 2023).

The Kaplan–Meier estimator provides the empirical survival curve from observed durations, including censored records (James et al., 2021).

SmartFlow NLEX fits a Cox model to estimate clearance time and uses Kaplan–Meier curves for the dashboard and for the scenarios of the clearance protocol.

2.1.4.2.11 Dirichlet Regression

Vehicle-class shares are compositional data that must sum to one. Dirichlet regression models such proportions jointly, without transforming them (Martínez-Minaya et al., 2023).

Modelling the shares jointly keeps the forecast composition consistent.

SmartFlow NLEX uses Dirichlet regression to forecast the daily shares of Class 1, 2 and 3 vehicles, which feed the emission views and the sandbox vehicle mix.

2.1.4.2.12 Monte Carlo Simulation and Bootstrap Resampling

Monte Carlo methods propagate uncertainty by repeated random sampling. Zhang et al. (2024) used Monte Carlo simulation in estimating network-wide traffic emissions. The bootstrap is a related resampling method that approximates the sampling distribution of a statistic by repeatedly resampling the observed data with replacement, and the percentiles of the resampled statistic give a simple uncertainty interval (Efron & Hastie, 2021).

SmartFlow NLEX uses a related resampling approach: a seeded bootstrap of 400 resamples of observed incidents gives 5th-to-95th-percentile ranges for its emission strategies. These ranges describe sampling uncertainty and do not guarantee a reduction.

2.1.5 Prescriptive Analytics in Traffic Systems

While descriptive analytics identifies historical patterns and predictive analytics forecasts future states, prescriptive analytics represents the pinnacle of the data maturity model. In the context of Intelligent Transportation Systems (ITS), prescriptive analytics merges predictive machine learning outputs with advanced mathematical optimization to recommend specific, optimal courses of action. The transition from prediction to action is the defining characteristic of modern prescriptive frameworks. According to Haydari and Yılmaz (2022) in their comprehensive review, the integration of Deep Reinforcement Learning (DRL) allows traffic systems to move beyond passive observation. By processing real-time predictive data, DRL agents can dynamically generate and disseminate traffic signal controls and routing decisions that actively mitigate real-world volatility.

SmartFlow NLEX applies prescriptive analytics through optimization models, rule-based advisories and scenario estimates (Section 3.11.4). Its recommendations are advisory, and operators decide whether to act on them.

2.1.5.1 Prescriptive Models and Decision Strategies

2.1.5.1.1 What-If Analysis and Scenario Simulation

Scenario simulation is a foundational decision-support mechanism that allows traffic operators to safely test hypothetical strategies in a virtual environment before real-world deployment. According to Rudskoy et al. (2021), it is heavily integrated through ITS virtual simulation platforms. These platforms maintain a synchronized virtual replica of the physical road system, enabling runtime "what-if" explorations. When a predictive model flags a sudden environmental shift, the virtual replica reconstructs the current traffic state to perform accelerated policy simulation.

Expanding on the environmental utility of simulation, microscopic platforms serve a vital role in validating sustainability interventions. Sagaama et al. (2024) demonstrate how simulation environments like SUMO are utilized for prescriptive emissions testing. By running "what-if" scenarios, researchers explicitly measure how dynamic eco-routing techniques impact vehicular energy consumption, proving that prescribing optimized routes significantly reduces network-wide emissions prior to physical execution.

SmartFlow NLEX applies this idea in its Scenario Sandbox, where operators compare an intervention with a no-intervention baseline in a microscopic simulation before deciding whether to apply it.

2.1.5.1.2 Rule-Based Decision Models

Knowledge-driven decision-support tools apply stored facts, rules and procedures to emulate expert reasoning with more consistent results (Shah et al., 2024).

Fuzzy rule-based systems evaluate inputs such as traffic volume and speed on a continuous scale rather than with a single threshold. Ali et al. (2021) showed that this produces graded responses that adapt to changing congestion in traffic-signal control.

SmartFlow NLEX uses rule-based advisories for heavy-vehicle time windows, peak toll differentials and patrol alert windows (Section 3.11.4.8).

2.1.5.1.3 Traffic Optimization Models (Linear Programming)

Traffic-flow problems can be expressed as mathematical programs in which intersections or interchanges are nodes and road segments are links with capacity constraints, so that an optimizer can find the best feasible allocation of flows.

Ma et al. (2022) formulated trajectory planning for connected and autonomous vehicles at freeway work zones as a mixed-integer linear program that minimizes delay and fuel use, showing how optimization can manage traffic around capacity drops.

Location models address where to station response resources. The maximal covering location problem places a fixed number of facilities to cover the largest weighted demand within a service distance (Baldomero-Naranjo et al., 2022), and Atik and Hajibabai (2023) applied a location-allocation model that maximises the coverage of incident demand to position incident responders on a road network. Toll-booth staffing is commonly treated as a queueing problem: Bari et al. (2022) modelled toll-plaza delay from queue waiting and service times that vary with approach volume and traffic composition and linked level-of-service thresholds to the number of tollbooths required, and P. Wang et al. (2020) combined a queueing-based service level with a neural-network forecast of queue length to schedule the number of open toll lanes. SmartFlow NLEX uses a maximal covering location model to position patrol units and a linear allocation for toll-booth staffing (Sections 3.11.4.2 and 3.11.4.3).

2.1.5.1.4 Fuzzy Inference Models

Fuzzy set theory represents vague categories such as “high congestion” with degrees of membership rather than hard thresholds, and Mamdani-type fuzzy systems combine such memberships through linguistic rules (Varshney & Torra, 2023). In the Takagi–Sugeno model, the consequent of each rule is a function of the inputs rather than a fuzzy set (Varshney & Torra, 2023). Its zero-order form, in which each consequent is a constant, is the simplest case.

Ali et al. (2021) and Tunc et al. (2021) applied fuzzy logic to adaptive traffic-signal control, where graded inputs produce smoother responses than fixed thresholds.

SmartFlow NLEX uses a zero-order Takagi–Sugeno advisory that grades the urgency of responding to a forecast congestion state (Monitor, Prepare or Act) from its probability and lead time (Section 3.11.4.5).

2.1.5.1.5 TOPSIS and Multi-Criteria Decision Analysis (MCDA)

Many operational decisions balance several criteria. TOPSIS ranks alternatives by their relative closeness to an ideal solution (Chakraborty, 2022) and is one of the most widely used multi-criteria methods (Çelikbilek & Tüysüz, 2020), including to prioritise safety improvements along a road section (Trivedi et al., 2024).

Li et al. (2025) combined a system-dynamics simulation of urban traffic with TOPSIS to rank policy combinations by their ability to reduce both congestion and emissions. TOPSIS ranks the alternatives by closeness to the ideal; decision makers then choose from the ranked list.

SmartFlow NLEX uses TOPSIS to rank event interventions and emission strategies, with weights stated by the proponents (Section 3.11.4.6).

2.1.5.1.6 Sustainability-Aware Traffic Optimization

Sustainability-aware traffic optimization serves as the ultimate prescriptive synthesis of modern intelligent transportation systems. Recognizing the environmental toll of vehicular congestion, researchers explicitly target the reduction of carbon emissions based on dynamic traffic distribution across urban networks. Zhang et al. (2024) demonstrate that investigating the effect of dynamic traffic distribution on network-wide emissions provides empirical evidence that traffic routing directly impacts regional carbon footprints.

Congestion and idling increase emissions, so measures that shorten delays also reduce them (Afrin & Yodo, 2020).

SmartFlow NLEX estimates the CO_2 that shorter lane blockages and faster response in the busiest hours would avoid, compares them with a what-if reduction in the heavy-vehicle share, and identifies where heavy-vehicle time windows or peak toll differentials would apply. Operators evaluate these options; nothing is applied automatically.

2.1.5.1.7 Two-Stage Stochastic Programming

Stochastic programming optimizes decisions when some parameters are uncertain by representing them as scenarios with probabilities. In a two-stage model, first-stage decisions are made before the uncertainty is resolved and recourse decisions afterwards, and the value of the stochastic solution measures the benefit of modelling the uncertainty (Li & Grossmann, 2021).

Incident clearance time is a natural uncertain parameter for such models, because incidents are stochastic and their durations are hard to predict (Grigorev et al., 2022). Y. Wang et al. (2021) formulated the pre-positioning of disaster supplies as a two-stage stochastic program in which facility location and supply decisions are made before uncertain demand is realised, and J. Wang et al. (2024) used discrete-event simulation of predicted incidents to allocate a limited fleet of safety service patrol vehicles across highway segments.

SmartFlow NLEX implements a two-stage stochastic mixed-integer clearance protocol that chooses how many response units to dispatch and when to escalate, using clearance scenarios drawn from the Kaplan–Meier curve (Section 3.11.4.4).

2.1.6 Geospatial Analysis and Traffic Visualization

Geospatial analysis serves as the foundational layer of modern transportation decision-intelligence systems, providing the visual framework needed to interpret large volumes of spatio-temporal traffic data. Bachechi et al. (2022) demonstrate that integrating traffic sensor observations into interactive dashboards using heatmaps and color gradients enables operators to efficiently pinpoint congested expressway sections and translate complex spatial data into actionable insights. Li, Z. and Li, D. (2022) further emphasize the integration of WebGIS to allow for dynamic interaction with urban traffic logistics, creating accessible interfaces for continuous infrastructure monitoring.

Berhanu et al. (2023) show that mapping the density of crash locations highlights corridors that need safety interventions. Visual summaries of this kind help operators see where and when congestion and incidents concentrate without reading raw tables.

2.1.7 Agent-Based Traffic Simulation

Agent-based models simulate individual travellers or vehicles that follow behavioural rules, so that system-level patterns emerge from their interactions. Bastarianto et al. (2023) review the use of agent-based models in urban transportation and the challenges of calibrating and validating them.

Rundel and De Amicis (2023) coupled the SUMO traffic simulator with a game engine to produce a low-cost digital twin that visualizes simulated traffic, showing how simulation can be made accessible to non-specialists.

Microscopic traffic simulation commonly pairs the Intelligent Driver Model for car-following with the MOBIL model for lane changing, both used to simulate traffic flow from individual driving behaviour (Sunberg & Kochenderfer, 2022; Treiber & Kesting, 2025). SmartFlow NLEX's Scenario Sandbox implements both models in TypeScript on an HTML5 Canvas, with demand drawn from NLEX toll data, so that operators can see how a lane closure or incident turns into a queue (Section 3.11.5.1).

2.1.8 Environmental Impact of Traffic Systems

2.1.8.1 Carbon Emissions and Traffic Congestion

Traffic congestion increases fuel consumption and emissions because of idling and stop-and-go driving (Afrin & Yodo, 2020). Zhang et al. (2024) showed that network-wide emissions depend on how traffic is distributed across a network, not only on total volume.

Tracking these emissions requires segmenting traffic by vehicle class, because energy use and emissions differ widely with engine type and load. Ignoring class differences would understate the contribution of heavy vehicles.

2.1.8.2 Theoretical Framework: National Emission Inventory Methodology

National emission inventories estimate road-transport emissions in tiers. At Tier 1, emissions are estimated from the quantity of fuel consumed and default fuel-based emission factors, while higher tiers multiply vehicle-kilometres by vehicle-category emission factors, which at Tier 3 also depend on average speed (Ntziachristos & Samaras, 2023).

The Philippines' plan for implementing its Nationally Determined Contribution sets the country's climate-mitigation roadmap for 2020–2030, including the transport sector. The plan identifies transport as one of the two largest emission sources in the energy sector, notes that most calculations in the national greenhouse-gas inventories use Tier 1 IPCC methods with aggregated data and default emission factors, and commits the Climate Change Commission to guiding agencies toward common emission factors (Climate Change Commission & Department of Environment and Natural Resources, 2023).

SmartFlow NLEX does not have fuel-sales data for the corridor, so it uses a distance-based estimate: the vehicle-kilometres of each toll trip, measured between its entry and exit plazas, multiplied by fixed per-class CO_2 factors. This makes its results a corridor activity estimate that is consistent in approach with distance-based inventory methods, not a Tier 1 inventory.

2.1.8.3 Baseline Emission Metrics and Vehicle Classification

Emissions differ greatly by vehicle type, so inventories stratify traffic by class. Emission models such as COPERT assign factors by vehicle category and speed (Ntziachristos & Samaras, 2023).

SmartFlow NLEX applies the per-class baseline formula given in Equation 3.9.3.5, using fixed factors of 196.1, 689.4 and 798.2 g CO_2/km for Classes 1, 2 and 3 (Section 2.1.8.5).

Applying class-specific factors ensures that heavy vehicles, which are a small share of traffic but a large share of CO_2, are weighted correctly in the corridor total.

2.1.8.4 Operational Efficiency and the Carbon "Idling Penalty"

Distance-only calculations understate the emissions caused by congestion, because vehicles queued behind an incident continue to emit while barely moving (Afrin & Yodo, 2020).

SmartFlow NLEX estimates this excess with a deterministic queueing model: while a lane is blocked for T minutes, arriving vehicles accumulate in a queue whose total delay grows with T^2, and the delay is multiplied by an idling rate weighted by the hour's vehicle-class mix (Equations 3.9.3.7 and 3.9.3.8). The idling fuel rates are planning values within the range reported by the U.S. Department of Energy (2015), so the estimate is indicative for the Philippine fleet. Because the queue is assumed to clear as soon as the lane reopens, the delay itself is understated.

Expressing incident delay as excess CO_2 links operational performance to environmental impact and gives operators an additional reason to shorten lane blockages.

2.1.8.5 Emission Factor Sources

Emission factors can be taken from national standards, international emission models or commercial emission-factor databases. P. Wang et al. (2021) show the value of fusing data from multiple sources for traffic analysis.

SmartFlow NLEX derives its emission factors from the EMEP/EEA guidebook's typical fuel consumption per kilometre by vehicle category and its CO_2 emission factors per kilogram of fuel (Ntziachristos & Samaras, 2023). Class 1 (light vehicles) uses a petrol passenger car, 61.9 g of fuel per km × 3.168 kg CO_2 per kg, or 196.1 g CO_2/km; Class 2 (buses and small trucks) uses the average heavy-duty diesel vehicle, 216.8 g/km × 3.180, or 689.4 g CO_2/km; and Class 3 (large trucks) uses a diesel truck over 32 tonnes, 251 g/km × 3.180, or 798.2 g CO_2/km. CO, NO_x and PM2.5 factors come from the guidebook's emission factors per kilogram of fuel multiplied by the same fuel use, and the table also stores a low and a high CO_2 value for each class from neighbouring guidebook categories. The factors are fixed values with no speed term, stored in a reference table and not retrieved at run time. Storing the factors keeps results reproducible and avoids depending on an external service at run time.

If NLEX adopts updated national factors, the reference table can be revised and the inventory recomputed.

2.1.9 Model Evaluation and Validation Techniques

2.1.9.1 Forecast Error Metrics (MAE, RMSE, MAPE, WMAPE)

Forecast error metrics quantify the difference between forecasts and observations (Hewamalage et al., 2021). The mean absolute error (MAE) measures the average error magnitude, while the root mean square error (RMSE) squares errors before averaging and so penalizes large errors more. Shang et al. (2022) report both when evaluating incident-duration predictions.

The mean absolute percentage error (MAPE) expresses error relative to actual values but is undefined at zero and unstable for small counts (Hewamalage et al., 2023). The weighted MAPE (WMAPE, also called WAPE), the ratio of total absolute error to total actual volume, is undefined only when all actual values are zero, so it avoids these problems and weights errors by volume (Hewamalage et al., 2023).

SmartFlow NLEX ranks regression models by WMAPE and also reports MAPE, RMSE and MAE for daily corridor totals.

2.1.9.2 Scale-Independent Metrics (MASE)

The mean absolute scaled error (MASE) divides a model's error by the in-sample error of a naive forecast, so it is scale-free, does not divide by individual actual values (which may be zero), and is comparable across series (Hyndman & Athanasopoulos, 2021). A value below 1 indicates that the model beats the naive forecast's in-sample error.

Because MASE is scale-free, it allows the same evaluation standard to be applied to targets with different units, such as daily vehicles and tonnes of CO_2 (Hewamalage et al., 2021).

SmartFlow NLEX computes MASE with a seasonal-naive (weekly) scale for its volume, CO_2 and incident models, so that targets with different units can be judged on the same basis. Because the scale is the naive forecast's error over the training period, a MASE above 1 can also mean that the scored days were harder to forecast than the training years. The volume models are therefore accepted when their MASE is below that of the seasonal-naive forecast on the same scored days.

2.1.9.3 Model Selection Criteria (AIC, BIC)

When developing statistical time-series models like ARIMA or SARIMAX, researchers must balance predictive accuracy with model simplicity. Introducing too many exogenous variables into an equation can lead to "overfitting," where the model memorizes historical noise rather than learning true underlying traffic patterns. Deretić et al. (2022) establish that the Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC) are employed as rigorous model selection techniques to prevent this mathematical bloat.

Both criteria reward goodness of fit while penalizing each additional parameter, which discourages overfitting among models of the same family fitted to the same data.

SmartFlow NLEX records AIC for its statistical volume models but selects among model families by held-out error, because information criteria are not comparable across different model types.

2.1.9.4 Validation Techniques

Standard machine learning validation techniques, such as randomized k-fold cross-validation, are fundamentally invalid for traffic analytics because they violate the sequential integrity of time-series data, leading to severe data leakage. Hyndman and Athanasopoulos (2021) establish that evaluating traffic models requires strict temporal validation methods to ensure that algorithms do not artificially look into the future during the training phase.

2.1.9.4.1 Chronological Train-Test Split Evaluation

To maintain temporal integrity, models are trained on earlier data and tested on later, unseen data. Rolling-origin evaluation repeats this at successive forecast origins so that every test day follows its training data (Hewamalage et al., 2023).

The size of the test window involves a trade-off: a larger window covers more seasonal conditions, while a smaller one leaves more data for training.

SmartFlow NLEX reports two evaluation windows, the last 20% of the record (80/20) and the last 10% (90/10), and uses the 80/20 window for the published metrics. Results that differ between the windows are reported as a limitation.

2.1.9.4.2 Walk-Forward Validation

To further validate performance in highly dynamic environments like expressways, researchers utilize Walk-Forward Validation, also known as Time-Series Cross-Validation. Hyndman and Athanasopoulos (2021) detail that in this approach, the training window progressively slides forward in time, continuously updating the model with recent data before forecasting the next immediate step. This technique accurately simulates the continuous data flow of a real-world operational environment.

SmartFlow NLEX uses rolling-origin evaluation for its volume, CO_2 and fleet-mix models, which simulates repeatedly forecasting the next one to two weeks as new data arrive.

2.1.9.5 Model Benchmarking and Performance Metrics

A complex, computationally expensive model such as an LSTM or XGBoost is justified only if it outperforms simpler statistical benchmarks on held-out data (Makridakis et al., 2022).

If a complex model does not achieve a meaningfully lower error than a simpler benchmark, its added complexity is not justified. Benchmarking against such standards confirms whether the extra cost buys a measurable gain in accuracy.

SmartFlow NLEX benchmarks every candidate against seasonal-naive and climatology forecasts. The daily LSTM volume and CO_2 models did not beat these baselines and were rejected, which shows why such benchmarking is necessary. The hourly stacked LSTM was adopted only after it beat both the same-hour-last-week forecast and the previous profile method on a full year of held-out hours.

2.1.9.6 System Performance and Load Stress Validation

Performance testing checks that a system remains responsive under load, and caching is one of the main techniques for reducing web application response times, so its effect should be part of that testing (Zulfa et al., 2020).

Kermo and Dželihodžić (2026) compare server-side, client-side and hybrid rendering in otherwise identical web front ends and find that server-side rendering cuts First Contentful Paint by about 65% relative to client-side rendering, which affects how quickly pre-rendered content reaches users.

SmartFlow NLEX plans load and stress tests of its API with a scripted load-testing tool against a separate copy of the system (Section 3.13.6).

2.1.10 Key Performance Indicators (KPIs) in Traffic Management

2.1.10.1 Alert Lead Time and Mean Time to Clear (MTTC)

Incident clearance time is a key measure of incident-management performance, and Islam et al. (2021) model the factors that influence it on freeways.

However, effectively reducing MTTC requires proactive notification systems that dispatch resources before localized congestion compounds. Grigorev et al. (2024) emphasize that maximizing predictive lead time provides traffic operators with the critical buffer window necessary to deploy tow trucks, redirect approaching vehicles, and implement temporary lane management strategies, directly mitigating the severity of secondary crashes.

SmartFlow NLEX reports the mean time to clear (MTTC) by incident type as a baseline (KPI K9). Alert lead time is a proposed KPI, because issued alerts are not yet logged against later conditions.

2.1.10.2 Environmental Metrics (CO_2 per class-km)

Emission KPIs should reflect the different contributions of vehicle classes. Road-transport emission guidance assigns factors by vehicle category (Ntziachristos & Samaras, 2023).

Reporting the share of CO_2 that each class contributes, relative to its share of vehicle-kilometres, shows where interventions would have the most effect.

SmartFlow NLEX reports corridor CO_2 intensity (grams per vehicle-kilometre) and the heavy-vehicle shares of vehicle-kilometres and CO_2 (KPI K7). Because its factors are fixed per class, intensity changes only with fleet mix.

2.1.10.3 Forecast Accuracy Metrics (WMAPE, MAPE and MASE)

Forecast accuracy KPIs should be interpretable and robust. Hyndman and Athanasopoulos (2021) present scaled errors such as MASE as an alternative to percentage errors, which are undefined when an actual value is zero and extreme when it is close to zero.

Accuracy should be reported with the forecast grain, horizon and evaluation window, so that users know what the figure refers to.

SmartFlow NLEX reports WMAPE and MASE against naive baselines for its forecasts (KPIs K1, K3 and K5). For the 48-hour hourly forecast it reports WMAPE per plaza, direction, class and hour, with MAE, RMSE and MASE against the same hour one week earlier. A weather-advisory accuracy KPI was dropped because the weather-triggered advisory was withdrawn.

2.1.10.4 Thresholds, Categories and Reference Baselines

Many outputs of a decision-support dashboard turn a continuous estimate into a category, colour or alert, and the scales behind them should follow recognized definitions wherever these exist. The Waze for Cities data feed reports each jam with a congestion level from 0 (free flow) to 5 (blocked) and gives each user alert a reliability score from 0 to 10 and a confidence score based on other users' reactions (Waze, 2023). The Philippine Atmospheric, Geophysical and Astronomical Services Administration (PAGASA) classifies rainfall by hourly rate, with light rain up to 2.5 mm/h, moderate rain above 2.5 and up to 7.5 mm/h, and heavy rain above 7.5 mm/h (PAGASA, n.d.), and the OpenWeather Air Pollution API reports air quality on an index from 1 (Good) to 5 (Very Poor) (OpenWeather, n.d.).

Turning predicted probabilities into a displayed state requires calibrated probabilities and a decision threshold. Post-hoc methods such as isotonic regression make predicted probabilities match observed frequencies, and the threshold for declaring a class should reflect the relative cost of false alarms and misses, with 0.5 optimal only when the two cost the same (Silva Filho et al., 2023). Forecast quality is most informative relative to a simple reference: a skill score expresses the proportional improvement over a benchmark (Hyndman & Athanasopoulos, 2021), and in short-term traffic forecasting the persistence forecast is the minimum baseline a model must beat (Manibardo et al., 2022). Survival models are usually judged by the concordance index, where 0.5 corresponds to random ordering and 1 to perfect ranking (Hartman et al., 2023), although its value depends on the follow-up horizon (Longato et al., 2020). Differences between shares that sum to one are properly measured in logratio geometry, through the Aitchison distance (Greenacre et al., 2023), and monitoring schemes flag abnormal behaviour when a statistic moves a fixed number of standard deviations beyond its reference level (Bayram et al., 2022).

Operational thresholds also have published reference points. The Highway Capacity Manual places basic freeway capacity at about 2,250 to 2,400 passenger cars per hour per lane, depending on free-flow speed (National Academies of Sciences, Engineering, and Medicine, 2022; Texas Department of Transportation, n.d.-a), while about half of the U.S. state transportation agencies surveyed require a minimum capacity of 1,100 to 1,900 passenger cars per hour per lane for short-term lane closures (Thapa et al., 2021). Daily volumes are converted to peak-hour demand with a design-hour factor, typically 8 to 12% of daily traffic on urban roads (Texas Department of Transportation, n.d.-b). Traffic incident management programs measure roadway and incident clearance times (Federal Highway Administration, n.d.-a) and national self-assessments ask whether programs have set clearance-time targets (Federal Highway Administration, 2021). Agencies schedule roadwork in off-peak periods and coordinate concurrent lane closures (Federal Highway Administration, n.d.-b, n.d.-c), and shifting truck trips out of peak periods has been associated with lower congestion and emissions (Kim et al., 2026). Road-safety network screening ranks segments by risk and groups them into risk levels so that limited resources go to the highest-risk locations first (Afghari et al., 2020; Bonera et al., 2024), simulation results are reported as the mean of independent replications with a Student-t confidence interval (Law, 2023), and usability is commonly judged against a mean System Usability Scale score of 68 (Hyzy et al., 2022).

Published sources rarely fix the exact cut-offs a particular system needs. SmartFlow NLEX therefore follows these sources for its scales and methods and sets the remaining cut-offs itself. Section 3.11.6 lists each one with its basis.

2.1.11 Feedback Loops and Model Retraining (MLOps Framework)

2.1.11.1 Feedback Loop Mechanisms in Machine Learning Systems

Machine-learning operations (MLOps) practices keep deployed models reliable as data change, through automated pipelines for training, validation, deployment and monitoring (Ruf et al., 2021).

A feedback loop compares forecasts with the outcomes that later occur and uses the differences to decide when a model needs retraining.

SmartFlow NLEX logs every served congestion forecast and scores it against the jams later observed. A feedback table for the other forecasts exists, but it holds no records yet, because the toll record ends before the period of the published forecasts.

2.1.11.2 Continuous Model Monitoring and Performance Tracking

To successfully manage an expressway, system administrators cannot wait for a catastrophic failure to realize a model is outdated; they must continuously track model degradation in real-time. Bayram et al. (2022) demonstrate that continuous monitoring frameworks achieve this by utilizing performance-aware drift detectors that constantly calculate the model's real-time error metrics, such as MAE and RMSE.

This continuous surveillance provides immediate alerts when a system's calculations become statistically unreliable. The authors explain that if the tracking system detects that the AI is suddenly failing to accurately predict the volume of heavy-duty vehicles, and the forecast error breaches a pre-defined acceptable threshold, the monitoring module automatically flags the system for concept drift.

SmartFlow NLEX applies a simple performance-based drift check: after each weekly retraining it compares the error of the forecasts it published with that of a same-weekday-last-week forecast over the same days, and marks the forecast as drifted when it does worse over at least seven scored days. The retraining batch also re-evaluates every model against baselines and restores the previous model if a retrained one performs worse (Section 3.11.3.6).

2.1.11.3 Model Retraining and Adaptive Learning (MLOps)

Ruf et al. (2021) review open-source MLOps tools that automate model retraining and validation across the pipeline stages.

Validating a retrained model against held-out data before deployment protects the system from publishing a worse model.

SmartFlow NLEX applies this through its weekly retraining batch, which backs up the active models (their output tables and, where later runs load a saved model, the model files), retrains them, applies an accuracy gate and restores the backup when the gate fails.

2.1.11.4 User Interaction Feedback

Prescriptive systems should also learn from how their users respond, for example by recording whether operators accept, modify or reject recommendations (Miller et al., 2021).

Such records show whether recommendations are practical and help agencies decide when rules and parameters need to be recalibrated (Miller et al., 2021).

SmartFlow NLEX will record operator decisions on recommendations during user-acceptance testing (KPI K12). Capturing them routinely in the dashboard is a proposed extension.

2.1.11.5 System Improvement Cycle

An improvement cycle links data, models and decisions: new observations update the models, and the results of decisions become new observations.

Ruf et al. (2021) describe MLOps pipelines in which retraining is triggered by new data or degraded performance.

In SmartFlow NLEX, the cycle consists of loading new NLEX data, weekly retraining with an accuracy gate, and operator review of recommendations. Because the system is advisory, its effect on the corridor depends on operators' decisions.

2.1.12 Large Language Models and Conversational Interfaces

2.1.12.1 LLMs for Commuter Conversational Interfaces

Numerical forecasts can be hard for non-specialists to interpret. Attioui and Lahby (2025) review the growing use of large language models (LLMs) in traffic forecasting research, including their ability to work with textual and contextual information.

To keep LLM answers grounded in reliable data, retrieval-augmented generation supplies retrieved documents to the model (Lewis et al., 2020), and tool use lets a model call external functions or application programming interfaces (APIs) to obtain current values (Schick et al., 2023; Patil et al., 2024).

SmartFlow NLEX's mobile assistant uses a Qwen model with three tools that return current corridor status from the system's API, so every number it reports comes from the system. It does not use retrieval-augmented generation and does not provide forecasts.

2.1.12.2 LLM-Driven Scenario Generation in Simulation Sandboxes

LLMs can also translate natural-language instructions into structured inputs for software, provided that the output is validated against what the software accepts.

Constraining the model to a fixed set of tools or a structured output schema reduces errors in the generated parameters (Patil et al., 2024).

In SmartFlow NLEX's Scenario Sandbox, a GLM model converts an operator's typed command into a structured scenario plan. Every field is validated by the system, and nothing is applied until the operator confirms it (Section 3.11.5.2).

2.1.13 Methods Considered for Future Work

Several established techniques were reviewed during the design of SmartFlow NLEX but are not part of the current system, and they are retained here as directions for future work. For descriptive analysis, seasonal-trend decomposition using LOESS separates a series into trend, seasonal and remainder components (Hyndman & Athanasopoulos, 2021), and its extension to several seasonal cycles (Bandara et al., 2025) would separate the daily and weekly patterns in hourly toll data explicitly. Autocorrelation and partial autocorrelation analysis would guide the choice of autoregressive orders, as Deretić et al. (2022) showed for traffic-accident series. Mixture models could identify traffic regimes from speed and flow observations, and fitting a speed–flow relationship to NLEX segments, as the Highway Capacity Manual does for basic freeway segments (National Academies of Sciences, Engineering, and Medicine, 2022), would estimate segment capacities. Kernel density estimation and the Getis-Ord Gi* statistic would test whether incident concentrations are statistically significant rather than ranking segments by counts (Berhanu et al., 2023; Alam & Tabassum, 2023).

For prediction, quantile regression forests would provide prediction intervals instead of point forecasts (Yao & Graham, 2025), and gated recurrent networks trained separately on wet and dry conditions could capture weather-specific speed dynamics (Hewamalage et al., 2021). Forecast combinations, which performed well in large forecasting competitions (Makridakis et al., 2020), could be tested against the current single champion models. Finally, volume-to-capacity monitoring and capacity-based alerts would become possible once segment capacities and passenger-car equivalents are available, and heuristic rerouting of emergency vehicles and commuters could extend the prescriptive tier. Each of these methods would be adopted only if it improves on the current models when evaluated against the same baselines.

2.2 Review of Related Systems and Technologies

This section explores the existing systems, architectures and tools that inform the proposed decision-intelligence system. Integrated technologies support both traffic management and traffic forecasting: cloud computing provides the capacity to process the large volumes of sensor data that traffic management requires (Arthurs et al., 2022), and machine-learning pipelines automate how forecasting models are trained, deployed and monitored (Kreuzberger et al., 2023).

2.2.1 Intelligent Transportation Systems and Infrastructure

2.2.1.1 Cloud-Integrated ITS

Intelligent transportation systems increasingly rely on scalable platforms that store and process large volumes of traffic data. Bachechi et al. (2022) describe a big-data platform that integrates traffic sensor data, simulation and visual dashboards for traffic monitoring.

Afrin and Yodo (2020) argue that integrating congestion measures into operational systems supports more sustainable and resilient transportation networks and more efficient use of existing road space.

SmartFlow NLEX stores its integrated data in a managed PostgreSQL database with PostGIS on AWS, organized in bronze, silver and gold layers, so that historical and live data for the whole corridor are available to the analytics.

2.2.1.2 Multi-Layered Automated Systems

Traffic-management platforms are commonly organized in layers for data collection, analytics and presentation (Bachechi et al., 2022; Miller et al., 2021).

Separating these layers allows the presentation to be designed independently of the data processing behind it (Chen et al., 2025).

SmartFlow NLEX follows this layering: ingestion into the warehouse, offline model training and an API that serves the dashboard and mobile application (Section 3.2.1.2).

2.2.2 Traffic Monitoring and Navigation Frameworks

2.2.2.1 Crowdsourced Reporting

Large volumes of crowdsourced reports must be filtered and prioritized. In software testing, Ling et al. (2025) used an LLM with clustering to prioritize crowdsourced test reports and reduce redundancy.

A similar approach could be used to prioritize commuter road reports. In the current version, SmartFlow NLEX shows community reports in the mobile feed without automated prioritization, which is a proposed extension.

2.2.3 Data Engineering and ETL Pipeline Frameworks

2.2.3.1 Data Integrity and Cleaning

Layered warehouse designs separate raw, cleaned and analysis-ready data. The lakehouse pattern unifies data warehousing and advanced analytics on open platforms (Armbrust et al., 2021).

Clean, consistent data are also needed to calibrate traffic simulations. Manglano-Redondo et al. (2025) calibrated origin–destination demand for SUMO simulations with a spatio-temporal neural network.

SmartFlow NLEX's pipeline validates, cleans and layers its data before analysis, and its sandbox draws demand from the cleaned hourly toll data.

2.2.3.2 Batch and Real-Time Handling

Traffic data can arrive in scheduled batches, as in traditional extract-transform-load (ETL) processes, or continuously as feeds are polled or streamed, which requires real-time processing (Mehmood & Anees, 2020).

SmartFlow NLEX combines batch loading of NLEX exports with polling of the live Waze feed about every two minutes.

2.2.4 Statistical Analysis and Validation

2.2.4.1 Correlative Validation

Statistical validation checks that data and model results behave as expected, for example by screening incoming data for anomalies (Bachechi et al., 2022) and by testing models against simple baselines such as naive and seasonal-naive forecasts (Hewamalage et al., 2023). SmartFlow NLEX performs its statistical modelling and validation in Python (statsmodels, SciPy and scikit-learn).

Its validation consists of range and consistency checks in the ETL pipeline and baseline-relative evaluation of every model (Sections 3.9 and 3.11.3).

2.2.5 Machine Learning and Forecasting Engines (Python)

2.2.5.1 Deep Learning Frameworks

Recurrent neural networks such as LSTM and GRU are widely used for time-series forecasting (Hewamalage et al., 2021).

Python provides mature libraries for statistical and machine-learning forecasting.

SmartFlow NLEX trains its models in Python with statsmodels, Prophet, scikit-learn, XGBoost, TensorFlow/Keras, PyTorch, lifelines and mgwr, and stores their outputs in the warehouse.

2.2.6 Mathematical Optimization (Linear Programming)

2.2.6.1 Decision Intelligence

Decision support for traffic management follows an analytics path from descriptive and diagnostic to predictive and prescriptive analysis, helping operators monitor conditions, detect incidents and evaluate responses (Shah et al., 2024).

SmartFlow NLEX follows this path. Its prescriptive models are solved with the YALPS linear and mixed-integer programming solver and are presented with their assumptions so that operators can judge them.

2.2.7 Agent-Based Simulation

2.2.7.1 Autonomous Agent Simulation

Bastarianto et al. (2023) review agent-based models in urban transportation, covering their use for evaluating transport policies and services before implementation.

SmartFlow NLEX's sandbox applies agent-based simulation to the expressway, using IDM car-following and MOBIL lane changing (Section 3.11.5.1).

2.2.8 Database and Data Storage Technologies

2.2.8.1 High-Performance Architectures

Relational databases with spatial extensions support both analytical queries and geographic operations. PostgreSQL with PostGIS stores geometries such as road segments, jam lines and incident points with spatial indexes (Bachechi et al., 2022; Makris et al., 2021).

SmartFlow NLEX uses PostgreSQL 18 with PostGIS 3.6 on AWS RDS for its warehouse, and Supabase for user authentication.

2.2.8.2 Caching and Precomputation

Analytics dashboards stay responsive by caching frequently requested results (Zulfa et al., 2020) and precomputing expensive aggregates instead of recomputing them on every request (Chen et al., 2025).

SmartFlow NLEX keeps an in-memory cache of API responses that serves the last result while refreshing it in the background, preloads common responses at start-up, and stores model outputs in the warehouse so that pages read precomputed results.

2.2.8.3 Redis

In-memory key-value stores such as Redis keep data in main memory, which gives high throughput and low latency for real-time web workloads (Yang et al., 2021), but main memory is volatile, so data can be lost on failure unless a separate persistence mechanism is used (Gong et al., 2022). Each key can be given a time to live, after which Redis deletes it automatically, and writing a key with SET replaces both its value and any earlier expiry (Redis, n.d.-a, n.d.-b). Upstash offers Redis as a managed service reached through an HTTP REST interface, which suits serverless functions (Upstash, n.d.).

In SmartFlow NLEX, the Waze ingestion process polls the feed every two minutes, writes it to the PostgreSQL warehouse, and overwrites a snapshot of the active jams and alerts in an Upstash Redis store with a ten-minute expiry, so a stalled feed is not presented as current. Because memory is volatile, the warehouse remains the system of record, and the dashboard's live map reads it through a short in-memory cache.

2.2.9 API Integration and External Data Sources

2.2.9.1 Event Data Collection

Event schedules can be collected from venue websites; for example, Liang et al. (2024) scraped event listings automatically from an arena's official website to predict travel demand around it. SmartFlow NLEX collected the Philippine Arena event calendar once from the venue's published schedule and updates it manually. An automated scraper is a proposed extension.

2.2.9.2 Weather and Air-Quality Data

Musah et al. (2022) evaluated the accuracy and suitability of the OpenWeatherMap API as an alternative source for weather data by comparing its forecasts against actual measurements recorded by the Brazilian National Institute of Meteorology (INMET) across two cities. The study focused on two key meteorological parameters, temperature and relative humidity, examining both current-day readings and lead-time forecasts spanning one to four days ahead.

Their results inform the use of OpenWeather data. SmartFlow NLEX uses OpenWeather's air-pollution history as an estimate of air quality at each exit. It is a modelled concentration from all nearby sources, not a roadside measurement, and is shown separately from the toll-based emission estimates. Its hourly weather comes from Open-Meteo, ERA5 and JMA archives scored against reference sources.

2.2.10 Visualization and System Architecture

2.2.10.1 Proactive Management Visualization

Interactive, spatial visualizations help operators interpret traffic data. Rundel and De Amicis (2023) show how simulated traffic can be visualized in an accessible three-dimensional environment.

Bachechi et al. (2022) built a web-based urban traffic-monitoring dashboard with interactive maps and data visualization.

SmartFlow NLEX uses interactive charts, heatmaps and a live corridor map so that operators can see where and when congestion and incidents occur.

2.2.10.1.1 Next.js

Kermo and Dželihodžić (2026) compare server-side and client-side rendering in modern web applications, and Savenko and Babii (2025) evaluate Next.js optimization strategies, including server-side rendering and incremental static regeneration, against Core Web Vitals.

SmartFlow NLEX uses Next.js 15 with React 19 as a static export, which pre-builds the pages and lets them run inside the Electron desktop application without a web server.

2.2.10.1.2 TypeScript

According to Nguyen (2022), TypeScript is a strongly typed superset of JavaScript that improves web-application reliability through static type checking, catching errors such as type mismatches during development rather than at run time. Interfaces keep data structures, such as records returned by a database, consistent as they pass through an application.

SmartFlow NLEX uses TypeScript in both the dashboard and the API, so that data passed from the warehouse to the charts and maps follows shared type definitions. This reduces, but does not eliminate, run-time errors.

2.2.10.1.3 Apache ECharts

According to Yu and Yang (2024), Apache ECharts is a browser-based visualization library that aggregates diverse and imperfect data into clear charts and is used in applications such as smart-city displays and rail-transit monitoring.

By providing a clear solution for raw data aggregation and display, ECharts will enable the transformation of the system's raw backend data into actionable insights for the end user. This literature confirms that ECharts possesses the necessary technical robustness to render the complex time-series graphs and emission trends required by the decision-intelligence dashboard.

2.2.10.1.4 Mapbox

A key component of the platform, Mapbox GL JS, has been the subject of numerous performance studies. Zunino et al. (2020) conducted a comparative analysis of three widely used JavaScript web mapping libraries Leaflet, Mapbox GL JS, and OpenLayers using Argentina's Life Quality Index as a case study. Their findings revealed that Mapbox GL JS achieved the best overall performance on mid- and high-end devices when displaying raster or vector maps. This performance advantage was attributed to Mapbox GL JS's use of WebGL-based hardware acceleration, which enables efficient rendering even for data-heavy applications.

SmartFlow NLEX uses Mapbox GL JS to render the corridor, live Waze jams and alerts, forecast congestion states and maintenance closures.

2.2.10.1.5 GitHub

Version control records every change to a codebase and supports collaboration among developers, who commit and merge their work in a shared source-code repository such as GitHub (Kreuzberger et al., 2023).

The SmartFlow NLEX team uses Git and GitHub, with personal feature branches and backup branches created before each merge. GitHub Actions also runs the hourly congestion refresh and the weekly retraining batch, reading the database credentials from encrypted repository secrets.

2.2.10.1.6 Figma

Du et al. (2025) argued that the development of multi-user collaborative interfaces requires a design methodology that is both task- and role-oriented to maintain operational consistency across shared workspaces. Their research emphasized that activities and identities of different users should have uniform visual indicators, ensuring that collaborators remain synchronized while performing distinct functional duties within the same environment.

SmartFlow NLEX follows this role-oriented approach: each operator role sees only the modules it needs, with consistent layouts across the traffic, incident and emission pages.

2.2.10.1.7 Electron

Electron combines the Chromium engine and the Node.js runtime into a single runtime so that web technologies can run as a cross-platform desktop application (Scoccia & Autili, 2020).

SmartFlow NLEX packages its dashboard with Electron as a Windows desktop application that serves the static Next.js build locally and connects to the SmartFlow NLEX API.

2.2.10.2 End-to-End Pipeline Design

Data pipelines need monitoring so that failures and stale data are detected before they affect decisions (Kreuzberger et al., 2023; Miller et al., 2021).

Common practices include logging each load or training run, checking the availability and freshness of live inputs, and recording the outputs that were served so that performance degradation can be detected later (Bayram et al., 2022; Kreuzberger et al., 2023; Miller et al., 2021).

SmartFlow NLEX logs uploads and retraining runs, flags live data older than 30 minutes, and logs served congestion forecasts for later scoring.

2.2.10.2.1 PostGIS

PostgreSQL with PostGIS has been used as the spatial database of a web-based urban traffic dashboard (Bachechi et al., 2022) and outperformed MongoDB on almost all spatio-temporal queries over a large vessel-tracking dataset (Makris et al., 2021).

In SmartFlow NLEX, PostGIS stores corridor segments, Waze jam lines and alert points with spatial indexes, which the live map queries by distance from the corridor.

2.2.11 Mobile Application Development Frameworks

According to Nguyen (2025), the evolution of mobile development has shifted toward frameworks that balance high-performance native rendering with developer efficiency. The study highlights React Native as a premier choice for cross-platform development due to its ability to translate JavaScript components into native UI elements. However, to manage the complexity of modern app deployment, Nguyen (2025) emphasizes the role of the Expo ecosystem and Expo Application Services (EAS). These tools provide a managed infrastructure that automates the "heavy lifting" of mobile development, such as credential management and build optimization. By utilizing Expo on top of React Native, developers can maintain a unified codebase while ensuring the application remains scalable and responsive across different operating systems.

SmartFlow NLEX's mobile application is built with Expo and React Native from a single codebase for Android and iOS. It currently runs as development builds; store release through Expo Application Services is planned.

2.3 Synthesis

The literature reviewed in Section 2.1 shows that expressway management benefits from moving from reactive observation toward anticipation. Prior studies have established the value of combining multiple data sources (Leinonen et al., 2025; P. Wang et al., 2021), the usefulness and limits of crowdsourced Waze data (Hoseinzadeh et al., 2020; Liu et al., 2024), the effects of rainfall and planned events on traffic (Giuliano & Lu, 2021; National Academies of Sciences, Engineering, and Medicine, 2022), and a broad set of forecasting methods. Benchmark studies, however, warn that complex models do not reliably outperform simpler ones (Makridakis et al., 2020; Manibardo et al., 2022), and prescriptive studies tend to address one decision at a time, such as signal control, patrol location or incident duration.

Section 2.2 shows that the technologies for such a system are mature and accessible: layered warehouses with spatial databases, Python forecasting libraries, optimization solvers, microscopic traffic simulation with IDM and MOBIL, web dashboards and cross-platform mobile frameworks, and language models that can be grounded through tool use. In the Philippine context, recent published research on NLEX is limited: Gernale and Juanico (2023) examined overloaded trucks on the corridor, which addresses a single aspect of its operation.

The gap that SmartFlow NLEX addresses is therefore not a new algorithm but the integration of these elements for one Philippine expressway corridor. It combines NLEX toll and incident records with Waze, weather, holiday and event data in one warehouse. It evaluates every forecast against naive baselines before publishing it, and it links validated forecasts to recommendations for staffing, patrol positioning, incident response and emissions. Operators can test interventions in an agent-based simulation, and commuters receive the same information through a mobile application. The project's contribution lies in this decision-support workflow, its transparent reporting of model accuracy and limitations, and its application to NLEX. Its value to operations remains to be confirmed through user-acceptance testing and, eventually, live deployment.

CHAPTER 3

METHODOLOGY

This chapter presents the approach used to develop SmartFlow NLEX and documents the system as it has been built. It describes the CRISP-DM and Kanban-based Agile development process, compares the existing and proposed business processes, and explains the data pipeline that integrates NLEX toll and incident records with crowdsourced Waze data, historical weather, holidays and event calendars. It then sets out the project risks and feasibility, the system requirements and key performance indicators, the descriptive, predictive and prescriptive analytics models with their validation procedures, the agent-based simulation sandbox and language-model features, the quality assurance plan and the deployment plan. Throughout the chapter, each capability is identified as implemented, partially implemented or proposed, so that the methodology describes the system that was actually evaluated. Technical terms used in the analytics sections are defined in Appendix A.1 (Glossary).

3.1 Development Methodology

a.) Data Mining Process Model

CRISP-DM

The project follows the Cross-Industry Standard Process for Data Mining (CRISP-DM), which organizes analytics work into six phases: business understanding, data understanding, data preparation, modeling, evaluation and deployment. In SmartFlow NLEX, business understanding defined the operational questions of the NLEX Traffic Control Center; data understanding profiled the toll, incident, Waze, weather and event data; data preparation produced the bronze, silver and gold layers of the warehouse; modeling trained candidate forecasting, classification and optimization models; evaluation compared every candidate with naive baselines on chronologically held-out data; and deployment published the accepted model outputs to the dashboard and mobile application. Because new data and model results repeatedly sent the team back to earlier phases, the cycle was iterated rather than executed once.

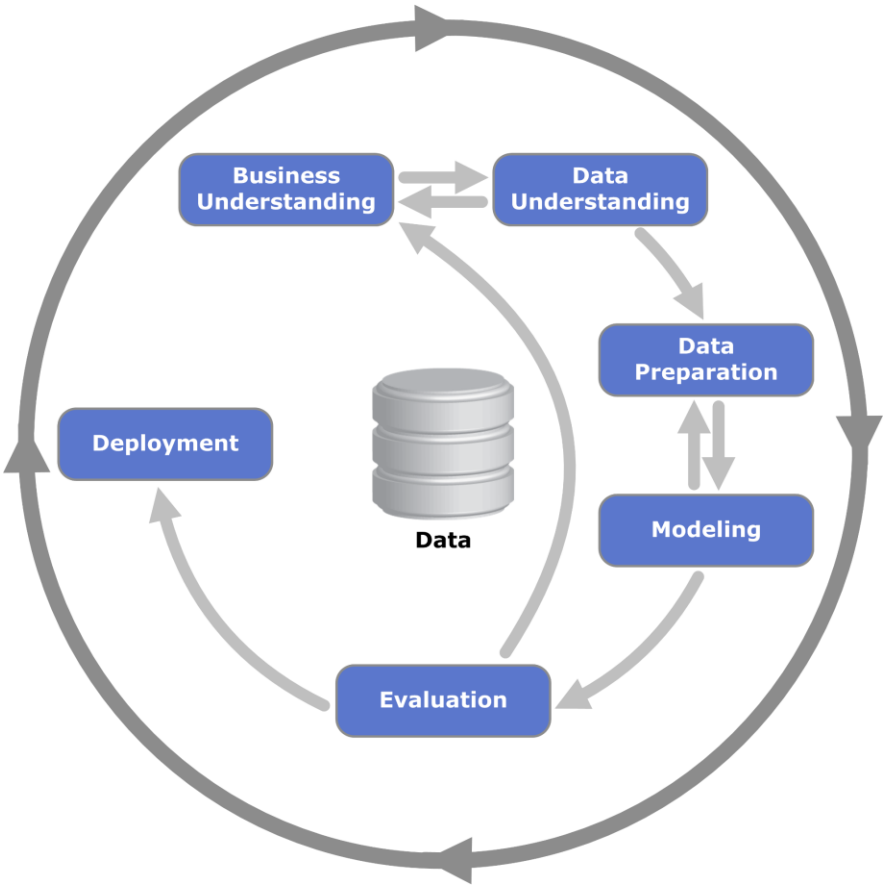


Figure 3.1.1 CRISP-DM Process for Data Mining

b.) Iterative Development Framework

Agile Methodology

Agile methodology is a management framework that promotes a flexible, incremental approach to developing system modules through continuous delivery, frequent testing, rapid prototyping and the ability to respond to change. Rather than organizing work into fixed-length sprints, the project adopts a Kanban-based approach in which tasks are visualized on a board and pulled through the stages Backlog, In Progress, Testing and Done. Work is grouped into five cadences that follow the CRISP-DM phases (Table 3.1.1). Each team member develops a module on a personal feature branch; before every merge into the shared Main-Dashboard branch, the integrator creates a backup branch so that a merge can be reverted, and the merged build is smoke-tested locally before it is pushed.



Figure 3.1.2 Agile Development Methodology

Table 3.1.1. Mapping Table

Cadence	CRISP-DM Phase	Analytics Focus	System Focus	Output

Cadence 1	Business and Data Understanding	Define operational questions and KPIs; profile toll, incident, Waze, weather and event data and record data gaps.	Provision the AWS RDS PostgreSQL/PostGIS database and Supabase authentication; define the warehouse layers.	Data inventory and exploratory analysis report
Cadence 2	Data Preparation	Load, clean and integrate the sources into bronze, silver and gold layers; derive time, incident, traffic, emission and spatial features.	Build the batch loaders and the dashboard upload pipeline with validation gates and undo.	Integrated multisource warehouse
Cadence 3	Modeling	Train candidate statistical, machine-learning and deep-learning models for volume, congestion, incidents, fleet mix and CO_2; build prescriptive models.	Build the dashboard analytics pages, the Live Map and the agent-based simulation sandbox.	Candidate models and prescriptive modules
Cadence 4	Evaluation	Evaluate models chronologically against naive baselines using WMAPE, MASE, RMSE and classification accuracy; select champions.	Integrate model outputs, accuracy displays and plain-language narratives; verify the sandbox engine.	Model evaluation results and verification logs
Cadence 5	Deployment	Publish accepted model outputs and schedule refresh and weekly retraining with an accuracy gate.	Package the dashboard as a desktop application, connect the mobile application and run system tests and user-acceptance testing.	Demonstration deployment of the decision-intelligence platform

3.2 Business Process

3.2.1 Comparison of Existing and Proposed Business Processes

Comparing the existing and proposed business processes identifies where the current process can be improved. The existing process serves as the baseline against which the proposed process and its supporting technologies are justified.

3.2.1.1 Process Map

3.2.1.1.1 Existing Business Process/es

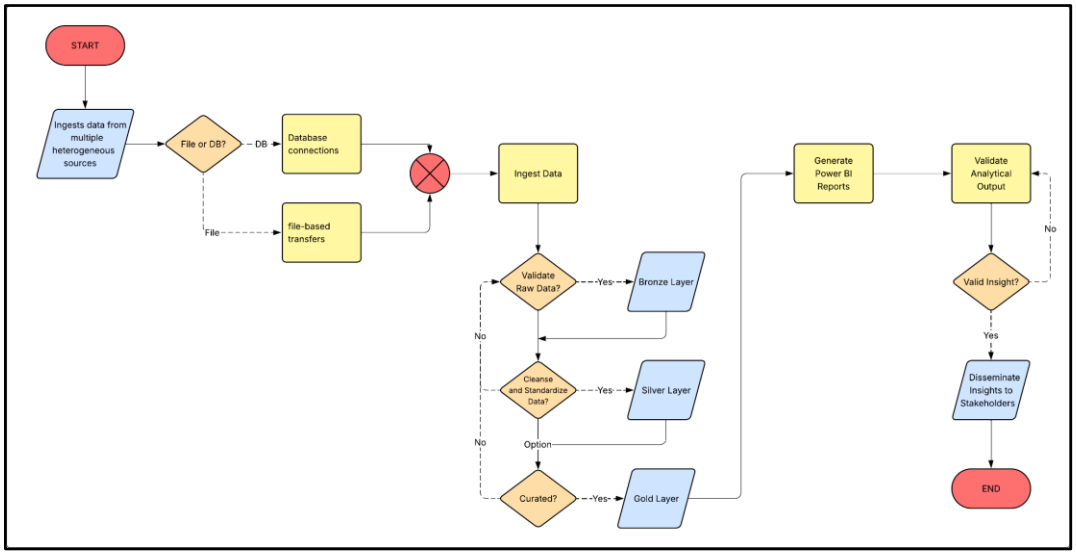


Figure 3.2.1.1.1.1 Flowchart Diagram (Existing Business Process)

The current NLEX business process involves The Central IT and Central Operations Maintenance and Management units at NLEX work in tandem to transform raw expressway data into operational intelligence through a structured data pipeline. Central IT manages the foundational infrastructure, using Apache NiFi and Kubernetes to ingest heterogeneous data from toll systems and lane alarms into a MinIO and Amazon S3 storage environment. This data is processed through a medallion architecture on Databricks, where the engineering team maintains the Bronze (raw) and Silver (cleansed) layers to ensure high data quality and reliability.

Transitioning to the analytical phase, the Big Data Analytics (BDA) team within Central Operations curates the Gold layer by applying specific business logic to create datasets optimized for traffic demand and road quality reports. The process culminates in the dissemination of validated insights via Microsoft Power BI and spreadsheet models, providing executive leadership with the necessary tools for data-driven decision-making.

This description of the existing process is based on the proponents' consultation with NLEX personnel and has not been independently verified against internal NLEX documentation. SmartFlow NLEX does not connect to the NLEX Databricks environment. For the capstone, NLEX data exports were loaded into a separate PostgreSQL warehouse that follows the same bronze, silver and gold layering, so the proposed process can be evaluated without modifying NLEX production systems.

3.2.1.1.2 Proposed Business Process/es

The proposed business process defines the users of the system and the actions each role may perform. Every user signs in with an email address and password followed by a six-digit time-based one-time password (TOTP) generated by an authenticator application. The second factor is enforced by the application programming interface (API), which rejects any session that has not completed it. Operators may mark a device as trusted for seven days when the server is configured with a device-trust key, and idle sessions are signed out automatically after 5, 10, 15, 30 or 60 minutes (15 minutes by default).

Table 3.2.1.1.2.1 System Users

User Role	Description	Responsibilities	Modules Access	Permissions	Authentication
SmartFlow NLEX Dashboard					
Data Analyst	Administrator of the analytics platform	Manage data uploads and undo, review model accuracy, audit system activity, and configure the mobile application	All modules: Overview, Traffic, Incidents, Emissions, Live Map, Maintenance, Mobile App, Scenario Sandbox, Data Management, Audit Log, Settings	Create, Read, Update and Delete	Email + password + authenticator (TOTP)
NLEX Traffic Control Center (TCC) Operator	Operations staff responsible for corridor-wide traffic flow	Monitor live corridor conditions, review forecasts and recommendations, run what-if scenarios, and schedule road maintenance	Overview, Traffic, Incidents, Live Map, Maintenance, Mobile App, Scenario Sandbox, Settings	Read; Create, Update and Delete for maintenance schedules	Email + password + authenticator (TOTP)
Incident Operator	Operations staff dedicated to incident management	Monitor incidents and live Waze alerts, and review incident analytics and response recommendations	Overview, Traffic, Incidents, Live Map, Maintenance (view), Mobile App, Settings	Read	Email + password + authenticator (TOTP)
SmartFlow NLEX Mobile					
Commuters / Drivers	End users of the mobile application	View corridor status and congestion forecasts, read alerts, post community reports, and ask the assistant	Dashboard, Corridor, Community, Assistant, Alerts	Read; Create for community posts	Email + password + authenticator (TOTP)

A. Dashboard Swimlane Diagrams

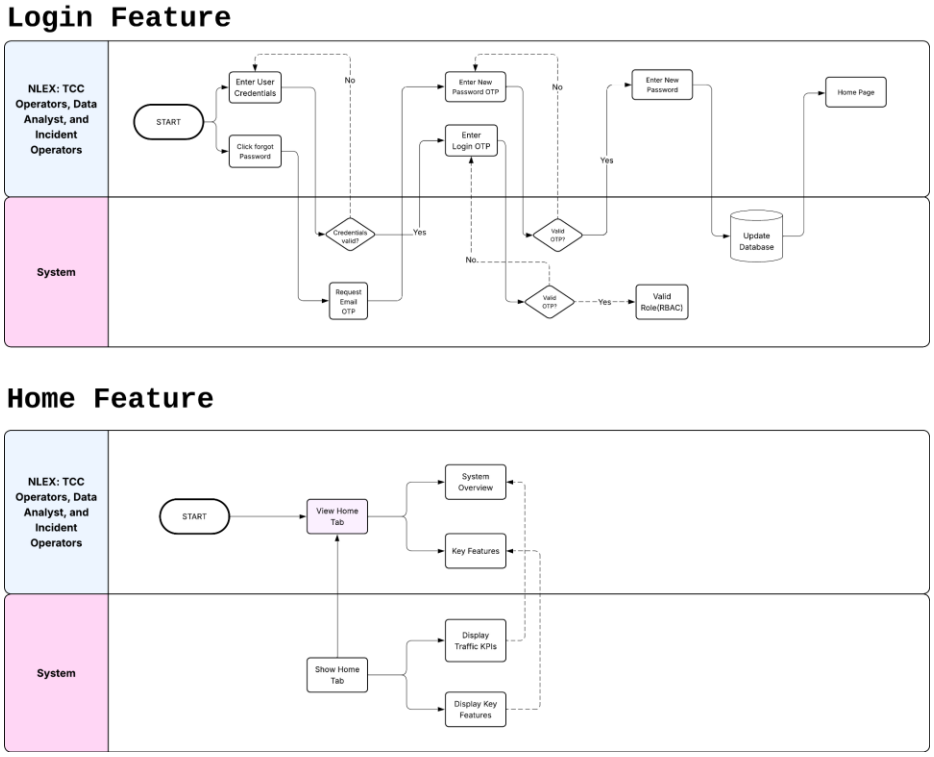
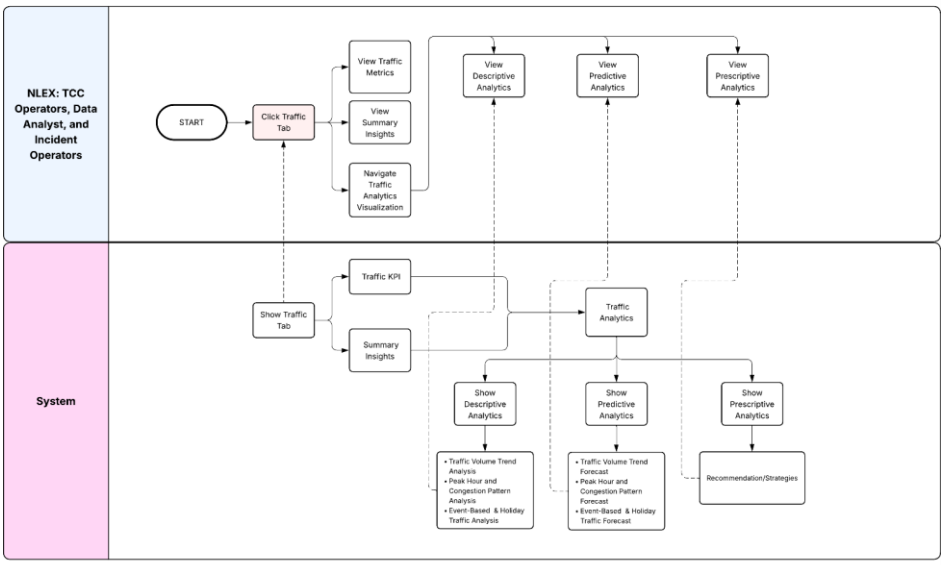


Figure 3.2.1.1.2.1 Dashboard Login and Home Feature Swimlane Diagram

(Proposed Business Process)

The Login Feature (Figure 3.2.1.1.2.1) is the system's security gateway. The user enters an email address and password or starts the Forgot Password sequence, which sends a reset link by email. After the credentials are accepted, the system requests the six-digit code from the user's authenticator application (shown as “OTP” in the figure). The API accepts only sessions that have completed this second factor, unless the device was marked as trusted within the last seven days. Role-based access control then determines which modules appear. The Overview page is the landing page and shows live corridor status by exit, derived from the Waze feed, together with navigation to every module the role may open.

Traffic Feature



Incident Feature

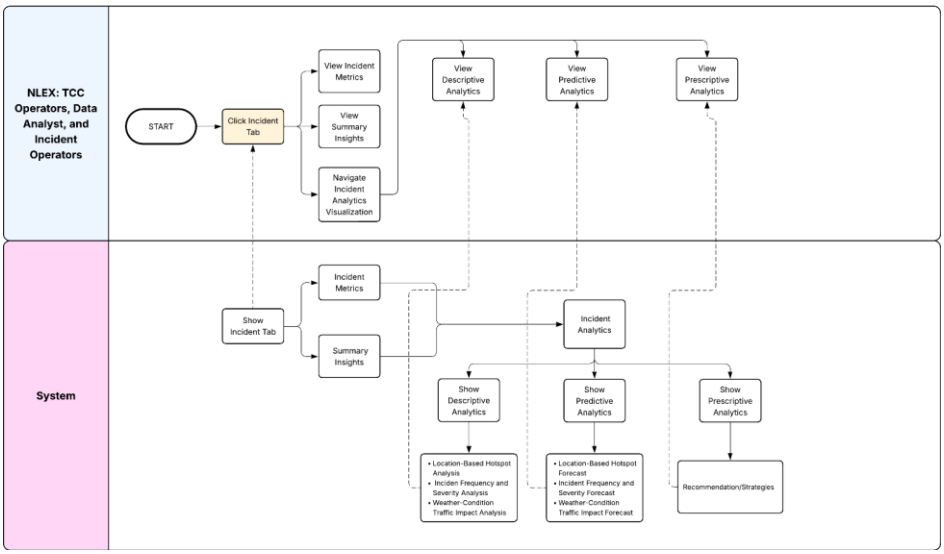
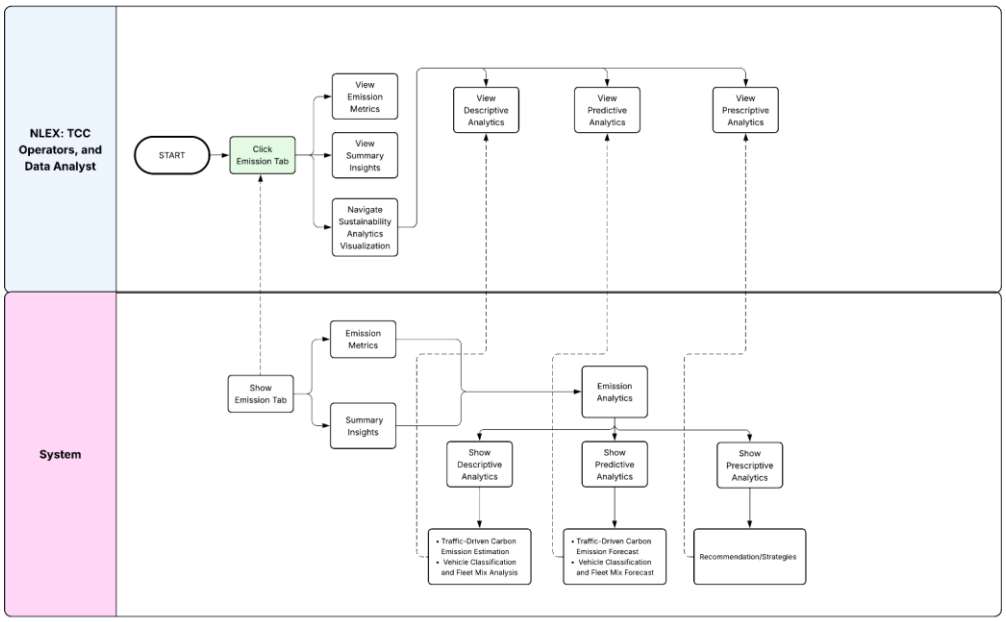


Figure 3.2.1.1.2.2 Dashboard Traffic and Incident Feature Swimlane Diagram

(Proposed Business Process)

Figure 3.2.1.1.2.2 shows that the Traffic and Incident features follow the same three-tier structure. Selecting a tab opens its Descriptive, Predictive and Prescriptive views. The descriptive view presents historical patterns computed from the warehouse: volume trends, hour-by-weekday profiles, holiday and event impacts, incident frequency, hotspots by kilometre post, wet-versus-dry crash rates and clearance times. The predictive view presents stored outputs of the champion models, such as daily volume, congestion state by exit and hour, incident counts and clearance-time estimates, together with each model's held-out accuracy. The prescriptive view presents recommendations computed from those forecasts: toll-booth staffing, congestion-response urgency, patrol positioning and the incident clearance protocol. Each chart can be accompanied by a plain-language narrative generated from the values already displayed.

Emission Feature



Agent Based Simulation Feature

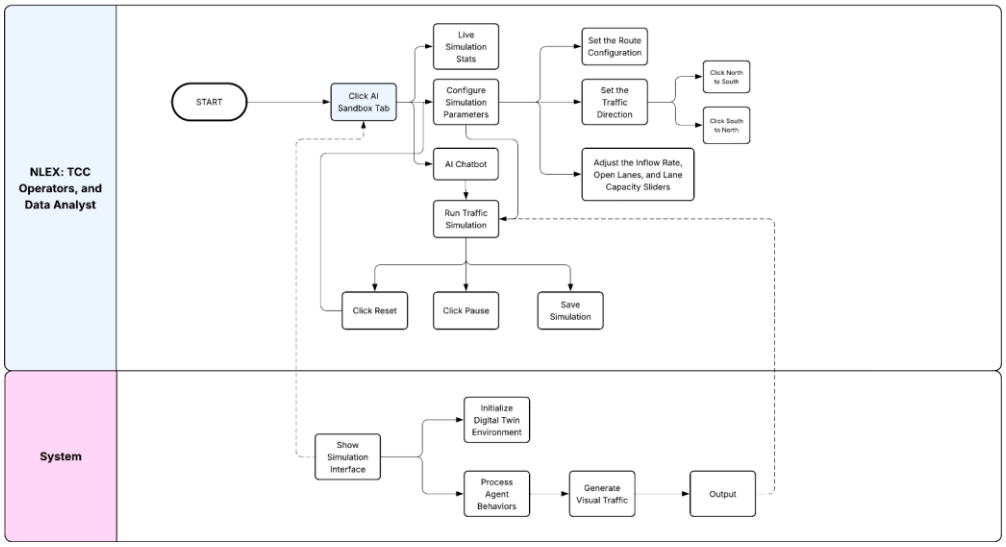


Figure 3.2.1.1.2.3 Dashboard Emission and Agent Based Simulation Feature

Swimlane Diagram (Proposed Business Process)

Figure 3.2.1.1.2.3 shows the Emission feature and the agent-based simulation feature. The Emissions tab presents descriptive estimates of corridor CO_2, NO_x and PM2.5 by vehicle class and exit, a CO_2 forecast and a fleet-mix forecast, and prescriptive emission strategies ranked by the Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) with bootstrap uncertainty ranges. The Scenario Sandbox (labelled AI Sandbox in the design figure) lets an operator choose a kilometre stretch, direction and time, stage interventions such as lane closures, speed limits, counterflow and incidents, either with on-screen controls or by typing a command that a language model converts into scenario settings, and then run a microscopic traffic simulation that compares the intervention with a no-intervention baseline.

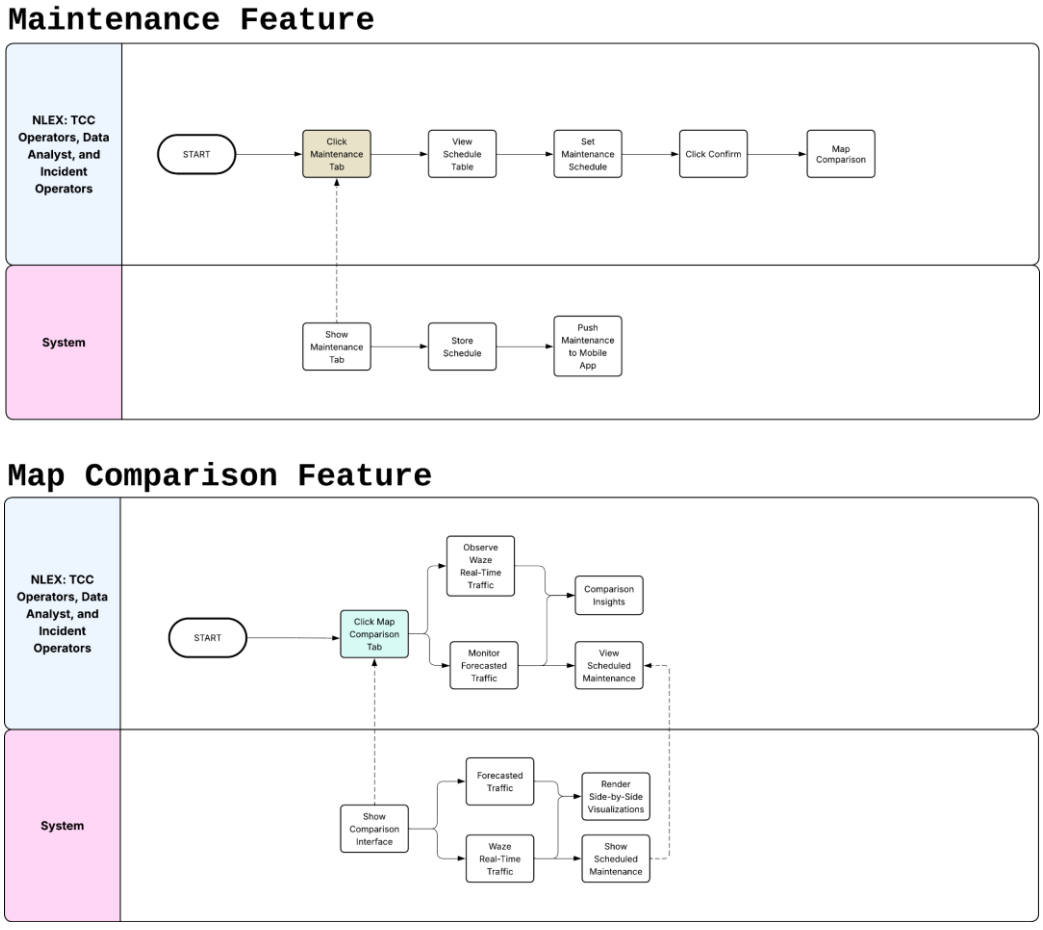
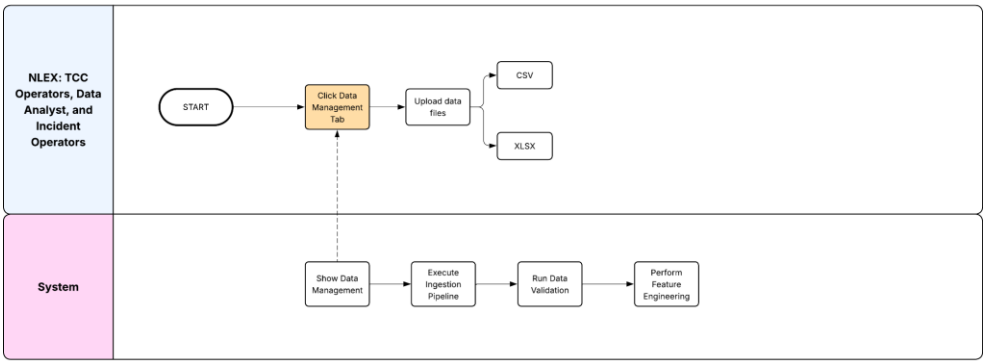


Figure 3.2.1.1.2.4 Dashboard Maintenance and Map Comparison Feature

Swimlane Diagram (Proposed Business Process)

Figure 3.2.1.1.2.4 shows the Maintenance and Live Map features. In the Maintenance feature, an authorized user views the schedule table, enters a road-work schedule by exit, direction and official kilometre posts, and confirms it. The schedule table lists ten schedules per page, and hovering over a bar in the timeline or corridor view shows the work's status, direction, kilometre range, nearest exit, time window and lane closure. The system validates and stores the schedule, the mobile application retrieves it and shows it to commuters as a notice. The Live Map (labelled Map Comparison in the design figure) keeps the side-by-side comparison: the left map shows live Waze jams and alerts from the last 60 minutes, and the right map shows the forecast congestion state for a chosen hour up to seven days ahead, together with the model that produced it. Active maintenance closures are drawn on both maps, and a corridor strip below them summarizes either view by exit.

Data Management Feature



Audit Log Feature

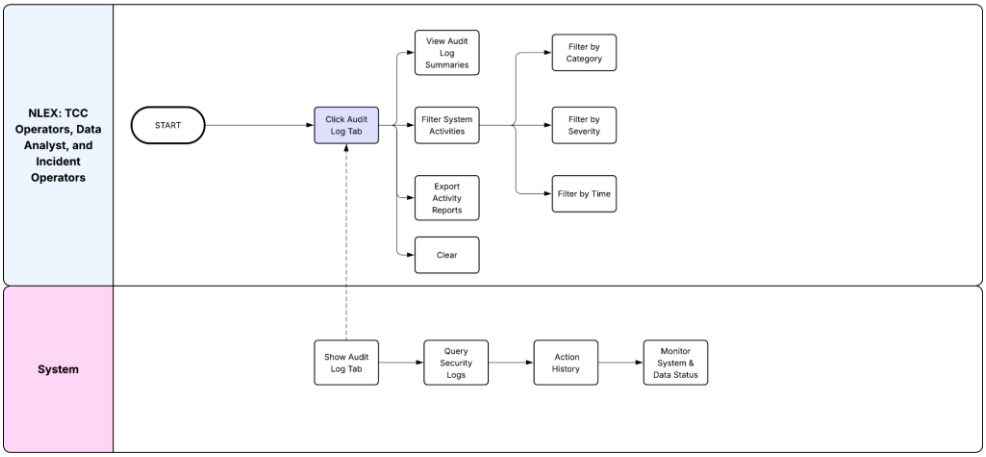


Figure 3.2.1.1.2.5 Dashboard Data Management and Audit Log Feature

Swimlane Diagram (Proposed Business Process)

Figure 3.2.1.1.2.5 shows the Data Management and Audit Log features. A data analyst uploads a CSV or XLSX export of toll, accident or breakdown records. The system classifies the file, runs it through the cleaning, transformation and validation gates, and either reports the result without saving (check mode) or loads it into the warehouse in a single transaction that can later be undone. Accepted data becomes available to the weekly retraining batch; uploading does not retrain models immediately. The Audit Log records sign-ins, sign-outs, idle time-outs, page views, uploads, maintenance changes, model retraining events and each hourly congestion-forecast refresh, and lets the analyst filter them by category, severity and time and export them. Routine page views and hourly refreshes are hidden from the list unless requested, and only signed-in data analysts can read or export the log.

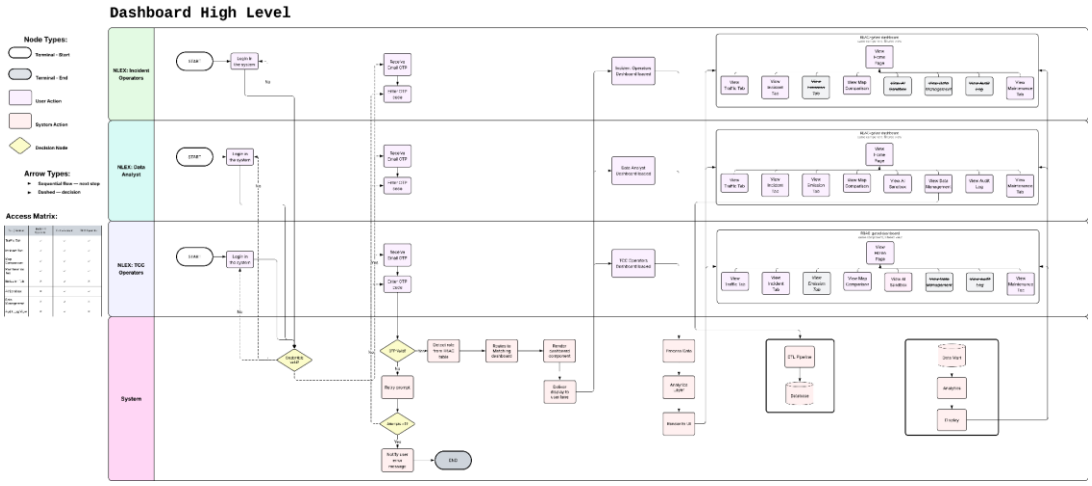
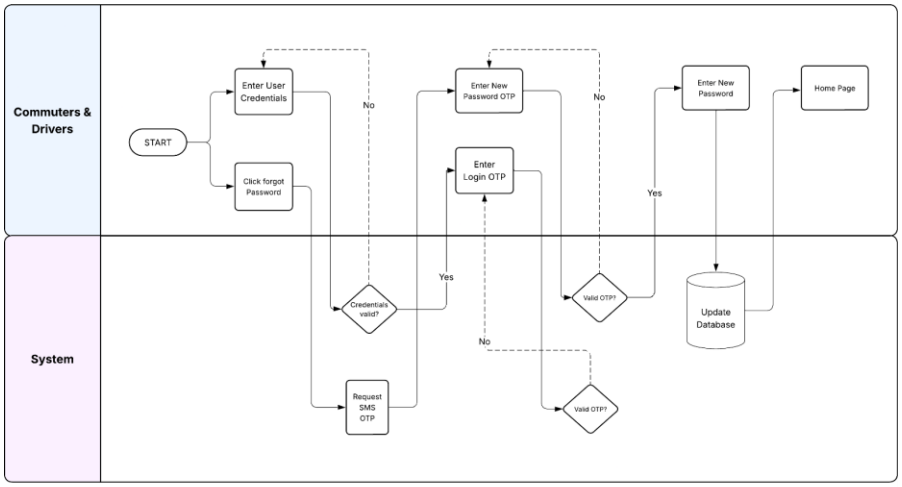


Figure 3.2.1.1.2.6 Dashboard High Level Swimlane Diagram (Proposed Business Process)

Figure 3.2.1.1.2.6 presents the high-level dashboard swimlane across the three operator roles. All users pass the same sign-in and authenticator check, after which role-based access control determines the visible modules. Incident Operators reach the analytics pages, the Live Map and the maintenance schedule in read-only form. TCC Operators additionally reach the Scenario Sandbox and may edit maintenance schedules. Data Analysts reach every module, including Emissions, Data Management and the Audit Log. All features draw on the same warehouse, which is populated by the ingestion pipeline and by the scheduled model jobs.

B. Mobile Swimlane Diagrams

Mobile Login Feature



Mobile Dashboard Feature

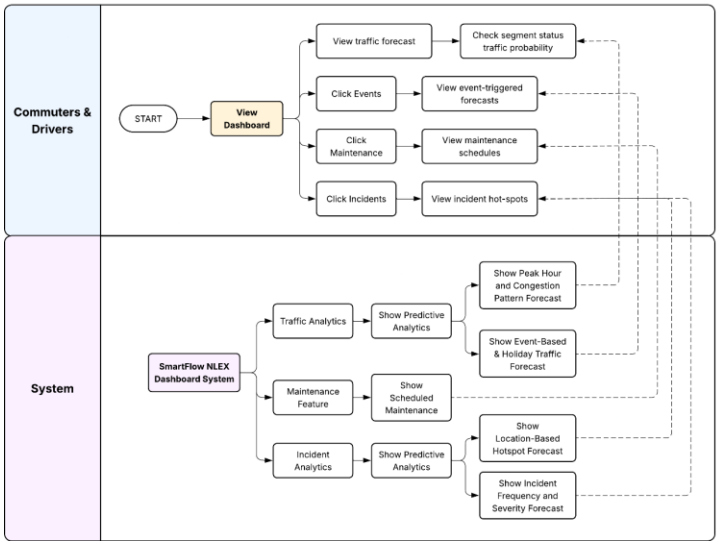


Figure 3.2.1.1.2.7 Mobile Login and Dashboard Feature Based Swimlane Diagram (Proposed Business Process)

Figure 3.2.1.1.2.7 shows the login and dashboard workflow of the mobile application across two swimlanes, Commuters & Drivers and System. A commuter signs in with an email address and password and then the six-digit code from an authenticator application (the figure shows this step as an SMS OTP; the implemented second factor is TOTP). The commuter may also reset a forgotten password through an emailed link.

After signing in, the Dashboard tab shows the corridor's live status, the volume and congestion outlook, upcoming maintenance notices and recent incidents, all retrieved from the SmartFlow NLEX API that also serves the operator dashboard. Operators control which sections appear in the application through the Mobile App page of the dashboard, so that advisories can be published to commuters without releasing a new version of the application.

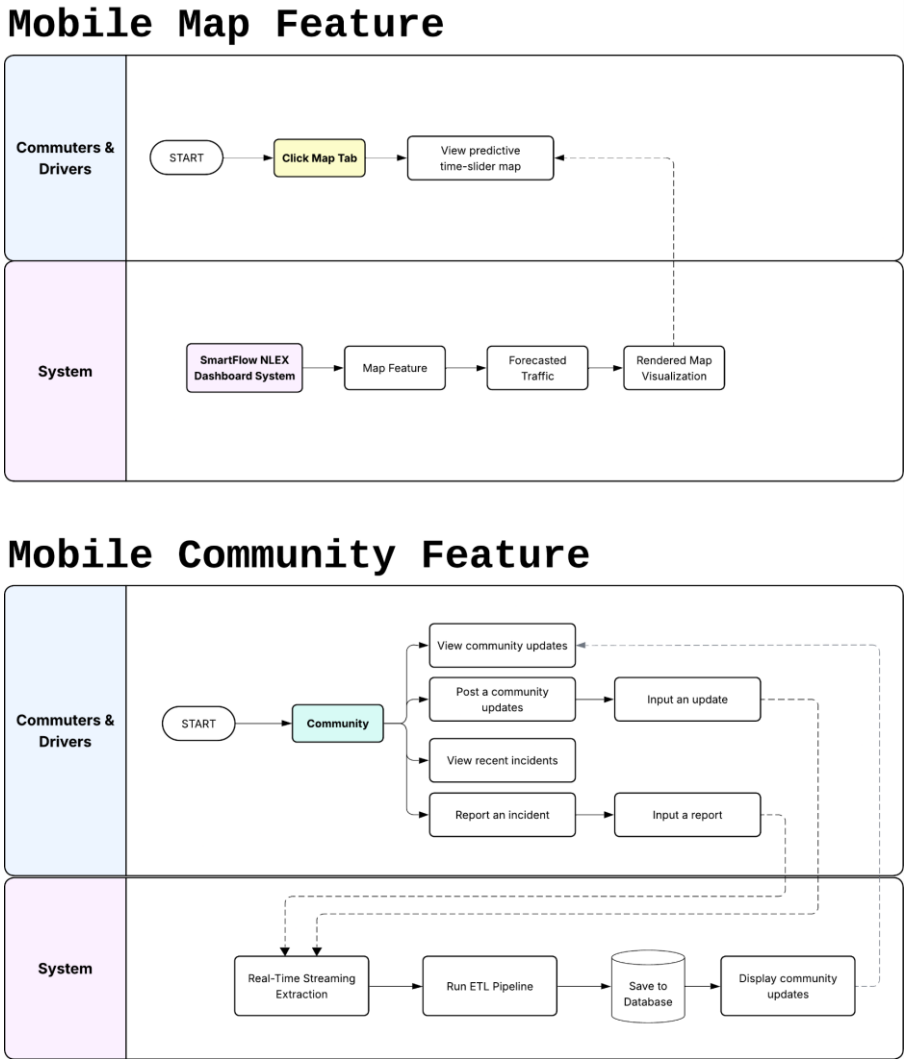


Figure 3.2.1.1.2.8 Mobile Map and Community Feature Based Swimlane Diagram (Proposed Business Process)

Figure 3.2.1.1.2.8 shows the Corridor (Map) and Community features. In the Corridor tab, the commuter switches between the live view and the forecast view and chooses a forecast hour. The application then retrieves the predicted congestion state of each exit for that hour, up to the model's published horizon of 168 hours. In the Community tab, the commuter reads and posts community updates and incident reports with optional photos. Community posts are stored in the application's own Supabase database and shown in the application feed. In the current version they are not loaded into the analytics warehouse and are not used by the forecasting models.

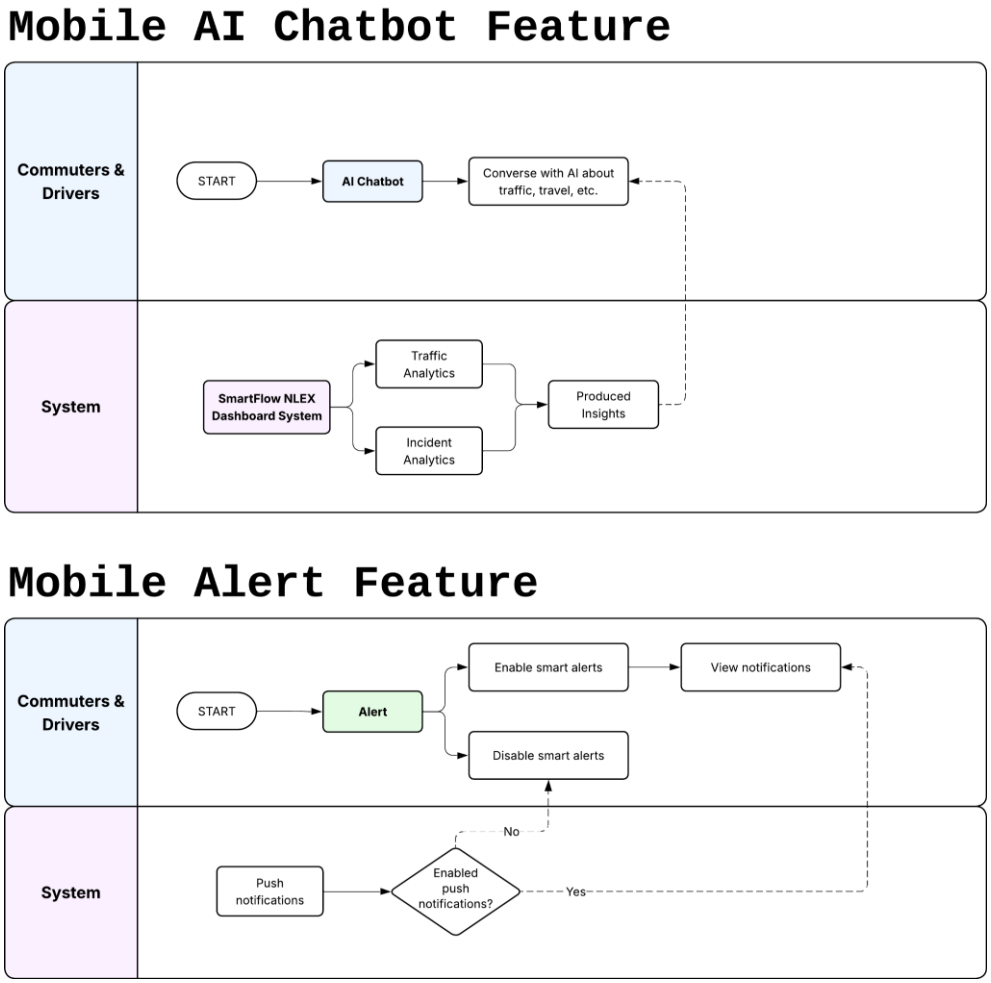


Figure 3.2.1.1.2.9 Mobile AI Chatbot and Alert Feature Based Swimlane Diagram

(Proposed Business Process)

Figure 3.2.1.1.2.9 shows the Assistant and Alert features. The Assistant answers questions about the NLEX corridor in English, Tagalog or a mix of both. A large language model (Qwen, accessed through OpenRouter) interprets the question and calls three tools that return live corridor status, a corridor overview and the list of exits from the SmartFlow NLEX API. It then phrases the answer in plain language. The assistant does not produce forecasts of its own and declines questions outside the NLEX corridor. In the Alert feature, the user enables or disables notifications. When notifications are enabled, the application schedules notifications on the device for operator advisories, scheduled maintenance and new incident reports while the application is in use. Server-sent push notifications are not implemented in the current version.

Mobile High-Level

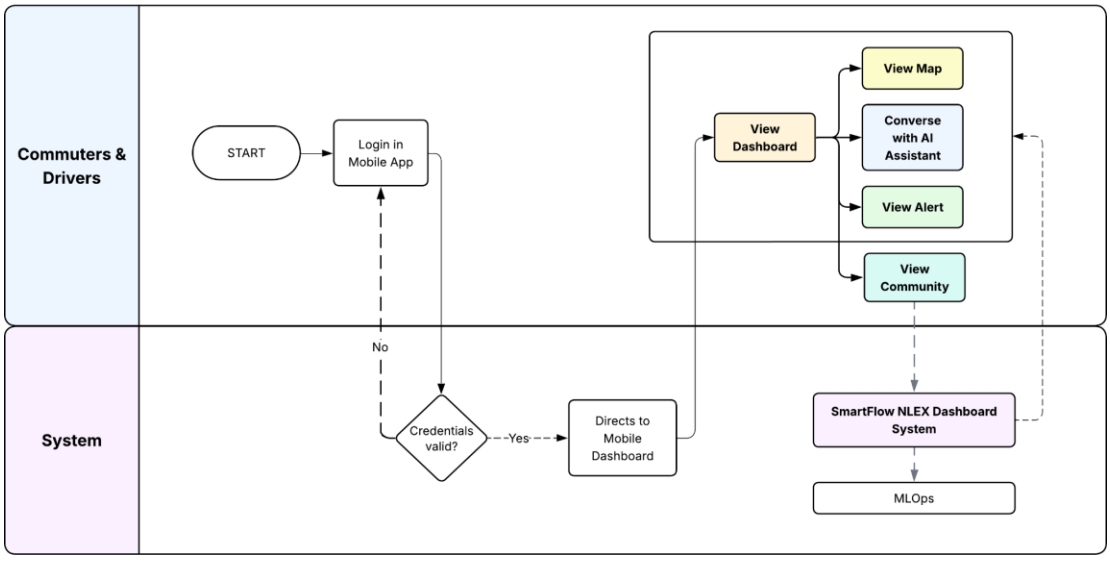


Figure 3.2.1.1.2.10 Mobile High Level Swimlane Diagram (Proposed Business Process)

The mobile high-level swimlane diagram (Figure 3.2.1.1.2.10) summarizes the end-to-end interaction of the mobile application across the Commuters & Drivers and System swimlanes. The workflow starts with sign-in, in which the system validates the credentials and the authenticator code and either returns the user to the login screen or opens the mobile dashboard.

From the dashboard, the user reaches five modules: Dashboard, Corridor, Community, Assistant and Alerts. The Dashboard, Corridor and Alert modules read live status, forecasts, maintenance schedules and advisories from the SmartFlow NLEX API. Community posts and authentication use Supabase. The Assistant uses a language model with tool access to the live corridor data. The machine-learning models themselves run in the operator-side pipeline and are retrained weekly. The mobile application consumes their stored outputs and does not train or modify models.

3.2.1.2 Data Flow Diagram

The SmartFlow NLEX data pipeline moves data from its sources through staging, transformation and a layered warehouse to the analytics models and the user interfaces. Figure 3.2.1.2.1.1 presents the pipeline as designed. The paragraphs below describe each layer as implemented and note where the implementation differs from the design. The differences are summarized in Table 3.2.1.2.1.

3.2.1.2.1 Proposed Data Pipeline

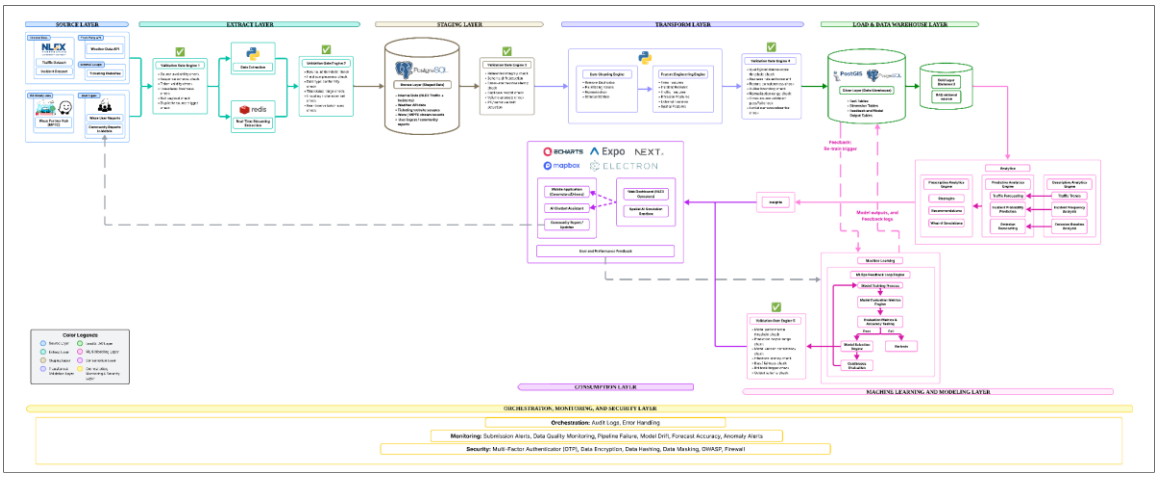


Figure 3.2.1.2.1.1 Data Pipeline

a.) Source Layer

The Source Layer gathers data from six origins. The first is NLEX Corporation exports: hourly toll transactions by plaza, direction and vehicle class, and accident, breakdown and crash records. The second is the Waze for Cities (Partner Hub) feed, both as a historical archive and as a live feed. The third is historical hourly weather from Open-Meteo, ERA5 and the Japan Meteorological Agency (JMA), with historical air quality from the OpenWeather Air Pollution service. The fourth is a Philippine Arena event calendar collected once from the venue's published schedule, and the fifth is the Philippine holiday calendar. The sixth is community reports submitted through the mobile application, which are stored separately and are not yet part of the analytics pipeline.

b.) Extract Layer

In the Extract Layer, historical data were loaded in batches. The 2022–2025 toll and Waze archives and the weather history were loaded by one-off scripts. New NLEX exports are loaded through the dashboard's Data Management upload, which parses the file and runs the validation gates. The live Waze feed is polled every two minutes by a separate ingestion process on AWS in the Asia Pacific (Singapore) region. Each cycle writes the feed to the database and also overwrites a short-lived snapshot of active jams and alerts in an Upstash Redis store, which expires after ten minutes so that a stalled feed is not presented as current. Redis therefore holds the latest snapshot rather than buffering records before they reach the database, and the dashboard's live map reads the more complete PostgreSQL copy through a 30-second in-memory cache.

c.) Staging Layer

Extracted records are stored first in the bronze schema of an Amazon Web Services (AWS) Relational Database Service (RDS) PostgreSQL 18 database with the PostGIS 3.6 extension, which keeps each source in its received form. Before records move on, the upload pipeline checks that the file parses and is non-empty, classifies it by its header signature, rejects records without a valid date, location or kilometre post, holds back dates after the end of the record, and skips duplicates by natural key. The design's personally identifiable information (PII) detection gate is not implemented. Breakdown records keep plate-number and driver fields in the warehouse, and masking them is listed as a required improvement.

d.) Transform Layer

The Transform Layer cleans and standardizes the staged records. Exit names are resolved to canonical names, dates and times are normalized, values outside physical ranges are removed (for example, clearance times outside 0–1,440 minutes and negative casualty counts), and the most reliable weather reading per hour is retained. Feature engineering derives calendar features (hour, weekday, month, weekend and holiday flags), traffic aggregates by exit, direction and class, incident durations and kilometre positions, per-class emission estimates and event and weather indicators. Imputation is applied only by the weather source, which flags imputed rows; the warehouse excludes those rows rather than interpolating.

e.) Load & Data Warehouse Layer

Validated records are published to the silver schema (cleaned records) and the gold schema (analysis-ready facts, aggregates and model outputs). The main fact table is gold.fact_traffic_hourly, which holds hourly toll entries and exits by plaza, direction and class. Supporting tables include the emission fact, incident event tables, Waze jam and alert facts with PostGIS geometries, and dimension tables for time, location and incident type. Model outputs and accuracy metrics are written to dedicated gold tables (Section 3.10). The gold layer serves the dashboard and the mobile application through the API. No retrieval-augmented generation (RAG) index was built on it.

f.) Analytics and Machine Learning Layer

The analytics layer reads the gold layer. Descriptive analytics are computed on request by SQL queries in the API. Predictive models are trained offline in Python, and their forecasts and held-out metrics are stored in gold tables. The congestion-state forecast is refreshed every hour: XGBoost and the GRU are refitted on the live Waze feed, the LSTM candidates (refitted in the weekly batch) are rescored on the same test hours, and the served model is chosen again. The other models, including the 48-hour hourly volume model, are retrained in a weekly batch that keeps a backup of the current models and publishes a new model only if it passes an accuracy gate against naive baselines; otherwise the previous model is restored. Prescriptive models (staffing, patrol positioning, clearance protocol, congestion urgency and emission strategy ranking) are computed by the API from the stored forecasts when a page is viewed.

g.) Consumption Layer

The Consumption Layer comprises two interfaces. Operators use a Next.js 15 and React 19 dashboard built as a static export and packaged as a Windows desktop application with Electron, which uses Apache ECharts for charts and Mapbox GL JS for the corridor map. Commuters use a mobile application built with Expo and React Native, which provides corridor status and forecasts, community reports, alerts and a conversational assistant.

h.) Orchestration, Monitoring, and Security Layer

The Orchestration, Monitoring and Security Layer supports the whole pipeline. Orchestration consists of GitHub Actions workflows that run the hourly congestion refresh at two minutes past each hour and the weekly retraining batch on Sunday at 22:00 (Philippine time). The weekly batch runs as three consecutive jobs because a hosted job may run for at most six hours, and trained model files are kept between runs in the workflow cache. The database credentials are read from encrypted repository secrets. A Windows scheduled task adds the latest finished hours of Open-Meteo weather every hour. Each hourly congestion refresh writes an audit-log entry recording whether it published, was skipped while the weekly batch was training, or failed. Monitoring consists of the audit log, the retraining batch log, a feed-freshness check that flags live data older than 30 minutes, and a log of served congestion forecasts that is later scored against observed jams. Security consists of authenticator-based sign-in, role-based access control on administrative and write endpoints, rate limiting and encrypted connections to the database. Gaps in the current implementation are discussed in Section 3.4.1 and Section 3.14.

Table 3.2.1.2.1. Data Pipeline Design Compared with the Implemented Pipeline

Layer	Design (Figure 3.2.1.2.1.1)	Implemented	Status
Source	NLEX data, weather API, ticketing scraper, Waze Partner Hub, community reports	NLEX exports; Waze archive and live feed; Open-Meteo/ERA5/JMA weather history with hourly Open-	Partial: no recurring scraper, community

		Meteo updates; OpenWeather air-quality history; one-off Philippine Arena calendar; holiday calendar	reports not ingested
Extract	Scheduled Python batch plus Redis real-time streaming; Gates 1–2	Batch loads; dashboard file upload and a command-line loader for yearly toll exports; live Waze polled every 2 minutes into PostgreSQL, with a 10-minute Redis snapshot of active jams and alerts	Partial: Redis holds a short-lived snapshot, not a buffer in front of the database; the dashboard reads PostgreSQL
Staging	PostgreSQL bronze layer; Gate 3 with PII detection	AWS RDS PostgreSQL 18 + PostGIS bronze schema for incident uploads (hourly toll exports are written directly to gold); parse, classify, required-field, date and duplicate checks	Partial: no PII detection or masking
Transform	Cleaning and feature engineering; Gate 4	Canonical exits, normalization, physical-range and business rules, calendar, traffic, incident, emission and spatial features	Implemented
Load / Warehouse	PostGIS silver warehouse; gold datamart used as RAG source	Silver and gold schemas; gold fact and model-output tables; PostGIS geometry for Waze and segments	Implemented; no RAG index
Analytics / ML	Descriptive, predictive and prescriptive engines; MLOps loop with Gate 5	SQL descriptive queries; offline Python training; hourly congestion refresh; weekly retrain with accuracy gate and restore; API-computed prescriptive models	Implemented (GitHub Actions scheduler)
Consumption	Next.js + Electron dashboard; Expo mobile app with AI chatbot	Next.js static export in Electron; Expo/React Native app; GLM narratives and sandbox commands; Qwen tool-calling assistant	Implemented

Orchestration, monitoring, security	Audit logs, drift monitoring, MFA, encryption, masking	GitHub Actions scheduled workflows (hourly and weekly); audit log including each model refresh; feed freshness; forecast log; TOTP MFA; role checks; TLS to database without certificate verification	Partial: no masking, no alerting

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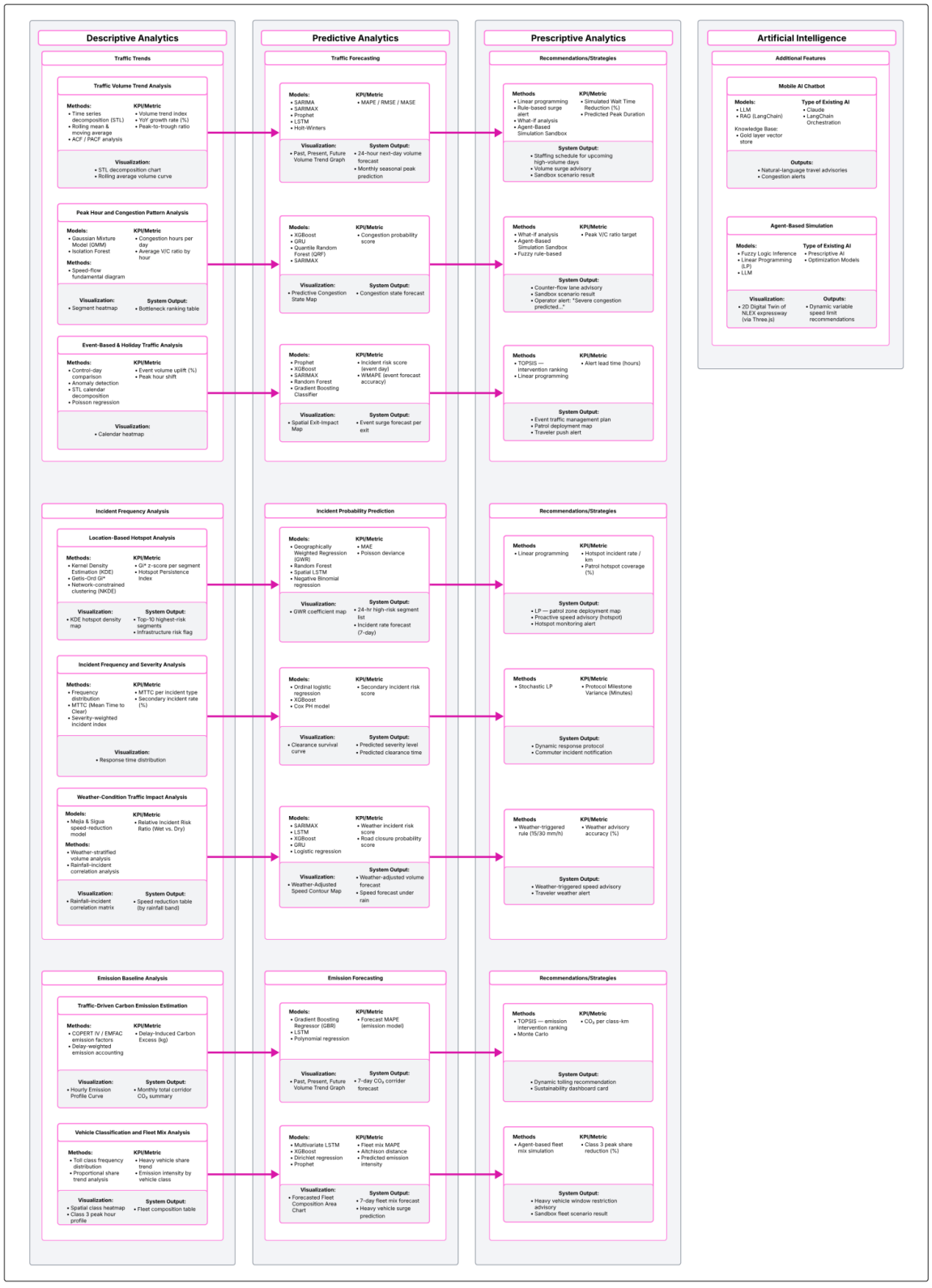


Figure 3.2.1.2.2.2 Data Pipeline - Modeling Layer

Figure 3.2.1.2.2.2 presents the modeling layer, which organizes the analytics into three tiers. Descriptive analytics establish historical baselines, predictive analytics anticipate future conditions, and prescriptive analytics recommend or evaluate interventions. The layer covers three operational domains. The figure reflects the original design. The methods that were implemented, partially implemented or deferred are listed in Section 3.11 and in Table 3.11.1.

In the Traffic Volume and Congestion domain, the descriptive tier aggregates hourly toll entries into daily, hourly and hour-by-weekday patterns and measures holiday and event effects against local control-day baselines. The predictive tier forecasts daily corridor volume with candidate statistical, machine-learning and deep-learning models evaluated by rolling-origin validation (Prophet is the current champion). It also classifies the congestion state of each exit for 1 to 168 hours ahead with the best of seven candidate classifiers, chosen again at every hourly refresh on the live Waze feed, and estimates event-day uplift at the Philippine Arena exit. It also forecasts hourly volume by plaza, direction and class for the next 48 hours with a stacked LSTM. The prescriptive tier converts these forecasts into an hourly toll-booth staffing plan, a fuzzy-rule congestion-response urgency (Monitor, Prepare or Act) and a TOPSIS ranking of event interventions.

In the Incident Management and Safety domain, the descriptive tier counts incidents by type, cause, hour and five-kilometre segment, compares crash rates in wet and dry hours, and summarizes clearance times with means, percentiles and Kaplan–Meier curves. The predictive tier forecasts daily incident counts with a negative binomial regression, estimates severity and clearance time (ordinal logistic regression, XGBoost and a Cox proportional hazards model), and models spatial variation with geographically weighted regression and a spatial LSTM. The prescriptive tier positions patrol units with a maximal covering location model and recommends how many response units to dispatch and when to escalate with a two-stage stochastic mixed-integer program that accounts for uncertain clearance times.

In the Environmental Impact domain, the descriptive tier estimates CO_2, NO_x and PM2.5 by multiplying the vehicle-kilometres of each toll trip by fixed per-class emission factors and estimates the additional CO_2 emitted by vehicles idling behind lane-blocking incidents. The predictive tier forecasts corridor CO_2 with gradient boosting regression and the vehicle-class mix with Dirichlet regression. The prescriptive tier compares three strategies: shorter lane blockages, faster response in the busiest hours and a lower heavy-vehicle share (a what-if only). It attaches bootstrap uncertainty ranges and ranks the strategies with TOPSIS, and also identifies the exits and hours where a heavy-vehicle time window or a peak toll differential would apply. Operators can test such interventions in the Scenario Sandbox before considering them on the road.

3.3 Business Solution

SmartFlow NLEX is a decision-intelligence system intended to help NLEX operators move from reactive monitoring to anticipatory traffic management, particularly for non-routine conditions such as incidents, adverse weather, holidays and large events. It integrates NLEX toll and incident records with crowdsourced Waze data, weather, holiday and event calendars in one warehouse. On top of that warehouse it provides descriptive baselines, validated forecasts and recommendations, and it reports the accuracy of every forecast so that operators can judge how far to rely on it.

The solution is delivered through two connected products. The first is an operator dashboard for the Traffic Control Center that presents traffic, incident and emission analytics, a live corridor map and maintenance scheduling. It also includes a scenario sandbox in which operators can simulate interventions such as lane closures or counterflow on a chosen stretch before applying them on the road. The second is a mobile application for commuters that shows live corridor status, congestion forecasts by exit for up to seven days, maintenance notices and community reports, and offers a conversational assistant for corridor questions. Together, the products are intended to help operators prepare for predictable and unpredictable demand and to help commuters plan trips around expected congestion. The system is advisory: all operational decisions remain with NLEX personnel.

3.4 Project Risk and Feasibility Studies

This section examines whether the SmartFlow NLEX project is realistic and feasible. It covers risks that could arise during development, including whether the required technology is available, whether the system aligns with NLEX's operations, and whether the costs are manageable. The goal is to ensure the project can be built within the capstone period and used beyond it.

3.4.1 Risk Management

Like any software project, SmartFlow NLEX faces a number of risks from issues with data sources and APIs in the Source Layer, to problems with the pipeline's Transform and Validation processes, and even team-related concerns. The proponents identified the most likely risks and planned how to handle each one before it became a bigger problem.

The risks are evaluated using two factors: how likely they are to occur (Probability) and how severe their impact would be on the project if they did (Impact). Both are rated on a 1–5 scale, where 1 is low and 5 is high. The Risk Level is then determined based on the combination of these two scores, as shown in the matrix, Table 3.4.1.1.

Table 3.4.1.1. Risk Management Impact

	LOW (IMPACT)	**MEDIUM (IMPACT)**	**HIGH (IMPACT)**
LOW (PROBABILITY)	LOW	LOW	MEDIUM
MEDIUM (PROBABILITY)	LOW	MEDIUM	HIGH
HIGH (PROBABILITY)	MEDIUM	HIGH	HIGH

Table 3.4.1.2 Risk Management Legend

LEGEND		
Impact	**Probability (Likelihood)**	**Impact (Severity)**
1 (Low)	**Unlikely:** Rare occurrence; less than 20% chance of happening during the development cycle.	**Minor:** Small problem; easy to fix without delaying the project.
3 (Medium)	**Possible:** Moderate likelihood; 20% to 50% chance of occurrence. May have happened in similar capstone projects.	**Moderate:** Will affect a specific part of the pipeline and may require adjusting the timeline or reassigning tasks.
5 (High)	**Likely:** Frequent or expected occurrence; over 50% chance. High probability due to external	**Critical:** Serious problem that could prevent the project from being completed or break a

	dependencies (APIs, web structures).	core layer of the pipeline.

Table 3.4.1.3. Risk Management Table

Risk ID	Potential Risk Factor	Prob. (1-5)	Impact (1-5)	Risk Level	Mitigation & Contingency Action Plan
R01	Event data becomes outdated: the Philippine Arena calendar was collected once and is not refreshed automatically.	5	3	HIGH	Mitigation: assign an owner to update the event table before each month. Contingency: rely on holiday and weekday seasonality, and label event forecasts as unavailable for events not in the calendar.
R02	Live data interruption: the Waze feed or its ingestion process stops, so live map and congestion inputs become stale.	3	5	HIGH	Mitigation: feed-freshness check flags data older than 30 minutes, and congestion forecasts are flagged as overdue after 2 hours and as expired when no forecast hour remains. Contingency: show the last available data with its timestamp and fall back to historical profiles.
R03	Incomplete or unverified NLEX data: missing clearance timestamps, gaps in Waze history, and unverified 2020–2021 records.	5	5	HIGH	Mitigation: restrict analyses to January 2022 onward, report coverage with each KPI, and exclude invalid durations. Contingency: confirm data provenance with NLEX and document remaining gaps as limitations.

R04	Low model accuracy: candidate models fail to beat naive baselines (as LSTM did for volume).	5	3	HIGH	Mitigation: rolling-origin evaluation against seasonal-naive and climatology baselines, and a weekly retraining gate. Contingency: keep the previously accepted model or present the baseline when no model passes.
R05	Team member unavailability or scheduler failure: the hourly and weekly jobs depend on a hosted scheduler and on cached model files.	3	3	MEDIUM	Mitigation: version-control all code and workflow definitions; the jobs run on GitHub-hosted runners and log each run to the audit log. Contingency: run the same scripts from a project workstation and redistribute tasks.
R06	Scope creep: new features are added beyond the approved scope.	5	3	HIGH	Mitigation: adviser approval before backlog additions. Contingency: freeze the scope after Cadence 3 and move non-essential items to the post-capstone roadmap.

R07	Cloud and API cost overrun: AWS RDS, language-model APIs, Render and Mapbox usage exceed the project budget.	3	3	MEDIUM	Mitigation: use free tiers where available, cache API responses, and monitor billing monthly. Contingency: downgrade the database instance or use a local PostgreSQL for development.
R08	Security weaknesses: user-editable roles with open sign-up, unauthenticated write endpoints, public repositories that expose infrastructure details, and unmasked PII.	3	5	HIGH	Mitigation: store roles in server-controlled metadata, disable public sign-up, require authentication on all write endpoints, mask PII, and restrict database network access. Contingency: rotate credentials and inform the NLEX focal person within 24 hours of any compromise.
R09	Shared-database write contention: several developers run jobs that delete and rewrite the same tables.	5	3	HIGH	Mitigation: give each job its own output table and use a staging database for development. Contingency: restore from the model backup schema.

R10	Language-model provider dependency: latency, credit limits or provider changes affect narratives, sandbox commands and the assistant.	3	3	MEDIUM	Mitigation: template fallbacks for narratives and validation of every model response. Contingency: switch providers through configuration.
R11	Data-sharing and privacy compliance: NLEX data and personal fields are processed outside NLEX systems.	3	5	HIGH	Mitigation: keep repositories private, mask personal fields, and document the data-sharing agreement. Contingency: remove affected data and notify the NLEX data owner.

3.4.2 Technological Feasibility

SmartFlow NLEX is built with widely used, well-documented and mostly open-source technologies, which makes it technically feasible within the budget and time of a capstone project. The technologies listed below are the ones actually used in the implemented system.

For analytics and machine learning, the system uses Python with pandas and NumPy for data processing, statsmodels for SARIMAX, Holt-Winters and generalized linear models, Prophet for decomposable time-series forecasting, scikit-learn and XGBoost for tree-based models, TensorFlow/Keras and PyTorch

for the recurrent neural network comparators, lifelines for survival analysis, and mgwr for geographically weighted regression. Optimization models are solved with the YALPS linear and mixed-integer programming solver in TypeScript.

For data access and caching, the API keeps an in-memory cache of route responses with stale-while-revalidate behaviour and preloads frequently used responses at start-up. Live Waze data are written to PostgreSQL every two minutes by a polling process on AWS, which also refreshes a ten-minute Upstash Redis snapshot of active jams and alerts. The dashboard reads the database copy.

For data storage, the system uses PostgreSQL 18 with the PostGIS 3.6 spatial extension on AWS RDS in the Asia Pacific (Singapore) region. The warehouse is organized into bronze, silver and gold schemas. Supabase provides user authentication, including authenticator-based multi-factor authentication, and stores the mobile application's community posts.

For data sources, the system uses NLEX toll and incident exports, the Waze for Cities (Partner Hub) feed, historical weather from Open-Meteo, ERA5 and JMA, historical air quality from OpenWeather, a Philippine Arena event calendar and the Philippine holiday calendar. Event data were collected once from the venue's published calendar; an automated ticketing-site scraper was planned but not built.

For the interfaces, the operator dashboard is built with Next.js 15, React 19 and TypeScript, uses Apache ECharts and Mapbox GL JS, and is packaged with Electron as a Windows desktop application. The API is built with Node.js, Express and TypeScript. The Scenario Sandbox runs a TypeScript traffic simulation on an HTML5 Canvas inside the dashboard. The mobile application is built with Expo and React Native for Android and iOS. Language-model features use GLM models from Z.ai (dashboard narratives and sandbox commands) and Qwen models through OpenRouter (mobile assistant), both through OpenAI-compatible APIs.

All of these components are free, open-source or available on free tiers during development, except the AWS RDS database and the language-model APIs, which are usage-billed. The team was able to learn and use all of them within the capstone timeline.

3.4.3 Organizational/Cultural Feasibility

SmartFlow NLEX will fit well with how NLEX Corporation already operates and where it is headed as a company. NLEX has already been moving toward digital solutions through tools like the Easytrip cashless payment system and the MPT DriveHub app under MPTC, so adding a data-driven traffic management system will be a natural next step for them.

The Web Dashboard will be designed specifically for NLEX operators at the Traffic Control Center (TCC). It will give them a centralized interface for monitoring current traffic conditions, reviewing historical trend data, and viewing predictive forecasts that anticipate how traffic will behave in the coming hours. Beyond visibility, the dashboard will also surface actionable guidance through the prescriptive analytics engine, which will generate recommended strategies such as lane management changes, and incident response protocols based on predicted traffic conditions. Operators will also be able to test scenarios before committing to any action using the What-If Simulation feature through the Agent-Based Simulation Sandbox, allowing them to evaluate the projected impact of different interventions in a risk-free environment. Throughout all of this, the system will only provide suggestions, and final decisions will remain with the operators managing the expressway. This design will make it easier for staff to trust and adopt the system without feeling like it is replacing their judgment.

The community feature of the mobile application builds on commuters' familiarity with crowdsourced reporting in applications such as Waze. In the current version, community posts are shown in the application feed and are not merged with the Waze feed or the analytics warehouse. Using them as an additional data source would first require moderation and validation rules.

The project responds to operational gaps that NLEX personnel described to the proponents, in particular the limited use of forecasts for non-routine conditions and the separation of toll, incident, weather and event data. By integrating these sources and adding forecasts and recommendations to a single interface, SmartFlow NLEX addresses those gaps without replacing existing NLEX systems.

3.4.4 Economic Feasibility

SmartFlow NLEX is inexpensive to build because most of its technology stack is open-source or available on free tiers during development. The main usage-billed items are the AWS RDS database and the language-model APIs.

After deployment, costs would rise because the system would need production hosting, monitoring, paid API tiers and staff time for maintenance. The figures below are planning estimates and have not been quoted by vendors. Items marked “to be confirmed” depend on the provider's billing tier and should be confirmed from the project accounts.

Table 3.4.4.1 will compare the expected costs during the development phase versus what it will cost to maintain the system after it is deployed.

Table 3.4.4.1. During and After Deployment Expected Cost Table

Expense Item	Estimated Cost	Notes
During Development (Capstone Period, 2025–2026)		
Python libraries (pandas, NumPy, statsmodels, Prophet, scikit-learn, XGBoost, TensorFlow/Keras, PyTorch, lifelines, mgwr)	Free	Open-source libraries for data processing, modeling and evaluation
Node.js, Express, Next.js, React, Electron, ECharts, YALPS	Free	Open-source frameworks for the API, dashboard and optimization
AWS RDS for PostgreSQL + PostGIS (Singapore)	To be confirmed	Managed database for the warehouse; billing tier to be confirmed from the AWS account
Supabase (authentication, mobile community posts)	Free tier	Two projects (dashboard and mobile)
Language-model APIs (Z.ai GLM; Qwen via OpenRouter)	To be confirmed	Usage-billed; free or low-cost models used during development
Waze for Cities (Partner Hub) feed	Free	Access through partner credentials; agreement to be documented
Weather and air quality (Open-Meteo, ERA5, JMA, OpenWeather Air Pollution)	Free	Historical data loaded once
Mapbox GL JS map tiles	Free tier	Development usage within free limits
Render (demonstration API for the mobile app)	Free tier	Free instance with cold starts
Expo (mobile application)	Free	Open-source framework for Android and iOS
Figma (UI/UX design)	Free	Free plan for student design work
GitHub (version control)	Free	Free plan for team repositories
Printing, documentation and travel	~₱3,000 total	Research materials, adviser meetings and stakeholder coordination

Estimated Development Total	~₱3,000 plus usage-billed items	Usage-billed items to be confirmed
After Deployment (Monthly and Annual Costs)		
Production database hosting (managed PostgreSQL with backups)	₱3,000 – ₱8,000/mo	Dedicated instance with automated backups
API and job hosting (container service and scheduler)	To be confirmed	Production API hosting; the scheduled jobs run on GitHub Actions
Language-model APIs (production volume)	To be confirmed	Depends on the number of narratives, commands and assistant queries
Mapbox (production map tile requests)	₱2,000 – ₱5,000/mo	Free tier insufficient for production usage
Google Play and Apple App Store fees	~₱6,500/year	Google Play ~₱1,500 one-time; App Store ~₱5,000/year
Windows code-signing certificate for the desktop installer	To be confirmed	Annual certificate
IT staff for maintenance and monitoring	₱10,000 – ₱20,000/mo	Updates, bug fixes, pipeline error handling and security patches
ML model retraining compute	₱5,000 – ₱12,000/mo	Weekly retraining batch and hourly congestion refresh on GitHub Actions hosted runners; cost depends on the plan's included minutes (to be confirmed)
Data storage and backups	₱1,500 – ₱4,000/mo	Growing historical traffic and incident data
Annual security audit	~₱20,000/year	Third-party check of authentication, access control and OWASP compliance
Estimated Monthly Total (Post-Deployment)	~₱21,500 – ₱49,000/mo plus items to be confirmed	To be covered by NLEX Corporation's IT budget upon deployment

As shown in Table 3.4.4.1, development costs are low because most components are open-source or on free tiers, with the database and language-model APIs as the main usage-billed items. Post-deployment costs are higher and depend on several vendor quotations that remain to be confirmed. Whether these costs are justified would need to be assessed by NLEX against measured operational benefits, which this capstone estimates but does not observe in live operation.

3.5 Business Requirements Overview

The business requirements for SmartFlow NLEX are categorized into functional requirements, the specific features and tasks the system must perform. And non-functional requirements, which define the system's performance and quality standards.

3.5.1 Functional Requirements

Table 3.5.1.1 Functional Requirements

ID	Functionality	Description	Status
FR-01	Multisource Ingestion	The system shall load hourly toll volume per plaza, direction and class, NLEX accident and breakdown records, Waze jams and alerts, historical weather, holidays and the event calendar into the warehouse, and record each load's source, row count and time.	Partial (live Waze automated; other sources batch)
FR-02	Event Calendar	The system shall maintain an event calendar (name, venue, date, category) for venues that affect the corridor, loaded from a maintained file or upload and validated against the event schema.	Partial (one-off scrape; no automated scraper)
FR-03	Data Validation	Each uploaded record shall be rejected or flagged if its plaza is not an NLEX plaza, its kilometre post is outside the corridor, a required field is missing, or a value fails its	Implemented

		type check; the upload report shall give counts per rejection reason.	
FR-04	Emission Estimation	The system shall estimate corridor CO_2, NO_x and PM2.5 per day, paying exit and vehicle class from the vehicle-kilometres of each toll trip (between its entry and exit plazas) and fixed per-class emission factors, report the CO_2 total with a low–high range, and estimate the additional CO_2 from vehicles idling in incident queues, showing the factors and date window used.	Implemented
FR-05	Volume Forecasting	The system shall produce a daily corridor volume forecast from the candidate model with the lowest held-out WMAPE that also beats the seasonal-naive baseline, report each candidate's accuracy, and refresh the forecast weekly.	Implemented
FR-06	Agent-Based Sandbox	The sandbox shall simulate lane closures, speed limits, counterflow and incidents on a chosen kilometre stretch, direction and time with vehicle agents whose demand is drawn from warehouse data, and report speed, throughput, queue length, travel time and CO_2 with and without the intervention.	Implemented
FR-07	Prescriptive Recommendations	The system shall recommend toll booths to staff per plaza and hour, patrol positions, the number of response units to dispatch for an incident, a congestion-response urgency and ranked emission strategies, showing the assumptions behind each.	Implemented
FR-08	Model Accuracy Display	Each predictive view shall display the held-out accuracy of the active model and its baseline, the evaluation window and the last retraining date.	Implemented
FR-09	Incident and Live Map	The dashboard shall plot Waze jams and alerts from the last 60 minutes and active maintenance closures on the corridor map, and provide historical NLEX incident analytics filterable by date, type and exit.	Implemented
FR-	Forecast	The mobile application shall let users choose	Implemented

10	Time Selection (Mobile)	a forecast hour up to the model's published horizon (168 hours) and show the predicted congestion state per exit.	
FR-11	Conversational Assistant (Mobile)	The mobile assistant shall answer NLEX corridor questions in English or Tagalog using tools that read live corridor status, state the data age, and decline questions outside its scope.	Partial (no forecast or departure-time advice)
FR-12	Travel Alerts (Mobile)	The mobile application shall show alerts for operator advisories, scheduled maintenance and new incidents while the application is in use.	Partial (on-device notifications only)
FR-13	Community Reports (Mobile)	Signed-in users shall post road reports with location and optional photo; cross-checking reports against Waze and forecasts is proposed future work.	Partial (posting implemented; cross-checking proposed)
FR-14	Authentication and Access Control	Users shall sign in with password and authenticator code (TOTP), may trust a device for seven days where the server enables it, are signed out after inactivity, and see only the modules permitted to their role.	Implemented
FR-15	Maintenance Scheduling	Authorized users shall create, edit, cancel and delete road-maintenance schedules by exit, direction and official kilometre posts, with validation of the kilometre range.	Implemented
FR-16	Data Management	Data analysts shall upload CSV or XLSX exports, run a check-only validation, load accepted records in one transaction and undo a previous upload.	Implemented
FR-17	Audit Log	The system shall record user sessions, page views, data, maintenance and model events (including each hourly congestion refresh), and let only data analysts view, filter and export them.	Implemented
FR-18	Mobile Configuration	Operators shall switch mobile application sections on or off and publish advisories from the dashboard without releasing a new	Implemented

		application version.	
FR-19	Model Narratives	The dashboard shall generate plain-language explanations of displayed model results using only the values shown, with a template fallback when the language model is unavailable.	Implemented
FR-20	Congestion-State Forecast	The system shall forecast the congestion state (low, medium, high) of each exit for 1 to 168 hours ahead and refresh the forecast hourly.	Implemented
FR-21	Scheduled Retraining	The system shall retrain the forecasting models weekly, keep a backup of the active models and publish a retrained model only if it passes the accuracy gate.	Implemented
FR-22	Hourly Volume Forecast	The system shall forecast hourly toll volume by plaza, direction and vehicle class for the 48 hours after the latest upload, with an 80% uncertainty band, and shall fall back to the previous profile method for any series whose recent history is incomplete.	Implemented

3.5.2 Non-Functional Requirements

Table 3.5.2.1 Non-Functional Requirements

Non-Functional Requirement	**Description**
Accuracy	Each published forecast shall have a lower held-out WMAPE than its naive baseline (seasonal naive or climatology) on the same evaluation window, and the congestion-state classifier shall exceed the accuracy of the exit-by-hour historical profile. Metrics shall be reported with their grain, horizon and evaluation window.
Performance	Cached analytics reads shall return within 3 seconds at the 95th

	percentile and uncached (cold) reads within 10 seconds with up to 25 concurrent users; the Live Map shall refresh its data at least every 30 seconds and shall flag the live feed as stale when its newest record is more than 30 minutes old.
Scalability	The system shall support at least 50 concurrent operator sessions with 95th-percentile response time under 3 seconds and an error rate under 1%; the degradation point shall be measured in stress testing.
Availability	The API health endpoint shall achieve at least 99.5% monthly availability measured by an external probe in production; planned system downtime shall be announced to users in advance.
Security	All client-server and server-database traffic shall use TLS; passwords shall be stored only by the identity provider; every write or administrative endpoint shall require an authenticated session with the second factor and an authorized role; personal fields shall be masked in the analytics layers; and security-relevant actions shall be audit-logged.
Usability	Operators and commuters participating in user-acceptance testing shall achieve a mean System Usability Scale (SUS) score of at least 68, and body text shall meet a contrast ratio of at least 4.5:1 and large text at least 3:1, following the WCAG 2.1 Level AA criterion (World Wide Web Consortium, 2025).

3.6 Key Performance Indicator (KPI) Catalogue

The KPI catalogue defines how the project measures whether SmartFlow NLEX meets its objectives. Each KPI states its link to an objective, its formula, unit, data source, target and whether it is leading or lagging. Targets that come from published references are cited. Targets the proponents set without an external reference are labelled design targets. Where the record does not yet allow a defensible target, the KPI first establishes a baseline. Tables 3.6.1 to 3.6.4 detail the four primary KPIs, and Table 3.6.5 lists the full catalogue. Results are reported in Chapter 4.

Table 3.6.1 Volume Forecast Accuracy KPI Catalogue

Volume Forecast Accuracy	
Linked Objective	Specific Objective 4 (multi-source predictive models deployed in the dashboard)
Strategic Objective	Ensure that the daily corridor volume forecast used for booth staffing and resource planning is more accurate than a naive forecast and accurate enough to plan with.
Formula	WMAPE = $\Sigma\lvert y_t - \hat{y}_t\rvert$ / $\Sigma\lvert y_t\rvert \times 100\%$, computed on rolling-origin holdout days (Equation 3.11.3.2). MASE is computed with the seasonal-naive (lag-7) scale. Grain: daily corridor total entries; horizon: 1–14 days.
Data Source	gold.fact_traffic_hourly aggregated to daily totals; gold.ml_model_metrics and gold.ml_horizon_accuracy (split 80/20)
Target (RAG)	Green: WMAPE below the seasonal-naive WMAPE on the same days and ≤ 12% (design target). Amber: below the baseline but > 12%. Red: not below the baseline.
Leading / Lagging	Lagging (measures model quality after the fact); the forecast it qualifies is a leading input to operations.

Table 3.6.2 Incident Clearance Time (MTTC) KPI Catalogue

Incident Clearance Time (MTTC)	
Linked Objective	Specific Objective 3 (recurring inefficiencies) and prescriptive incident response
Strategic Objective	Establish a defensible baseline of how long incidents occupy the carriageway, by incident type, against which future response improvements can be measured.
Formula	$MTTC_k = (1/\|I_k*\|)\ \Sigma\ T_i^{site}$ over $i \in I_k*$, where I_k* is the set of type-k incidents with valid start and site-cleared times, I_k is all type-k incidents, and T_i^{site} = site-cleared time − event start; reported with the median, 90th percentile and coverage $\|I_k*\| / \|I_k\|$. Lane-blockage duration is reported separately.
Data Source	silver.nlex_accident_events_clean and silver.nlex_breakdown_events_clean (NLEX event exports, 2022–2025)
Target (RAG)	Provisional design targets for the projected mean clearance time: accidents Green ≤ 30 min, Amber ≤ 45 min, Red above 45 min; breakdowns Green ≤ 10 min, Amber ≤ 18 min, Red above 18 min (indicative). These targets were set by the proponents, are shown on the Clearance Response Protocol, and will be replaced by targets agreed with NLEX; an improvement cannot be attributed to SmartFlow NLEX without a before-and-after deployment study.
Leading / Lagging	Lagging

Table 3.6.3 Corridor CO_2 Forecast Accuracy and Inventory Coverage KPI Catalogue

Corridor CO_2 Forecast Accuracy and Inventory Coverage

Linked Objective	General Objective (emissions) and Specific Objective 4
Strategic Objective	Ensure that the corridor CO_2 estimate covers the whole tolled corridor and that its forecast is reliable enough to plan emission interventions. SmartFlow NLEX has no roadside emission sensors, so the CO_2 inventory is a calculated estimate; the KPI therefore measures forecast accuracy against that inventory and the coverage of the inventory, not agreement with measured emissions.
Formula	Forecast: WMAPE and MASE of the daily CO_2 forecast against the inventory (Equation 3.9.3.5). Coverage = plaza-hours with class-resolved counts ÷ all plaza-hours × 100%.
Data Source	gold.fact_emissions_hourly; gold.fact_vkm_hourly; gold.open_plaza_km; silver.nlex_emission_factors; gold.ml_predictive_emissions; gold.ml_model_metrics
Target (RAG)	Green: forecast WMAPE below the seasonal-naive baseline and ≤ 15% (design target), and coverage = 100% (design target). Amber: below the baseline but > 15%, or coverage 95–99%. Red: not below the baseline, or coverage < 95%.
Leading / Lagging	Lagging

Table 3.6.4 Scenario Sandbox Run Time and Recommendation Acceptance KPI Catalogue

Scenario Sandbox Run Time and Recommendation Acceptance	
Linked Objective	Specific Objective 1 (evaluate response strategies in the agent-based sandbox) and Specific Objective 2 (usable decision support)
Strategic Objective	Confirm that a what-if scenario can be evaluated within the time available to an operator, and that operators judge the system's recommendations acceptable.

Formula	Run time = wall-clock seconds to complete a scenario comparison (intervention and baseline replications) at stated settings (stretch length, simulated duration, number of replications). Acceptance rate = accepted ÷ (accepted + modified + rejected) recommendations × 100%, recorded with a rubric during user-acceptance testing.
Data Source	Timed sandbox runs on the demonstration machine; UAT decision records from TCC and incident operators
Target (RAG)	Run time — Green ≤ 120 s, Amber 120–200 s, Red > 200 s (design targets). Acceptance — Green ≥ 90%, Amber 75–89%, Red < 75% (design targets). Acceptance indicates operator agreement, not correctness, and is reported with the reasons for rejection.
Leading / Lagging	Run time: system-performance measure (neither leading nor lagging). Acceptance: leading (adoption indicator).

Table 3.6.5 SmartFlow NLEX KPI Catalogue Summary

KPI	Linked Objective	Definition and Formula	Unit	Target	Lead /Lag
K1 Volume forecast error	SO4	WMAPE and seasonal MASE of the daily corridor volume forecast, 1–14-day rolling origin	%, ratio	Below seasonal naive; ≤ 12% (design)	Lagging
K2 Congestion-state skill	SO4	Accuracy, macro-F1 and skill score (acc − acc_{pro}^{f})/(1 − acc_{pro}^{f}) against the exit-by-hour profile, by horizon band (evaluated in Chapter 4; the dashboard shows accuracy against persistence by horizon, and the accuracy and severe- and heavy-hour recall of every candidate on the same test hours)	ratio	Skill > 0	Lagging
K3 Incident	SO4	WMAPE and MASE of the	%,	MASE < 1	Laggi

count forecast error		daily incident forecast	ratio		ng
K4 Clearance-time discriminatio n	SO4	Concordance index (C-index) and mean absolute error of the clearance-time model	ratio, min	C-index > 0.5	Laggi ng
K5 CO_2 forecast error	GO, SO4	WMAPE and MASE of the daily CO_2 forecast against the inventory	%, ratio	Below seasonal naive; ≤ 15% (design)	Laggi ng
K6 Emission inventory coverage	GO	Plaza-hours with class-resolved counts ÷ all plaza-hours	%	100% (design)	Laggi ng
K7 Corridor CO_2 intensity	GO	Σ CO_2 ÷ Σ vehicle-km; heavy-vehicle (Class 2 and 3) share of vehicle-km and of CO_2	g/vkm, %	Baseline to be established	Laggi ng
K8 Delay-induced CO_2	GO	Σ idle CO_2 from lane-blocking incidents (Equations 3.9.3.7 and 3.9.3.8); the queue model understates delay	t/yr, %	Baseline to be established	Laggi ng
K9 MTTC	SO3	Mean, median and p90 of site-clearance and lane-blockage times by type, with coverage	min, %	Baseline to be established	Laggi ng
K10 Sandbox intervention effect	SO1	Change in throughput, maximum queue, mean speed and travel time versus the baseline, mean of seeded replications with 95% CI	veh/mi n, m, km/h, s	Baseline per scenario	Leadi ng
K11 Sandbox run time	SO1	Wall-clock seconds per scenario comparison at stated settings	s	≤ 120 s (design)	Syste m perfo rman ce
K12 Recommend ation acceptance	SO1, SO2	Accepted ÷ all presented recommendations in UAT	%	≥ 90% (design)	Leadi ng

K13 API latency	SO2 (NFR)	95th-percentile response time at stated concurrency	s	< 3 s cached (design)	Lagging
K14 Usability	SO2	Mean System Usability Scale score from UAT participants	score 0–100	≥ 68 (design target; the commonly used SUS benchmark, Hyzy et al., 2022)	Lagging
K15 Incident response time	SO3	Mean, median and p90 time from report to first responder, with coverage	min, %	Baseline to be established	Lagging
K16 Secondary incident rate	SO3	Accidents within 2 km (either direction) of a primary accident, from its start until 30 min after it was cleared ÷ primary accidents	%	Baseline to be established	Lagging

3.7 Mock-ups

The mock-ups below were produced in Figma during the design stage and show the intended layout of each screen. The implemented interfaces follow the same module structure but differ in places. The AI Sandbox became the Scenario Sandbox, Map Comparison became the Live Map, a Mobile App configuration page and a Settings page were added, and sign-in uses an authenticator code instead of a one-time password sent by email or SMS. Screens of the implemented system are presented in Chapter 4.

3.7.1 Dashboard

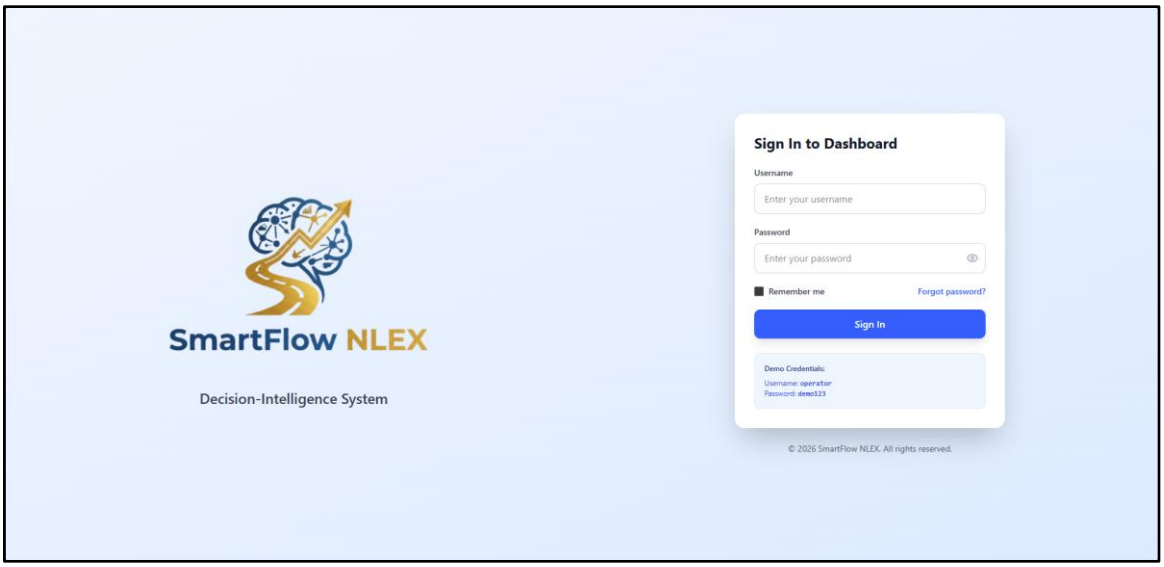


Figure 3.7.1.1 Login Page

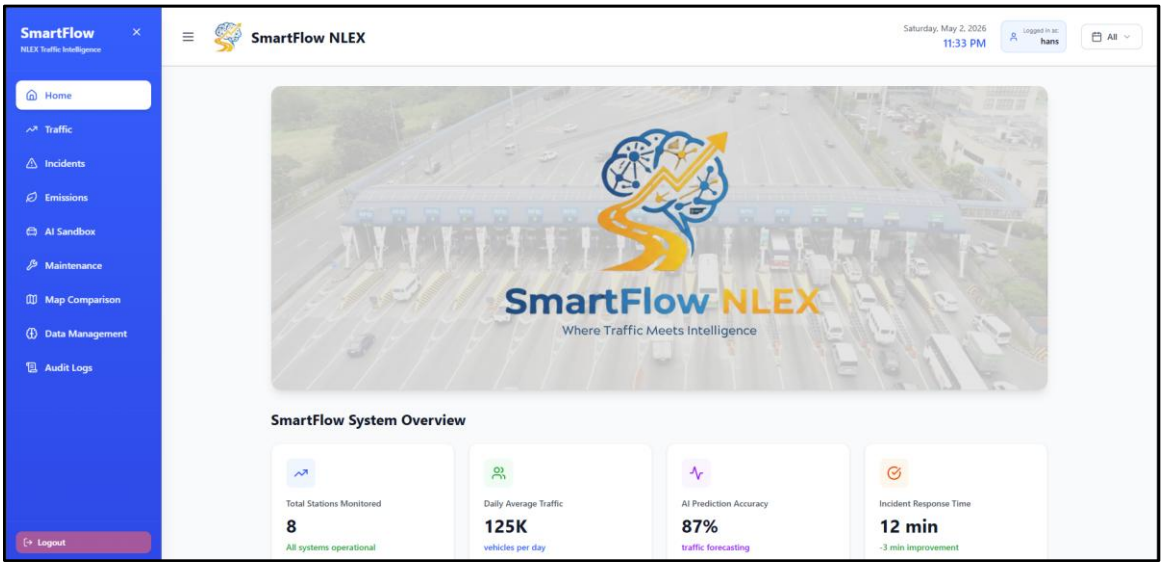
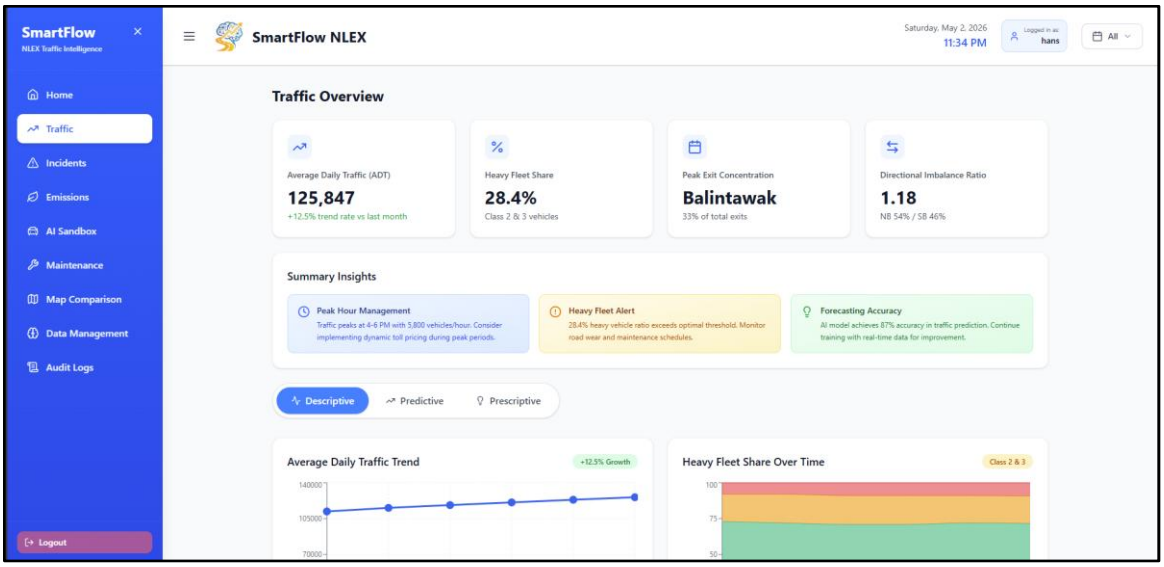


Figure 3.7.1.2 Home Page



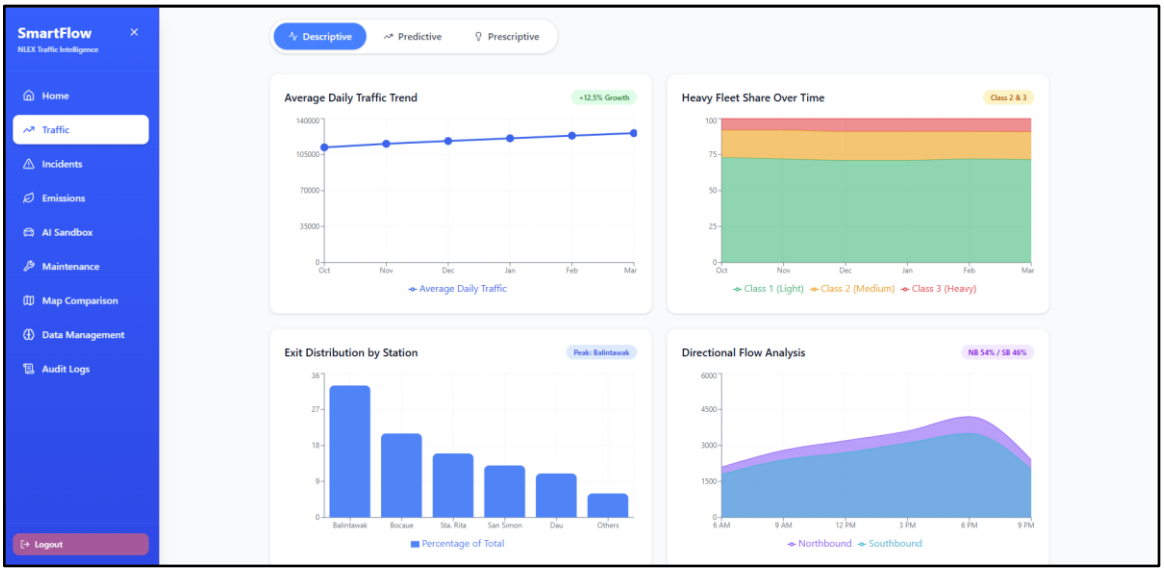


Figure 3.7.1.3 Traffic Page Descriptive

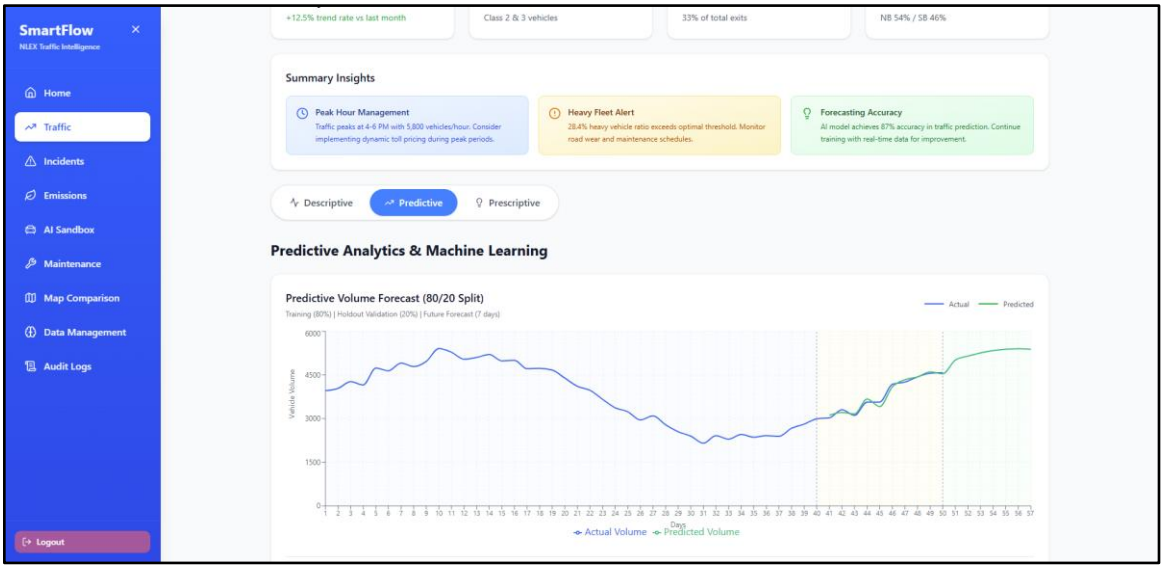


Figure 3.7.1.4 Traffic Page Predictive

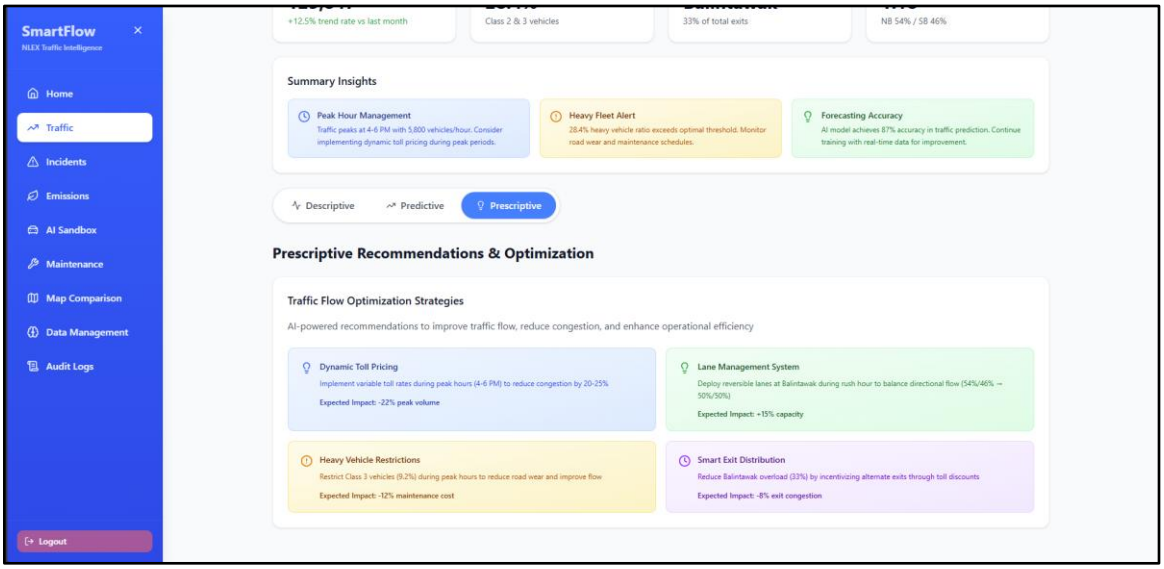
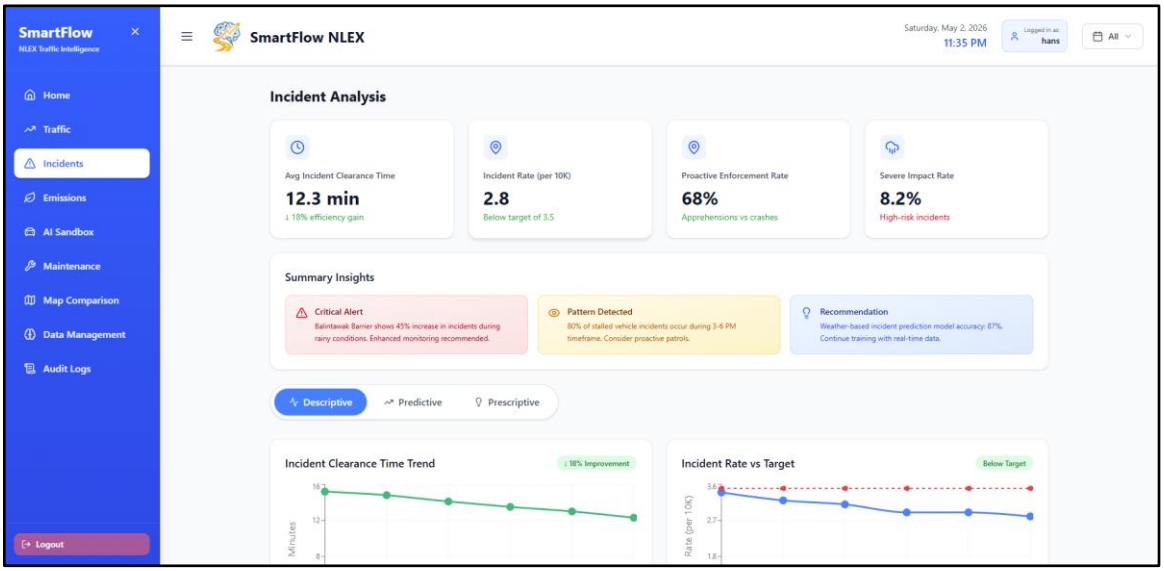


Figure 3.7.1.5 Traffic Page Prescriptive



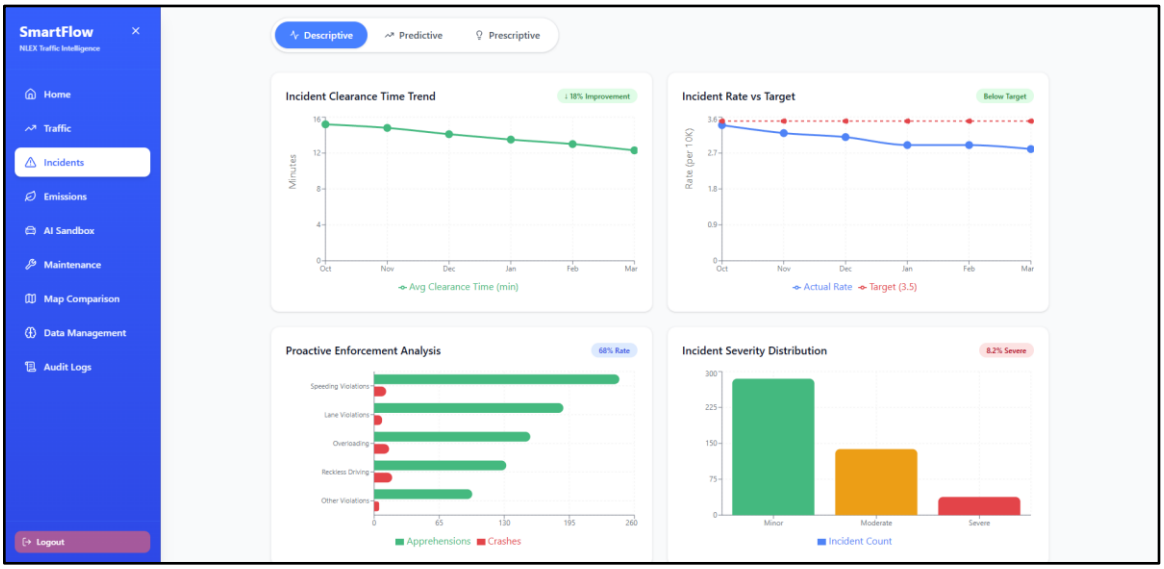


Figure 3.7.1.6 Incident Page Descriptive

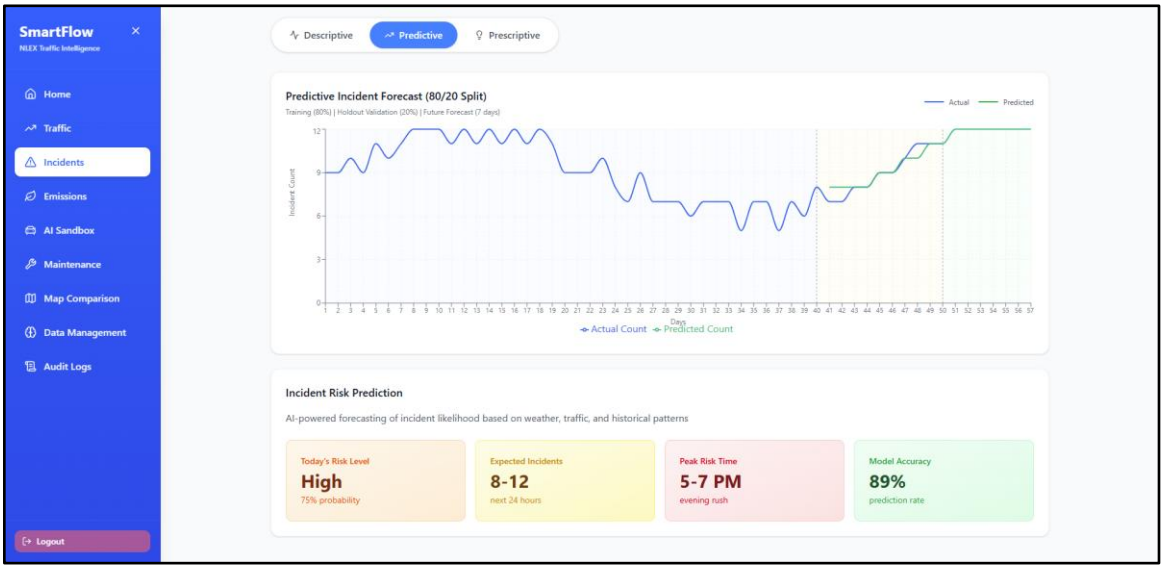


Figure 3.7.1.7 Incident Page Predictive

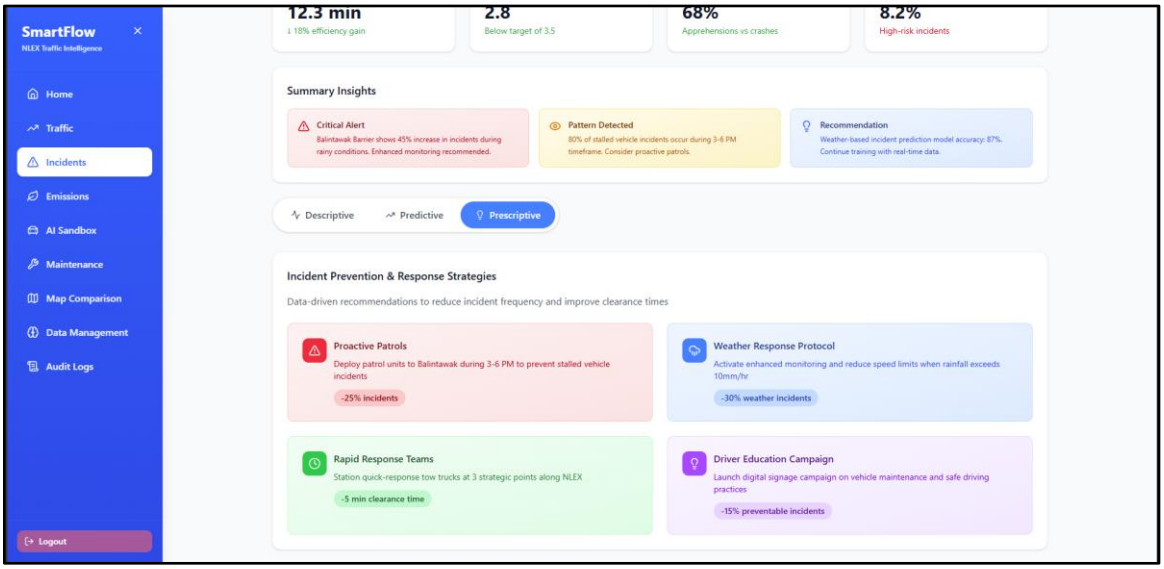
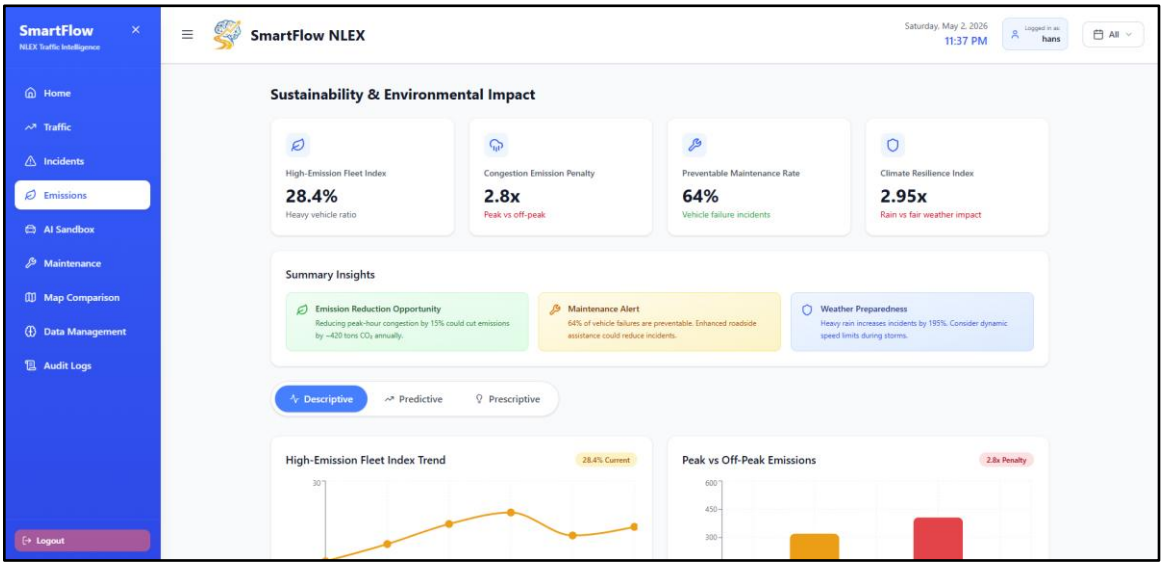


Figure 3.7.1.8 Incident Page Prescriptive



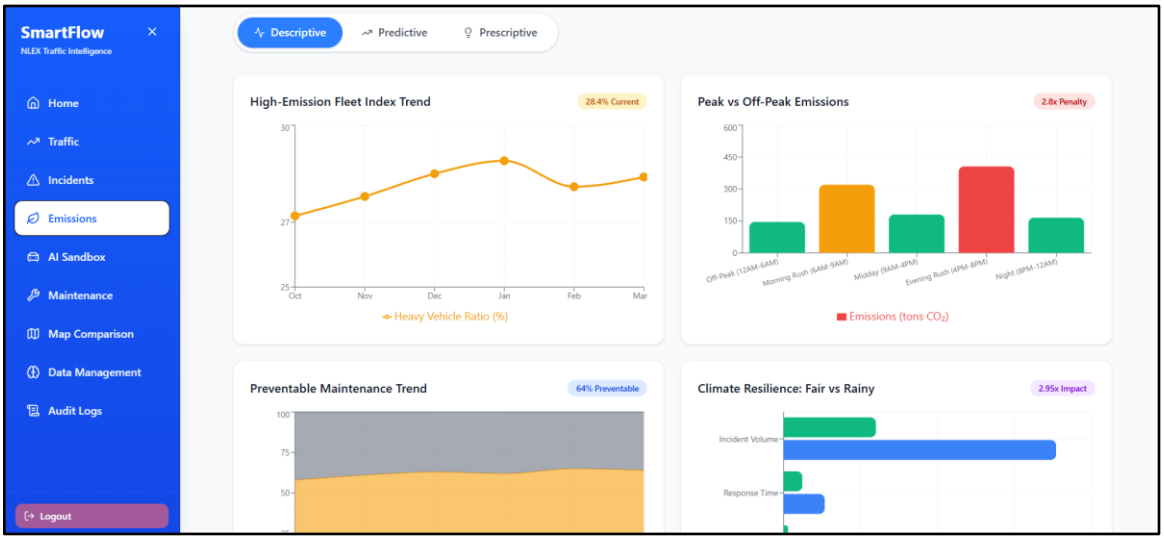


Figure 3.7.1.9 Emission Page Descriptive

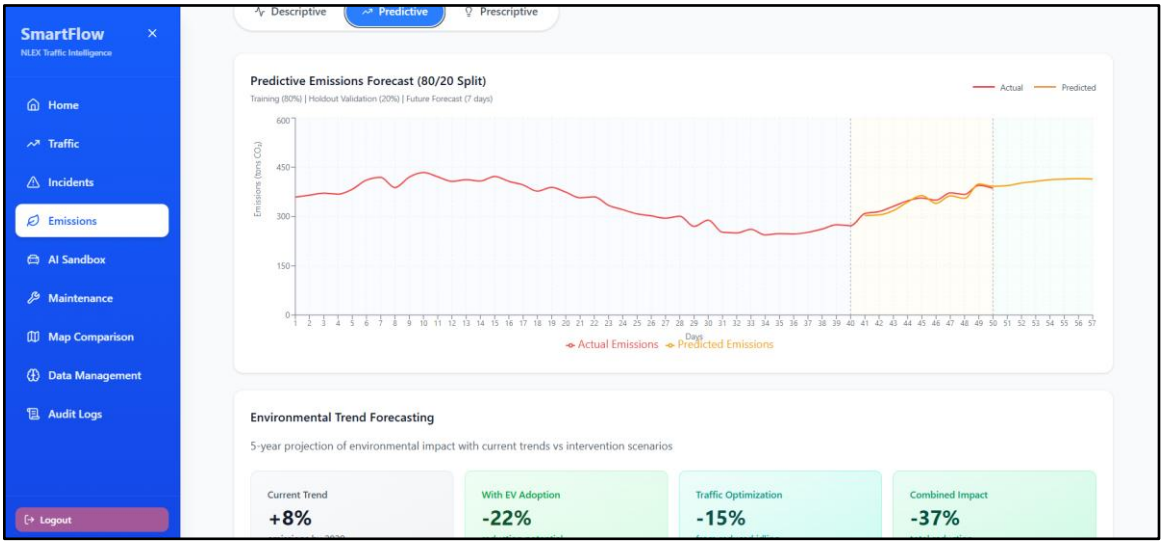


Figure 3.7.1.10 Emission Page Predictive

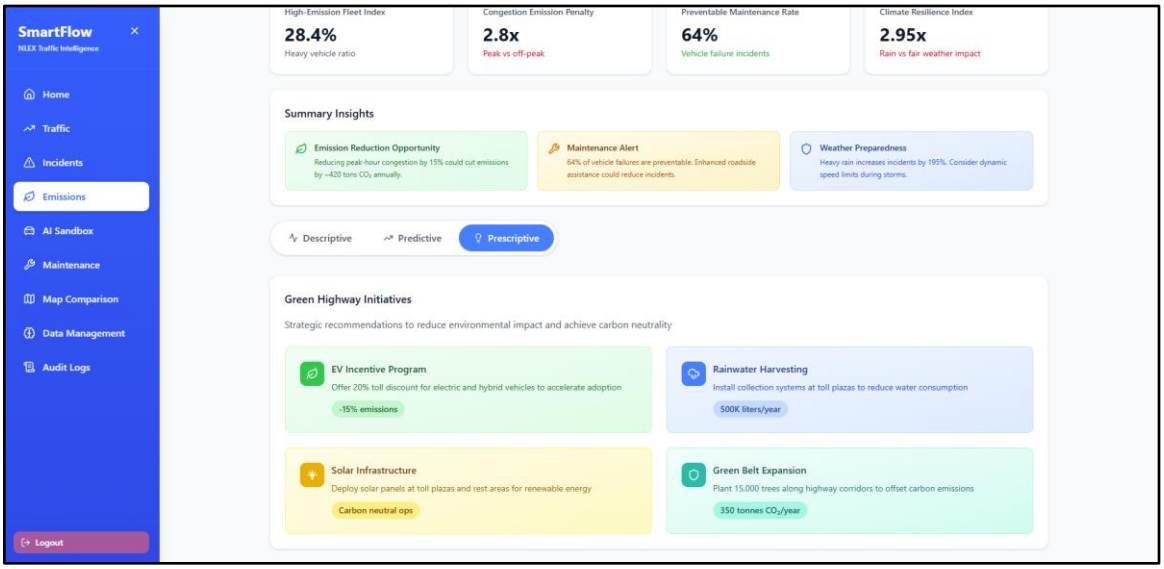


Figure 3.7.1.11 Emission Page Prescriptive

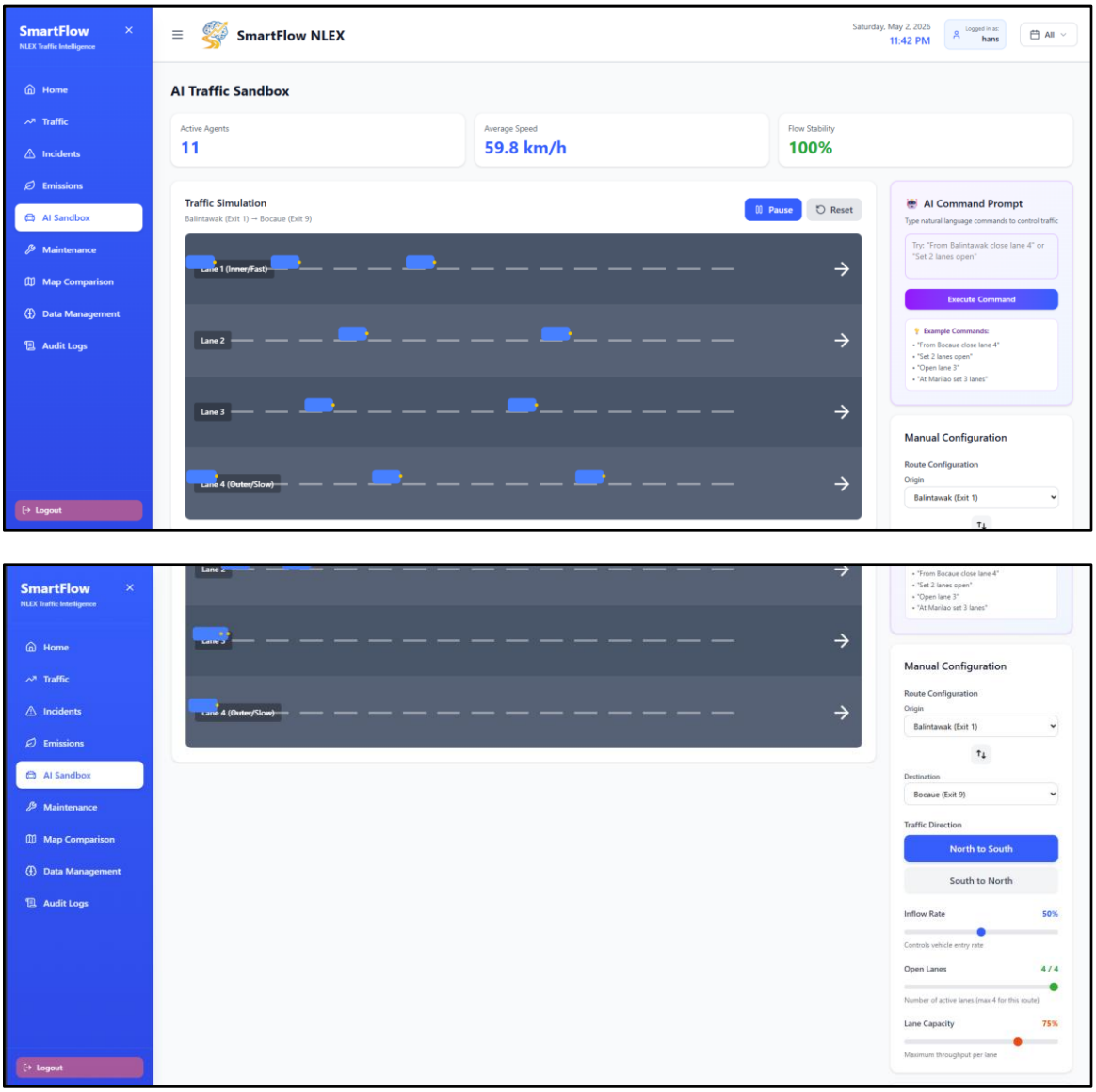


Figure 3.7.1.12 AI Sandbox Page

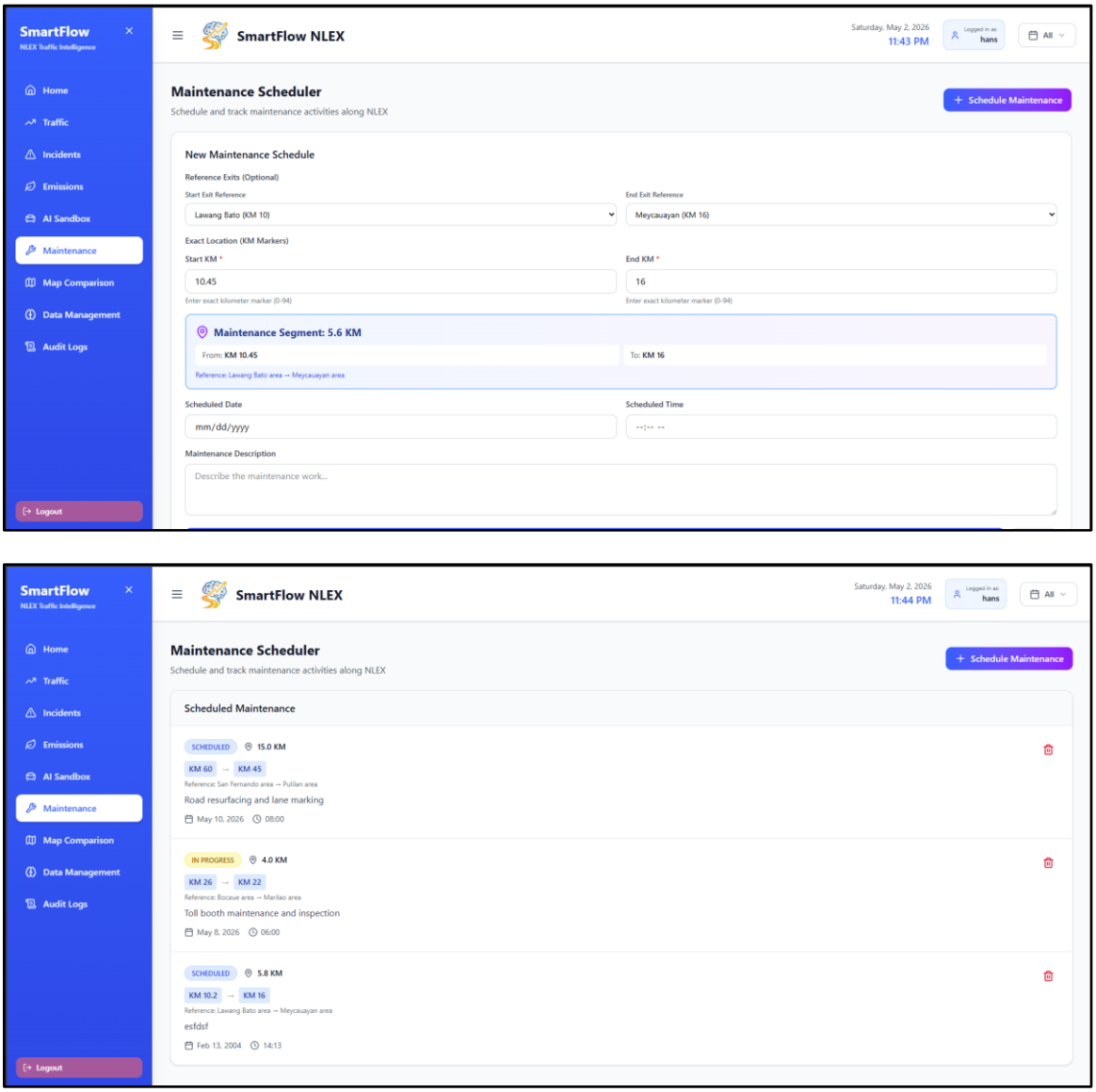


Figure 3.7.1.13 Maintenance Page

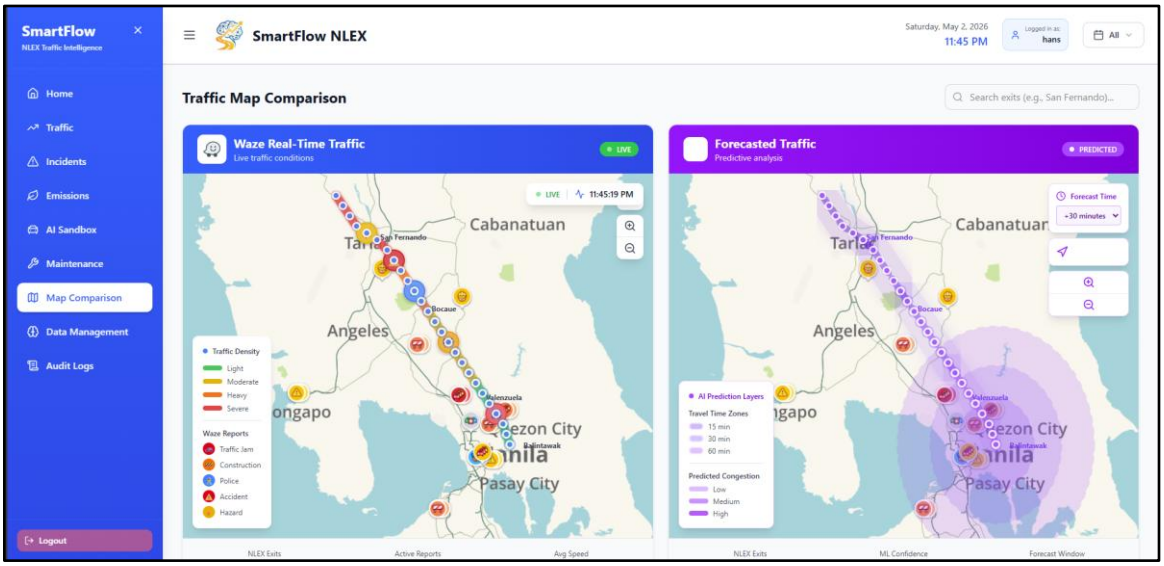


Figure 3.7.1.14 Map Comparison Page

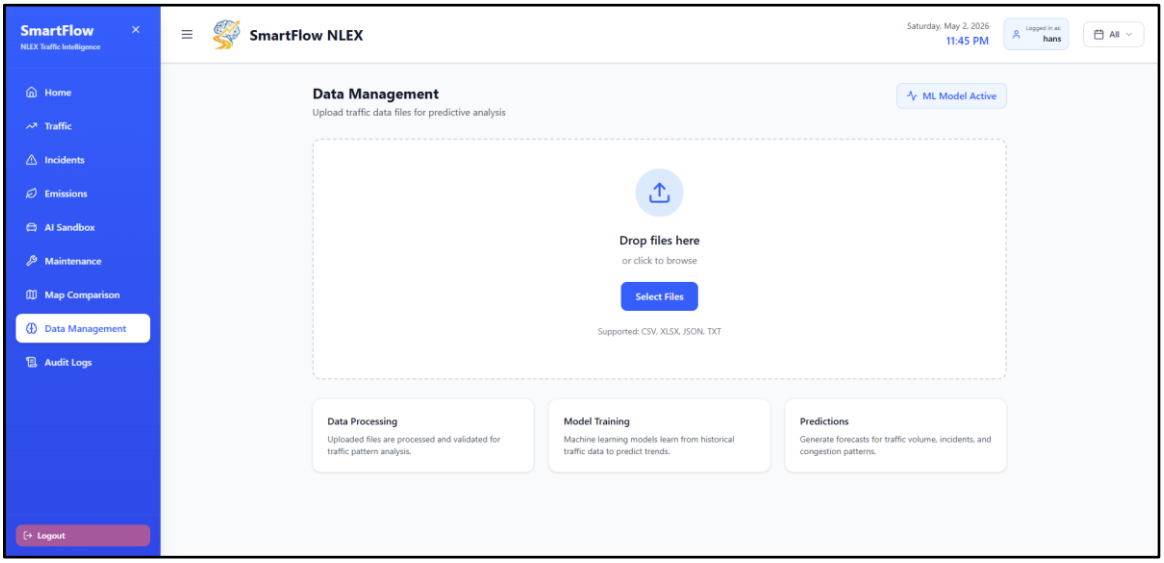


Figure 3.7.1.15 Data Management Page

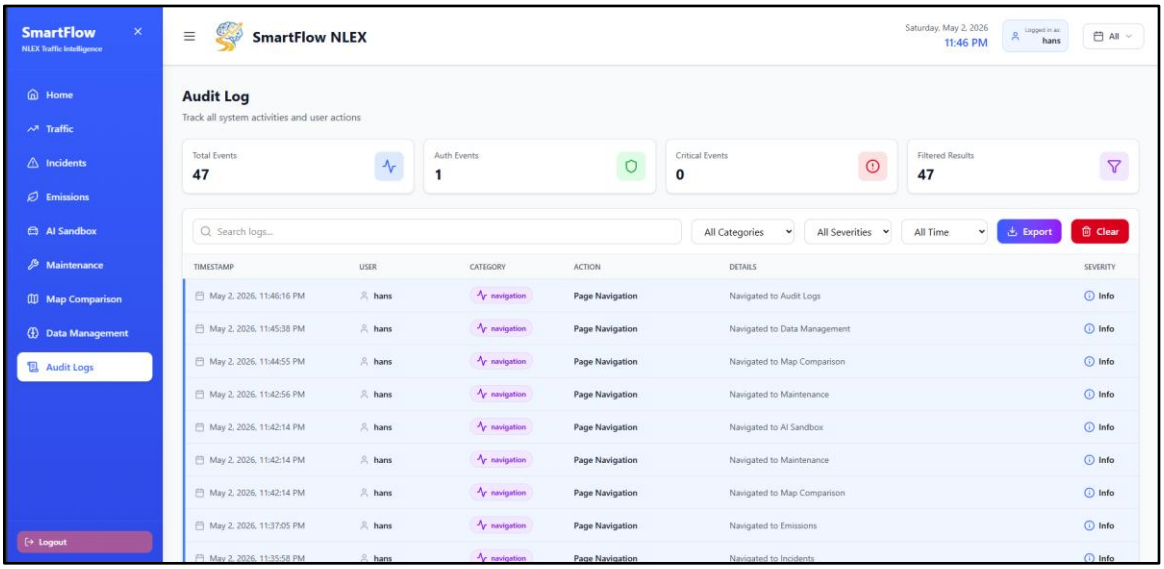


Figure 3.7.1.16 Audit Log Page

3.7.2 Mobile Application

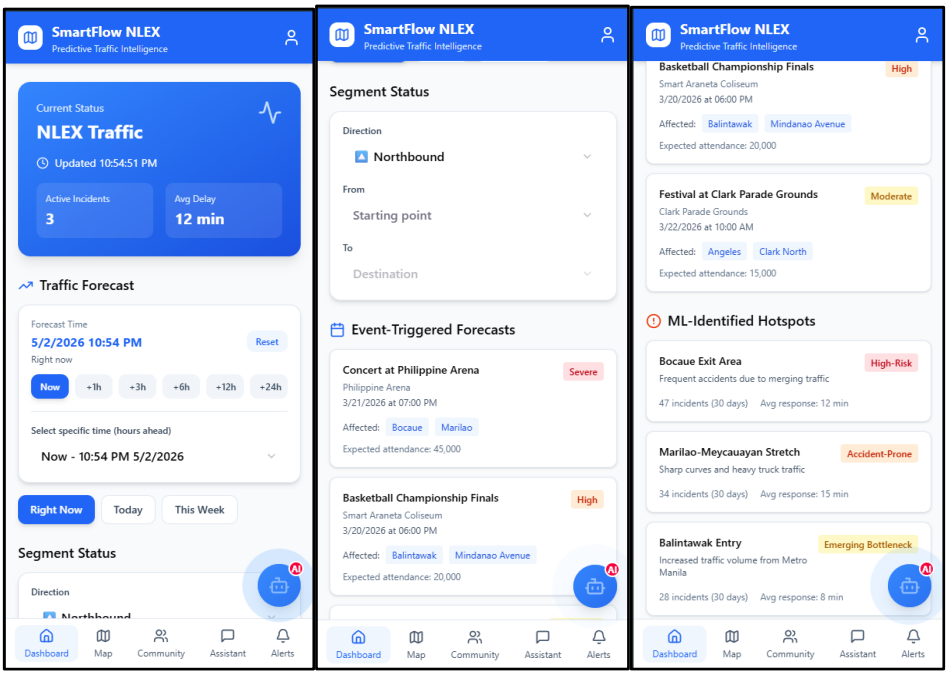


Figure 3.7.2.1 Home Page

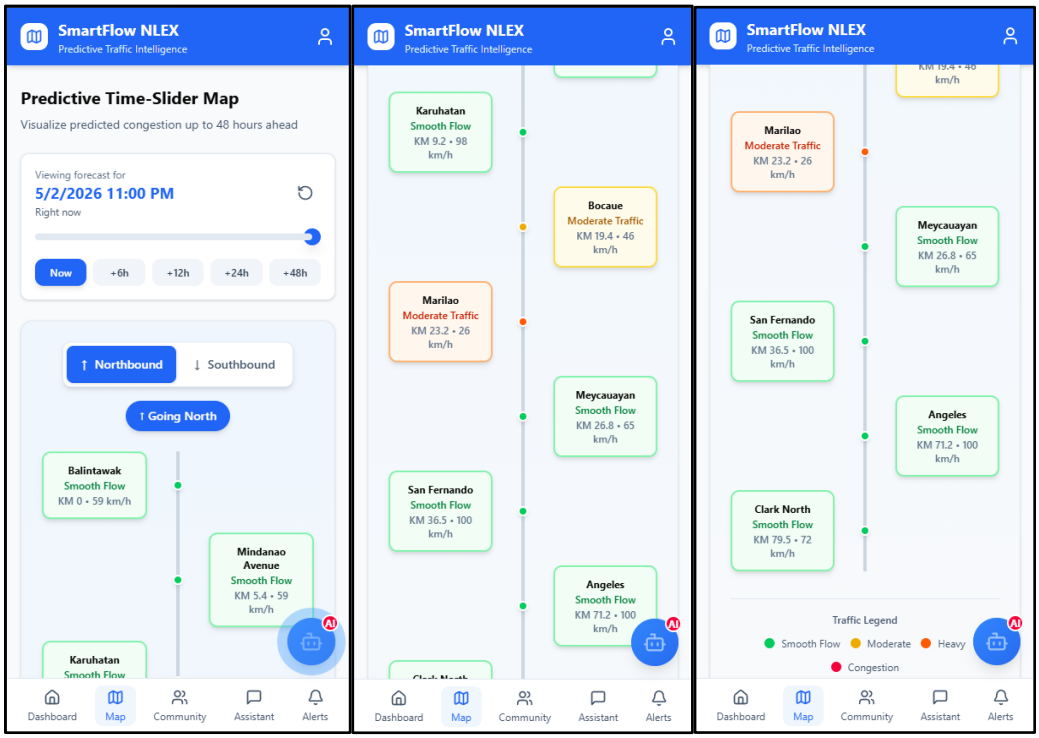


Figure 3.7.2.2 Map (Corridor) Page

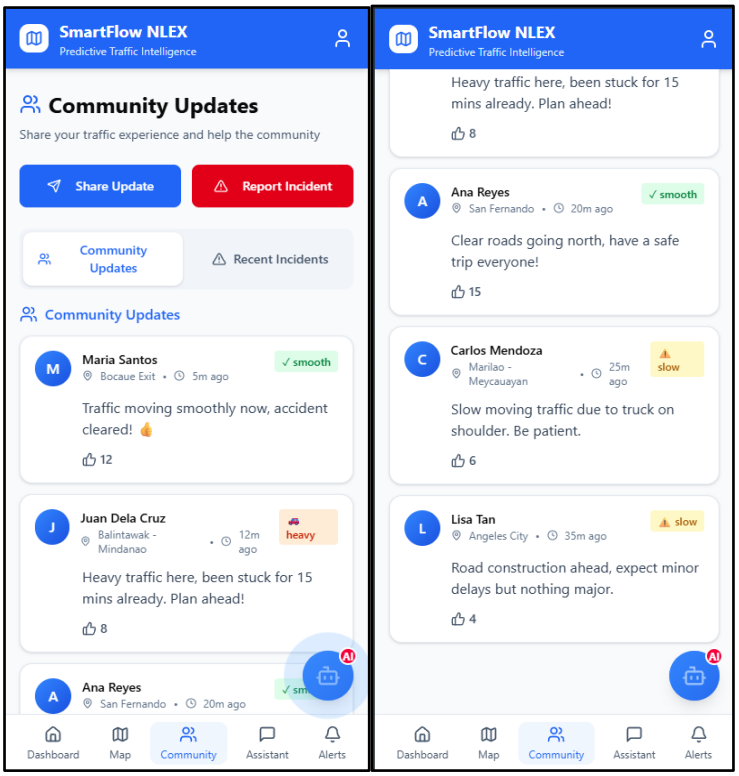


Figure 3.7.2.3 Community Page

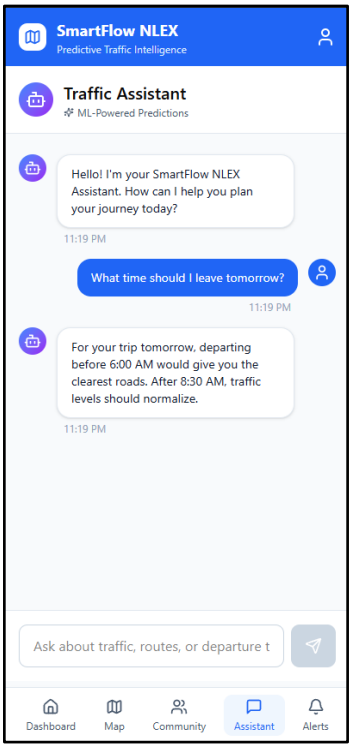


Figure 3.7.2.4 Assistant Page

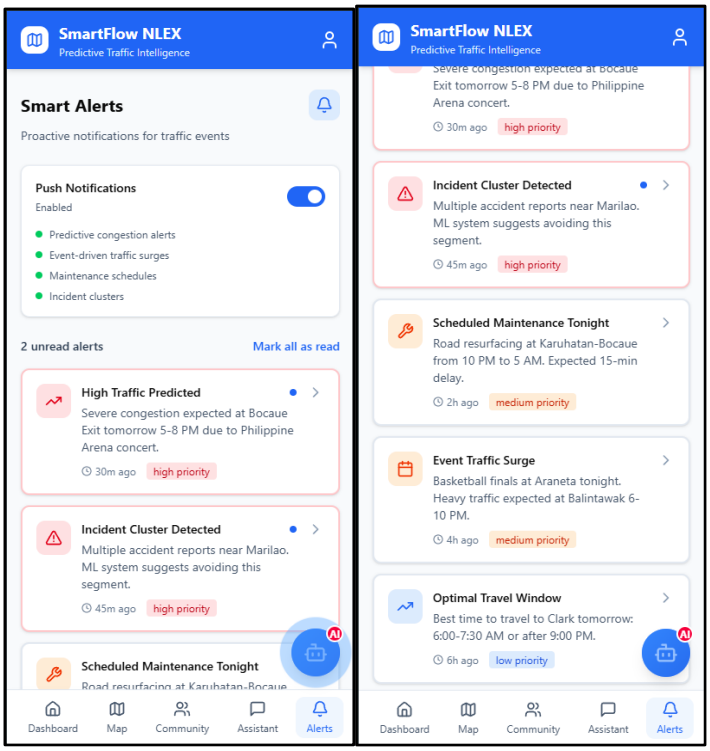


Figure 3.7.2.5 Alert Page

3.8 System Generated Outputs

This section describes the outputs generated by SmartFlow NLEX and identifies how each is produced. Each output is either computed from historical records, predicted by a model, recommended by a prescriptive model or simulated in the sandbox.

a.) Traffic Situation and Incident Summary

This descriptive output combines hourly toll records, NLEX incident records and the live Waze feed. It presents volume trends, hour-by-weekday heatmaps, holiday and event effects against local baselines, incident counts by type, cause and five-kilometre segment, wet-versus-dry crash rates, clearance times, and a live corridor map of Waze jams and alerts. It allows TCC operators to see recurring congestion periods and incident concentrations along the corridor from Balintawak (Km 12) to Sta. Ines (Km 87).

b.) Predictive Traffic Outlook (Volume and Congestion Forecasts)

This predictive output presents the daily corridor volume forecast from the champion model (currently Prophet), selected weekly by held-out accuracy against naive baselines. It also presents the congestion-state forecast (low, medium or high) for each exit and hour from 1 to 168 hours ahead, produced by the best of seven candidate classifiers, chosen again at every hourly refresh, and the expected uplift on Philippine Arena event days. It also presents an hourly volume forecast by plaza, direction and vehicle class for the 48 hours after the latest upload, produced by a stacked LSTM. Each forecast is shown with its evaluation window and accuracy so that operators can judge its reliability. The output supports planning of toll-booth staffing and patrol deployment ahead of expected demand.

c.) Carbon Emission and Environmental Sustainability

This output estimates the corridor's CO_2, NO_x and PM2.5 emissions by vehicle class and exit. The estimates are calculated from the vehicle-kilometres of each toll trip, measured between its entry and exit plazas, and fixed per-class emission factors (196.1, 689.4 and 798.2 g CO_2/km for Classes 1, 2 and 3); they do not depend on speed, and the CO_2 total is shown with a low–high range. The output adds the CO_2 emitted by vehicles queued behind lane-blocking incidents and shows the CO_2 and fleet-mix forecasts. OpenWeather's modelled air-quality estimates are shown separately from the calculated emissions.

d.) Smart Response Recommendation

This prescriptive output is computed from the stored forecasts when an operator opens the Traffic or Incident prescriptive view. It is not issued automatically as an alert. It comprises an hourly toll-booth staffing plan per plaza, a congestion-response urgency (Monitor, Prepare or Act) for each exit and hour, recommended patrol positions from a maximal covering location model, and an incident clearance protocol that states how many response units to dispatch and when to escalate. Each recommendation is shown with its assumptions so that operators can judge it before acting.

e.) Commuter Trip Advisory

This output is generated for the mobile application. It translates the congestion-state forecast into a corridor view in which the commuter chooses an hour, up to 168 hours ahead, and sees the expected condition at each exit, alongside live corridor status, maintenance notices and community reports. Travel-time estimates and “best time to leave” recommendations are proposed features and are not part of the current version.

f.) Scenario Comparison Report

This simulated output is produced by the Scenario Sandbox. For a chosen kilometre stretch, direction and time, it compares an intervention with a no-intervention baseline. The comparison covers average speed, throughput, longest queue, travel time, unmet demand and CO_2 rate, reported as the mean of seeded replications with a 95% confidence interval. The output estimates the likely effect of an intervention and does not observe it.

g.) Model Narratives

For each predictive and prescriptive chart, the dashboard can generate a short plain-language explanation using a language model. The explanation is built only from the values already displayed, falls back to a fixed template when the language model is unavailable, and never changes a model's result.

3.9 Data Gathering (with ETL)

Data gathering follows the Extract, Transform and Load (ETL) process shown in Figure 3.9.1, which is the design of the pipeline. Historical records were loaded in batches, new NLEX exports are loaded through the dashboard's upload pipeline with validation gates, and the live Waze feed is polled continuously. All data are stored in a layered PostgreSQL warehouse (bronze, silver and gold) that supports the descriptive, predictive and prescriptive analytics. The subsections below describe the sources, the extraction and validation that are implemented, the transformations and the warehouse.

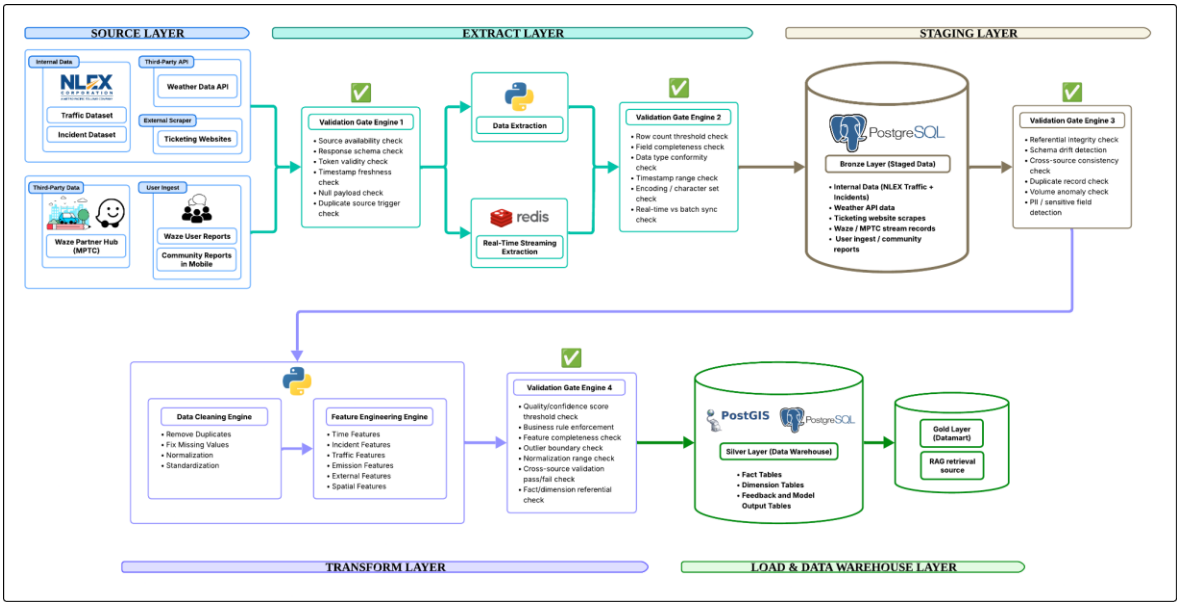


Figure 3.9.1 ETL Process Diagram

3.9.1 Data Sources

Internal data were provided by NLEX Corporation as data exports. The traffic data are NLEX's hourly toll export, which gives for each date, hour, entry plaza, exit plaza and vehicle class (Class 1, 2 or 3) the number of paid trips. It is aggregated to hourly counts per plaza, direction (northbound or southbound), entry or exit, and class for 19 of the corridor's 20 exits from 1 January 2022 to 31 December 2025 (1,461 days, with no missing days), and its entry–exit pairs give each trip's distance for the emission estimate. Trips paid at SCTEX, SFEX and NLEX Connector plazas are not NLEX exit transactions and are kept out of the traffic record, which is why the SCTEX interchange has no toll volume of its own. Average daily traffic and all other volume measures are derived from them. The incident data comprise the accident event export (start, lane-blockage-cleared and site-cleared times, kilometre post and cause), the breakdown event export (vehicle class, cause and response deployments), and road and motorcycle crash reports (date, hour, kilometre post, cause, type, weather and casualties). Records dated 2020–2021 in some incident tables could not be verified and are excluded from the analyses. The Apprehension and Undercap table described in the design was not loaded into the warehouse.

External data comprise the Waze for Cities (Partner Hub) feed and contextual sources. The Waze data consist of an hourly historical archive of jams and irregularities from January 2022 to 18 April 2026, filtered from the raw Waze dumps to the corridor, and a live feed of jams and alerts polled about every two minutes since 4 August 2026. Weather is an hourly dataset for about 20 locations along the corridor from 2020 onward, combining Open-Meteo, ERA5 and JMA data scored against NASA POWER and NOAA; a scheduled task adds the latest finished hours from Open-Meteo every hour. Historical air quality comes from the OpenWeather Air Pollution service. Event data come from a calendar of 159 events at the Philippine Arena (119), the Philippine Sports Stadium (33) and Clark Global City (7), collected once from the venues' published schedules in August 2026 and since extended by hand; it records each event's capacity and, for 63 events, a stated attendance. Holidays come from the Philippine holiday calendar. Community reports from the mobile application are stored separately in Supabase and are not part of the analytics data. Table 3.9.1 summarizes the sources.

Table 3.9.1 Data Sources of SmartFlow NLEX

Source	Provider	Coverage	Granularity	Refresh	Status
Toll transactions (traffic volume)	NLEX Corporation	1 Jan 2022 – 31 Dec 2025; 19 of 20 exits (trips paid at SCTEX, SFEX and NLEX Connector plazas kept out of the traffic record)	Hourly × entry plaza × exit plaza × class (NLEX export), aggregated to plaza × direction × entry/exit × Class 1–3	Upload pipeline (yearly exports through the dashboard or a command-line loader)	In use

Accident events	NLEX Corporation	1 Jan 2022 – 31 Dec 2025	Event (start, blockage-cleared, site-cleared, km post)	Dashboard upload	In use
Breakdown events and stalled vehicles	NLEX Corporation	1 Jan 2022 – 31 Dec 2025	Event (class, cause, deployments)	Dashboard upload	In use (contains personal fields)
Road and motorcycle crash reports	NLEX Corporation	2022 – Dec 2025 (2020–2021 rows excluded)	Event (date, hour, km post, cause, casualties)	Dashboard upload	In use
Live Waze jams and alerts	Waze for Cities (Partner Hub)	4 Aug 2026 – present	Poll about every 2 min; jam lines and alert points	Continuous polling	In use
Historical Waze	Waze raw archive filtered to the corridor	Jan 2022 – 18 Apr 2026	Hourly × exit	One-off batch	In use (gap 19 Apr – 3 Aug 2026)
Weather	Open-Meteo, ERA5, JMA	2020 onward; about 20 locations	Hourly (temperature, rain, wind, humidity, pressure)	Historical load, then hourly Open-Meteo updates by a scheduled task; Open-Meteo forecast for future rain	In use
Air quality	OpenWeather Air Pollution	27 Nov 2020 – 2 Aug 2026; 20 exits	Hourly (AQI, CO, NO_2, O_3, SO_2, PM2.5, PM10)	One-off load	In use (modelled AQI estimates, not roadside

					measure ments)
Emission factors	System reference table derived from the EMEP/EEA guidebook (Ntziachristo s & Samaras, 2023)	Static	Per vehicle class (g/km)	Static	In use
Events	Calendar of the Philippine Arena, Philippine Sports Stadium and Clark Global City (159 events)	Dec 2020 – Dec 2027	Event-day, venue, category, capacity; attendance for 63 events	One-time collection (Aug 2026); later additions by hand	In use
Holidays	Philippine holiday calendar and proclamation s	2020 – 2026	Day	Seed script	In use
Corridor geometry	Derived from exit coordinates and official km posts	Km 12 – Km 87; 20 exits, 19 segments (≈ 76 km)	Segment	Manual	In use
Community reports	SmartFlow mobile users	—	Post	Real time (app)	Not ingested into the warehou se
Apprehensio n and undercap	NLEX Corporation	—	—	—	Not loaded

3.9.2 Extract

Extraction follows three paths. The Waze archive, the weather and air-quality histories, the event calendar and the emission factors were loaded by batch scripts. NLEX toll exports, including the yearly 2022–2025 exports, pass through the ETL pipeline described below. Data analysts load new exports through the Data Management page, and the yearly files, which are too large to send from a browser, are run through a command-line loader that records each load as an upload that can be undone. The live Waze feed is polled about every two minutes by a separate ingestion process that writes directly to the bronze tables.

The upload pipeline applies four gates. Gate 1 (source availability) confirms that the file can be read and is not empty; NLEX's hourly toll export, which holds about 9.5 million rows a year, is recognised from its header and read as a stream. Gate 2 (field completeness) classifies the file as toll, accident or breakdown data from its column headers: toll files must contain every required column, and accident and breakdown files at least two signature columns. Gate 3 (NLEX corridor filter) validates each record: dates and times must be valid, required fields (date, location and kilometre post) must be present, vehicle class must be 1, 2 or 3, toll plazas must resolve to one of the 20 NLEX exits, incident kilometre posts must lie inside the corridor (Km 12 to 87, checked on the equivalent measured-kilometre scale), and dates after the end of the record or after the current date are held back. For the hourly toll export, Gate 3 also applies the client's processing rules for that export. Gate 4 (schema transformation) converts the accepted records to the warehouse schema and, for toll files, computes the vehicle-kilometres of each trip and rebuilds the emission estimate for the days loaded. Before loading, the file is compared with data already loaded, and a warning is shown when plaza-days that already hold at least 200 vehicles would change by more than 25%. An analyst can run these checks in check-only mode, which reports the results by rejection reason without saving any data.

Incident records that pass are written to the bronze layer of the AWS RDS PostgreSQL warehouse and published to silver. Toll records, which arrive already aggregated by hour, are written directly to the gold traffic, vehicle-kilometre and emission tables, and an hourly toll export replaces the whole days it covers. Every load is a single transaction, with duplicates skipped or replaced by natural key, and is recorded; the ten most recent loads can be undone. The live Waze feed bypasses the upload gates. Its records are written as received and are filtered to the corridor when they are published to the silver layer.

Several checks in the original gate design are not implemented. These are source token validation, encoding checks, schema-drift detection, automated cross-source consistency checks beyond the upload comparison warning, and detection and masking of personally identifiable information. Cross-layer consistency is instead checked manually with a verification script that reconciles row counts and key uniqueness across the bronze, silver and gold layers. Masking of the plate-number and driver fields in breakdown records is a required improvement before any production use.

3.9.3 Transformation

Validated bronze records are transformed in two steps, cleaning and feature engineering, before they are published to the silver and gold layers.

3.9.3.1 Data Cleaning Engine

The cleaning step resolves every plaza and exit name to a canonical name, normalizes date and time formats, parses kilometre posts, and removes duplicates by natural key. It applies business rules such as non-negative casualty counts, clearance times between 0 and 1,440 minutes, AQI values between 1 and 5, physically plausible weather values and finalized event status. Missing values are not imputed in the warehouse. Weather rows flagged as imputed by the source are excluded, and records without valid timestamps are left out of the duration-based KPIs, which report their coverage.

3.9.3.2 Feature Engineering Engine

The feature-engineering step derives the analytical variables used by the models. The variables are grouped into six categories: time, incident, traffic, emission, external and spatial features.

a.) Time Features

Temporal variables are derived from timestamps and the time dimension: hour of day, day of week, month, quarter, weekend flag and holiday flag. The time dimension covers every hour from 2020 to 2027.

b.) Incident Features

Incident variables describe each event's type, cause, severity class (property damage only, injury or fatal, derived from casualty counts), kilometre position, start time, lane-blockage duration and site-clearance duration. Waze alerts are mapped to a common incident-type dimension.

c.) Traffic Features

Traffic variables are derived from the hourly toll records. The average daily traffic (ADT) for plaza e, vehicle class c and direction g is the mean daily count over the days with valid data:

$$ADT_{e,c,g} = \frac{1}{N_e \sum_{d \in D_e} V_{e,c,g,d}} \quad (3.9.3.1)$$

where V_e,c,g,d is the number of class-c vehicles entering plaza e in direction g on day d, D_e is the set of days with valid data for plaza e, and N_e is the number of those days. The values are toll-plaza entries, not cross-section counts. The total ADT of a plaza sums the three classes, and the total plaza entries per direction sum all plazas:

$$ADT_{e,g} = \sum_{c=1}^{3} ADT_{e,c,g} \quad (3.9.3.2)$$

$$V_{corr,g} = \sum_{e \in E_g} ADT_{e,g} \quad (3.9.3.3)$$

where E_g is the set of plazas recorded for direction g. Plazas whose records carry no direction are summed separately, so the total of northbound and southbound entries does not equal all entries. Holiday and event effects are measured as the percentage deviation of a day's volume from a local control-day baseline:

$$\Delta_d = \frac{V_d - \bar{V}_d^{base}}{\bar{V}_d^{base}} \times 100\% \quad (3.9.3.4)$$

where $\bar{V}_d^{base}$ is the mean volume of the same weekday within ±45 days of day d, excluding holidays and event days, with at least four such days required. The design also specified an exit traffic share and a Volume Deviation Index (a z-score against the full-period mean). These were implemented only in an early prototype pipeline. The local baseline in Equation 3.9.3.4 replaced them because a full-period mean includes future data and ignores weekly seasonality.

d.) Emission Features

Emission variables estimate the CO_2 released on the corridor. Baseline emissions are calculated with a distance-based method that multiplies the vehicle-kilometres driven on NLEX by fixed per-class emission factors (Ntziachristos & Samaras, 2023). The vehicle-kilometres are taken from each toll trip's entry and exit plazas:

$$E^{base} = \sum_{e}\sum_{h}\sum_{g}\sum_{c=1}^{3}\sum_{j\in J_{e,h,g}} n_{j,c} d_{j,c} F_c \times 10^{-6} \quad (3.9.3.5)$$

where J_e,h,g is the set of entry–exit plaza pairs whose trips paid at plaza e in hour h and direction g, n_j,c is the number of class-c trips on pair j, d_j,c is the distance of those trips on NLEX in kilometres (Equation 3.9.3.6), F_c is the fixed CO_2 emission factor of class c (196.1, 689.4 and 798.2 g CO_2/km for Classes 1, 2 and 3, derived from the EMEP/EEA guidebook as described in Section 2.1.8.5), and 10^{-6} converts grams to tonnes. A trip that paid on SCTEX, SFEX or the NLEX Connector is not part of the NLEX traffic record, but the kilometres it drove on NLEX are included and are assigned to its NLEX entry plaza. Because each trip's whole NLEX distance is assigned to the plaza where it paid, emissions by exit describe the trips that paid at that exit, not the CO_2 emitted on the road section beside it. The factors are stored in the warehouse and are not retrieved from an API at run time, and the estimate has no speed term. NO_x and PM2.5 are estimated in the same way, with per-class factors of 0.246, 5.626 and 6.513 g/km (NO_x) and 0.00124, 0.119 and 0.138 g/km (PM2.5) for Classes 1, 2 and 3 from the same reference table, which also holds CO factors.

The distance d_j,c depends on what the toll record shows about the two ends of the trip:

$$d_{j,c}$$

$$= \begin{cases} \left|k(x_j) - k(a_j)\right|, & \text{entry and exit known and distinct} \\ \dfrac{\sum_{x \in X_j} \left|k(x) - k(a_j)\right|}{\left|X_j\right|}, & \text{paid on entry at an open-section plaza} \\ \dfrac{\sum_{i \in K_{e,c}} n_{i,c} d_{i,c}}{\sum_{i \in K_{e,c}} n_{i,c}}, & \text{no usable entry} \end{cases} \quad (3.9.3.6)$$

where a_j and x_j are the entry and exit plazas of pair j, and $k(\cdot)$ is a plaza's measured kilometre position on NLEX. A plaza on the NLEX Connector takes the position of the Harbor Link junction (measured Km 13.63), and a plaza on SCTEX or SFEX takes the position of the SCTEX junction (measured Km 85.23), so only kilometres on NLEX are counted; a trip whose two ends join NLEX at the same junction is given zero distance. X_j is the set of open-section exits ahead of the entry plaza in the direction of travel, up to the Bocaue Barrier where the closed system begins, or the exits on both sides where the record gives no direction. Because an open-section toll is paid on entry and no exit is recorded, the trip is assumed to leave equally at each of these exits (a proponent assumption; inferring destinations from entry-only records is discussed by Lei et al., 2021), and the nearest and farthest of these exits bound its distance. K_e,c is the set of class-c trips at the same plaza whose distance is known. Its mean is used for trips with no recorded entry (entry 0 or "No Ticket") and for trips whose entry and exit are the same plaza outside the open section.

The additional CO_2 emitted by vehicles idling in the queue behind a lane-blocking accident i is estimated from the lane-blockage duration and an idling rate weighted by the vehicle-class mix of the hour in which the blockage started:

$$E_i^{idle} = \varphi_{h_i} \cdot \frac{\lambda_{h_i}}{60} \cdot \frac{\left(t_i^{cleared} - t_i^{start}\right)^2}{2} \quad (3.9.3.7)$$

$$\varphi_h = \sum_{c=1}^{3} s_{h,c} \cdot \frac{q_c f_c}{60} \quad (3.9.3.8)$$

where $t_i^{cleared}$ is the time the blocked lane was cleared and t_i^{start} the event start time ($0 < t_i^{cleared} - t_i^{start} \leq 1{,}440$ minutes; accidents that blocked no lane are excluded), h_i is the hour of day in which the blockage started, λ_h is the corridor-average hourly vehicle flow [veh/h] per exit segment and direction at hour h over the selected period (total vehicles ÷ days ÷ exits ÷ 2 carriageways), and φ_h is the CO_2 emitted per idling vehicle-minute at hour h [kg/veh-min]. In Equation 3.9.3.8, s_h,c is the share of class-c vehicles among the vehicles recorded at hour h, and q_c is the fuel burned while idling [US gal/h], taken as 0.3 for a car, 0.8 for Class 2 and 1.0 for a large truck, planning values set by the proponents within the range reported by the U.S. Department of Energy (2015), from 0.16–0.17 gal/h for compact cars to nearly 1 gal/h for a transit bus. f_c is the CO_2 emitted per gallon of fuel burned: 8.887 kg for gasoline (Class 1) and 10.180 kg for diesel (Classes 2 and 3) (U.S. Environmental Protection Agency, 2026). This gives 0.044, 0.136 and 0.170 kg CO_2 per vehicle-minute for Classes 1, 2 and 3. The term $\lambda(t^{cleared} - t^{start})^2/120$ is the total vehicle-minutes of delay in a deterministic queue that builds while the lane is blocked (D/D/1 triangle), with E_i^idle in kg CO_2. The idling rates are United States figures, because no Philippine values were found, and the Class 2 rate comes from heavy-duty trucks, so it may overstate idling by buses and small trucks. The queue is assumed to clear as soon as the lane reopens, which understates the delay. Delay-induced CO_2 is reported as a KPI and is not used as an input feature of the forecasting models.

The CO_2 total is reported with a low–high range, because the toll record cannot fix where open-section trips leave or which vehicles make up each toll class. The lower value is:

$$E^{low} = \sum_{c=1}^{3} \left(VKM_c + \sum_{o \in O} n_{o,c} (d_o^{low} - d_o) \right) F_c^{low} \times 10^{-6} \quad (3.9.3.9)$$

where VKM_c is the class-c vehicle-kilometres of Equation 3.9.3.5, O is the set of open-section entry trips, d_o is their central distance, d_o^low the distance to the nearest exit ahead, and F_c^low the lowest CO_2 factor among the guidebook categories that fit class c (180.6, 492.9 and 667.8 g/km). The upper value E^high replaces d_o^low with the distance to the farthest exit ahead and F_c^low with F_c^high (251.2, 836.3 and 944.5 g/km). The range therefore combines the two inputs that the toll record cannot fix.

e.) External Features

External variables include hourly temperature, rainfall, wind and humidity, a wet-hour flag (mean corridor rainfall above 0.3 mm in the hour), holiday flags and Philippine Arena event-day flags. For forecasts, future weather is replaced by the day-of-year climatology of the training data, or by the Open-Meteo forecast for near-term rain, so that no observed future value enters a model.

f.) Spatial Features

Spatial variables encode each plaza's official kilometre post (Km 12 at Balintawak to Km 87 at Sta. Ines), the measured position along the corridor centreline (which also gives each toll trip's distance on NLEX), the open-section exit distances used for trips paid on entry, and the PostGIS geometries of the 19 corridor segments, Waze jam lines and Waze alert points. Kilometre positions are stored as measured distances and shown to users as official posts. Lane counts used by the simulation sandbox come from OpenStreetMap.

Before publication, the silver layer applies the business rules listed above, and the gold layer is rebuilt from the silver records. Referential integrity between fact and dimension tables is not enforced by foreign keys. It is checked by the manual verification script.

3.9.4 Load and Data Warehouse Layer

The cleaned records are stored in AWS RDS PostgreSQL 18 with the PostGIS 3.6 extension, organized into a medallion architecture. The bronze schema keeps each source as received. The silver schema holds cleaned and standardized records, including the corridor segments, Waze jam and alert facts with PostGIS geometries (with spatial indexes) and the NLEX incident event tables. The gold schema holds analysis-ready facts and aggregates, such as hourly traffic, hourly emissions, daily weather, jam profiles and volume profiles, together with model outputs, accuracy metrics, retraining logs and the log of served forecasts. Serving views in the public schema expose these tables to the API.

The gold layer is the single source of truth for the dashboard and the mobile application. The API computes descriptive analytics from the gold facts on request, reads stored forecasts and metrics for the predictive views, and computes prescriptive recommendations from those forecasts. The design proposed the gold layer as a retrieval-augmented generation (RAG) source for the chatbot. That retrieval index was not built. The language-model features instead receive the relevant values directly or call API tools (Section 3.11.5).

3.10 Presentation of Schema

During design, the operational data were modelled as a fact constellation schema with two fact tables (traffic volume and incident log) sharing six dimension tables (Figure 3.10.1). In the implemented warehouse, the same analytical roles are filled by the tables listed in Table 3.10.1, organized by medallion layer.

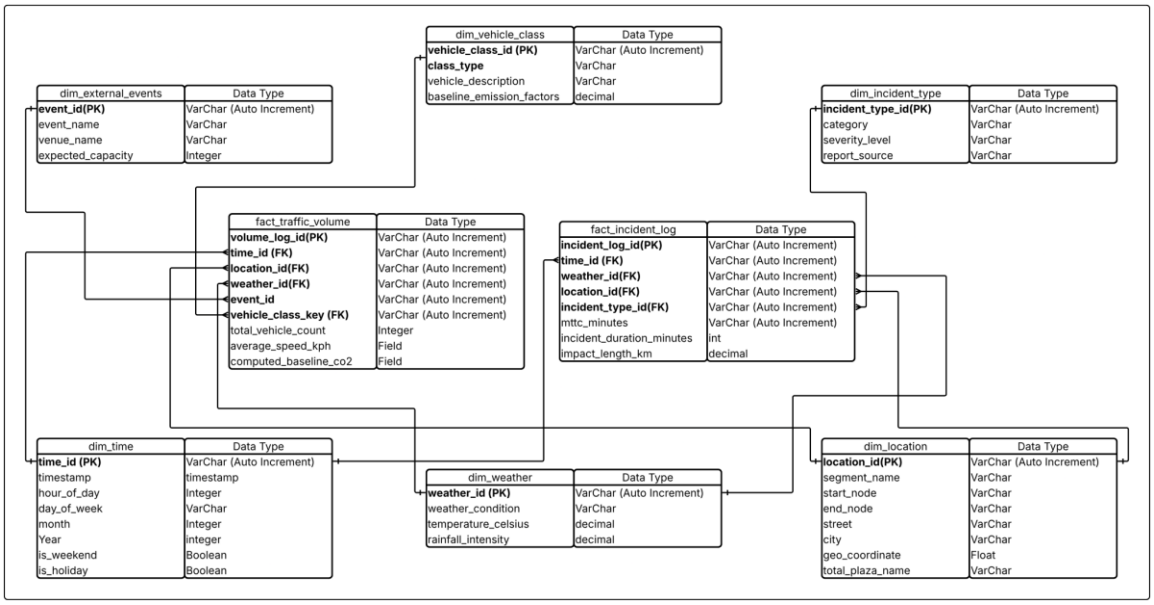


Figure 3.10.1 Conceptual Dimensional Model of the SmartFlow NLEX Warehouse (Design)

The conceptual model separates continuous traffic measurements from discrete incidents. Keeping them in separate fact tables prevents incident records from distorting the volume series used for forecasting while still allowing both to be analysed by time, location and weather.

In the implementation, the traffic fact is gold.fact_traffic_hourly. It records entries and exits per plaza, hour, direction and vehicle class, and vehicle-kilometres and emissions are stored in parallel tables, gold.fact_vkm_hourly and gold.fact_emissions_hourly. Speeds are not stored with the toll counts because they come only from Waze. The incident fact is split by source: silver.fact_incident_log holds live Waze alerts with PostGIS points, while NLEX accident and breakdown events are held in silver.nlex_accident_events_clean and silver.nlex_breakdown_events_clean, with their timestamps and kilometre posts.

The six design dimensions map to the implemented warehouse as follows:

1. dim_location: implemented as silver.dim_location, with 19 corridor segments stored as PostGIS LINESTRING geometries with spatial indexes, supported by the official kilometre-post and segment-length tables.
2. dim_time: implemented as public.dim_time, with one row per hour from 2020 to 2027 and hour, weekday, month, quarter, weekend and holiday attributes.
3. dim_vehicle_class: represented by silver.nlex_emission_factors, which holds the three vehicle classes and their emission factors.
4. dim_weather: represented by the hourly weather tables (public.hourly_weather, silver.hourly_weather_clean) and the daily summary gold.daily_weather_summary.
5. dim_external_events: represented by public.philippine_arena_events and the holiday calendar public.ph_holidays.
6. dim_incident_type: implemented as silver.dim_incident_type, which currently classifies Waze alert types; NLEX incident types are stored as attributes of the event tables.

A star-style view, public.v_nlex_galaxy_hourly, joins the time, location, weather, jam, incident and event data at hourly grain and is the closest implemented equivalent of the conceptual constellation. Model outputs and evaluation results are stored in dedicated gold tables, including the volume, congestion, emission, fleet-mix and incident forecasts, the model metrics, the horizon accuracy, the retraining batch log and the served-forecast log, with backups of every model output kept in a separate schema.

Table 3.10.1 Implemented Warehouse Objects by Layer

Layer	Main objects	Role
Bronze	nlex_accident_data, nlex_breakdown_data, nlex_incidents, waze_raw_jams, waze_raw_alerts, waze_hourly_jams, nlex_emissions, nlex_emission_factors, philippine_arena_events	Raw records as received from each source
Silver	nlex_accident_events_clean, nlex_breakdown_events_clean, nlex_incidents_clean, fact_waze_jams (LINESTRING), fact_incident_log (POINT), waze_jam_hourly_exit, hourly_weather_clean, nlex_emissions_clean, dim_location, dim_incident_type, nlex_emission_factors, philippine_arena_events_clean	Cleaned, standardized and spatially indexed records
Gold (facts and aggregates)	fact_traffic_hourly, fact_traffic_hourly_origin, fact_vkm_hourly, fact_emissions_hourly, daily_traffic_volume_corrected, daily_weather_summary, daily_emissions_summary, exit_km_post, exit_segment_km, open_plaza_km, congestion_jam_profile, exit_volume_profile	Analysis-ready facts, profiles and reference tables
Gold (model outputs)	ml_predictive_volume, ml_forecast_48h_runs, ml_forecast_hourly_48h, ml_congestion_forecast, ml_predictive_emissions, ml_predictive_fleet_mix, ml_event_surge_forecast, incident and weather-speed outputs, ml_model_metrics, ml_horizon_accuracy, ml_batch_runs, ml_congestion_forecast_log	Forecasts, accuracy metrics, retraining log and served-forecast log
Public	dim_time, ph_holidays, hourly_weather, philippine_arena_events, audit_logs, data_uploads, nlex_maintenance_schedules, nlex_mobile_config, serving views	Shared dimensions, application tables and views used by the API

3.11 Business Analytics Models and Testing

This section documents the analytics models of SmartFlow NLEX in three tiers: descriptive, predictive and prescriptive. It also covers model training and testing, the agent-based simulation sandbox and the language-model features. For every method, it states the purpose, inputs, formulation, procedure, output, interpretation and the system feature that uses it. Table 3.11.1 classifies every method named in the design and gives its implementation status, so that methods proposed in the design but not yet built are kept and identified as proposed.

Table 3.11.1 Analytics Method Inventory and Implementation Status

Tier	Method	Classification	Status	System feature
Descriptive	Volume aggregation and hour-by-weekday profiles	Data transformation / KPI	Implemented	Traffic – Descriptive
Descriptive	Local control-day comparison (holidays, events)	Analytical technique	Implemented	Traffic – Descriptive
Descriptive	Incident frequency, severity and 5-km segment counts	KPI / aggregation	Implemented	Incidents – Descriptive
Descriptive	Wet-vs-dry crash rate ratio (per hour)	Descriptive metric (rate ratio)	Implemented	Incidents – Descriptive
Descriptive	Clearance-time summary and Kaplan–Meier estimator	Statistical estimator	Implemented	Incidents – Descriptive, Prescriptive
Descriptive	Emission inventory and fleet mix	Deterministic accounting model	Implemented	Emissions – Descriptive
Descriptive	STL/MSTL, ACF/PACF, KDE/NKDE, Getis-Ord Gi∗	Analytical techniques /	Proposed	—

		statistical tests		
Descriptive	GMM regimes; speed–flow (fundamental diagram) fit; log-normal fit	Statistical models	Proposed	—
Descriptive	Volume Deviation Index; V/C ratio	Derived anomaly score; KPI	Proposed	—
Predictive	Prophet (champion), Holt-Winters, Holt's linear, SARIMAX, LSTM; seasonal-naive and climatology baselines	Statistical / deep-learning models	Implemented (LSTM, SARIMAX rejected)	Traffic – Predictive
Predictive	Congestion-state candidates (1–168 h): XGBoost, GRU, single-layer, stacked and dual-path LSTMs, dual-path LSTM with XGBoost, MSTL with exponential smoothing; served model chosen again every hour	Machine-learning, deep-learning and statistical models	Implemented	Live Map, Traffic, Mobile Corridor
Predictive	48-hour hourly volume forecast: stacked LSTM (single-layer, dual-path and hybrid candidates) with a split-conformal 80% band; same-hour-last-week and profile baselines	Deep-learning models	Implemented	Traffic – Predictive
Predictive	Per-exit event uplift with bootstrap interval	Statistical estimator	Implemented	Traffic – Predictive
Predictive	Negative binomial incident counts (Poisson, RF, XGBoost, SARIMAX, LSTM, GRU comparators)	Statistical / ML models	Implemented	Incidents – Predictive
Predictive	Ordinal logistic and XGBoost severity; Cox PH clearance time; logistic secondary-incident risk	Statistical / ML models	Implemented	Incidents – Predictive

Predictive	Geographically weighted regression; spatial LSTM	Statistical / deep-learning models	Implemented	Incidents – Predictive
Predictive	Gradient boosting CO_2 forecast (polynomial, LSTM, derived comparators)	Machine-learning model	Implemented	Emissions – Predictive
Predictive	Dirichlet regression fleet mix (XGBoost, Prophet, LSTM comparators)	Statistical model	Implemented	Emissions – Predictive, Sandbox
Predictive	XGBoost volume model; quantile random forest; GRU wet/dry speed models; Bi-LSTM with attention; multiple linear regression; forecast ensembles	ML / statistical models	Proposed (not implemented)	—
Prescriptive	Toll-booth staffing allocation	Optimization (linear allocation, exact greedy solution)	Implemented	Traffic – Prescriptive
Prescriptive	Maximal covering location model for patrol positions	Optimization (binary integer program)	Implemented	Incidents – Prescriptive
Prescriptive	Two-stage stochastic clearance protocol	Optimization (stochastic mixed-integer program)	Implemented	Incidents – Prescriptive
Prescriptive	Fuzzy congestion-response advisory	Rule-based inference	Implemented	Traffic – Prescriptive
Prescriptive	TOPSIS ranking of event interventions and emission strategies	Multi-criteria decision analysis	Implemented	Traffic, Emissions – Prescriptive
Prescriptive	Emission strategy estimates with seeded bootstrap ranges	Scenario estimation / uncertainty analysis	Implemented	Emissions – Prescriptive
Prescrip	Weather-triggered speed	Rules /	Proposed	—

tive	advisory; capacity-based surge alert; heuristic rerouting; variable speed limits; price-optimized tolling	optimization	(not implemented)	
Simulation	Agent-based microsimulation (IDM car-following, MOBIL lane changing)	Simulation methodology	Implemented	Scenario Sandbox
Language model	GLM command interpretation and narratives; Qwen tool-calling assistant	Interface (natural-language processing)	Implemented	Sandbox, dashboard narratives, mobile Assistant

3.11.1 Descriptive Analytics

The descriptive tier summarizes the historical record to establish baselines for traffic, incidents and emissions. All descriptive measures are computed by SQL queries in the API on request for the date range the operator selects, so they always reflect the current warehouse.

3.11.1.1 Data Overview

The descriptive tier uses the hourly toll fact (19 of the 20 exits, January 2022 to December 2025), the NLEX accident, breakdown and crash event tables, the historical and live Waze jam and alert facts, hourly weather, the holiday and event calendars, and the emission inventory (Section 3.9). Analyses default to the most recent twelve months of the record and can be set to any range within it.

3.11.1.2 Volume Aggregation and Temporal Profiles

Purpose: to show how much traffic the corridor carries and when demand peaks. **Inputs:** hourly entries by plaza, direction and class. **Procedure:** hourly counts are summed to daily and monthly totals by direction, averaged by hour of day and day of week, and ranked by plaza; each figure is compared with the preceding period of equal length. **Output: volume trend charts, an hour-by-weekday heatmap, plaza rankings and average daily traffic by class (Equation 3.9.3.1). Plazas whose records carry no direction are shown as a third part of the volume trend, so that the trend's total equals the Total Volume KPI. The plaza ranking also shows SCTEX as the vehicles that entered NLEX through the Mabiga toll plaza; this bar is not part of the corridor total. Interpretation:** recurring peaks identify when staffing and patrol coverage are most needed. **Feature:** Traffic – Descriptive and Overview.

3.11.1.3 Holiday and Event Impact by Local Control-Day Comparison

Purpose: to measure how holidays and Philippine Arena events change traffic. **Inputs:** daily entries at the relevant plaza, the holiday and event calendars. **Formulation:** the percentage deviation of the day's volume from the mean of comparable control days (Equation 3.9.3.4). Control days are the same weekday within ±45 days, excluding holidays and other event days, with at least four required. Days with several events are merged into one record. **Output:** a deviation per holiday or event and its average by year. **Interpretation:** a positive deviation indicates added demand. A local baseline is used because a global average would mix pandemic-period traffic into every comparison. The control-day design follows Giuliano and Lu (2021). **Feature:** Traffic – Descriptive (event and holiday panels).

3.11.1.4 Incident Frequency, Severity and Spatial Concentration

Purpose: to identify when and where incidents occur. **Inputs:** accident and breakdown events and crash reports with kilometre posts, causes and casualty counts. **Procedure:** incidents are counted by day, hour and weekday, type and cause; injuries and fatalities are summed; and incidents are counted in five-kilometre bins of official kilometre posts to rank segments. **Output:** incident trends, an hour-by-weekday heatmap, cause and type breakdowns and a ranked list of high-incident segments. **Interpretation:** segments and hours with persistently high counts are candidates for patrol positioning. **Feature:** Incidents – Descriptive.

3.11.1.5 Weather-Stratified Crash Rate

Purpose: to compare crash risk in wet and dry conditions. **Inputs:** crash counts and hourly rainfall. An hour is wet when the mean corridor rainfall exceeds 0.3 mm. **Formulation:**

$$RR = \frac{\frac{n_{wet}}{H_{wet}}}{\frac{n_{dry}}{H_{dry}}} \quad (3.11.1.1)$$

where n is the number of crashes and H the number of hours in each condition. **Output:** the crash rate per hour in wet and dry conditions and their ratio, with an approximate Poisson 95% confidence interval exp(ln RR ± 1.96√(1/n_wet + 1/n_dry)), which is too narrow if wet hours are clustered. **Interpretation:** a ratio above 1 indicates a higher crash rate per hour of exposure in wet conditions. The ratio is normalized by hours of exposure, not by traffic volume (Romanowska & Budzyński, 2022). **Feature:** Incidents – Descriptive (Crash Rate in Rain).

3.11.1.6 Clearance-Time Summary and Kaplan–Meier Estimation

Purpose: to describe how long incidents take to clear. **Inputs:** event start, lane-blockage-cleared and site-cleared times. **Procedure:** for the mean time to clear (MTTC), durations outside 0 to 1,440 minutes are excluded, and the MTTC, median and 90th percentile are reported with the share of records that have valid timestamps. For survival analysis, durations of 0 or less are excluded and the survival function is estimated with the Kaplan–Meier estimator, with durations above 1,440 minutes treated as censored at 1,440 minutes:

$$\hat{S}(t) = \prod_{t_i \le t} \left(1 - \frac{d_i}{n_i}\right) \quad (3.11.1.2)$$

where d_i is the number of incidents cleared at time t_i and n_i the number still uncleared just before t_i. **Output:** MTTC by incident type and survival curves. **Interpretation:** the curve gives the probability that an incident is still uncleared after t minutes and supplies the clearance scenarios used in the prescriptive clearance protocol (James et al., 2021). **Feature:** Incidents – Descriptive and Prescriptive.

3.11.1.7 Emission Inventory and Fleet Mix

Purpose: to estimate the corridor's emissions and the contribution of each vehicle class. **Inputs: hourly toll trips by entry plaza, exit plaza and class, the measured kilometre posts, the emission factors, and the lane-blockage times of accidents. Formulation: Equations 3.9.3.5 to 3.9.3.9. Output: CO_2, NO_x and PM2.5 by day, paying exit and class; the CO_2 total with a low–high range; each class's share of trips, vehicle-kilometres and CO_2; corridor CO_2 per vehicle-kilometre; and delay-induced CO_2. For 2025 the estimate is 834,674 t of CO_2 (range 610,173 to 1,150,245 t), or 318.9 g per vehicle-kilometre, and delay-induced CO_2 is 1,338 t (0.16%). Interpretation: because the factors are fixed per class, changes in the estimate reflect changes in trips, trip length and fleet mix, not in driving speed; emissions by exit are those of the trips that paid there. Feature:** Emissions – Descriptive.

3.11.1.8 Live Corridor Status

Purpose: to show current conditions. **Inputs:** Waze jams and alerts from the last 60 minutes. **Procedure:** jams and alerts within the corridor tolerance are assigned to the nearest exit and summarized by jam level, speed and alert type; the congestion index is the mean Waze jam level (0 to 5). **Output:** status by exit on the Overview and the Live Map. **Feature:** Overview, Live Map and the mobile Dashboard.

3.11.1.9 Proposed Descriptive Methods

The design also specified several descriptive methods that were not implemented in the current version. They remain relevant to the research problem and are retained as proposed extensions:

1. **STL/MSTL decomposition and rolling means** to separate trend, weekly and annual seasonality and irregular variation in daily volume (Hyndman & Athanasopoulos, 2021). The champion volume model currently captures these components within its own structure (Section 3.11.2.2).

2. **ACF and PACF analysis** to identify candidate autoregressive orders. The current SARIMAX specification uses fixed orders with a weekly period.

3. **Gaussian mixture models** to identify traffic regimes from speed–flow data. Congestion states are currently defined by Waze jam-speed thresholds of 10 and 20 km/h.

4. **Speed–flow (fundamental diagram) fitting to estimate free-flow speed and capacity by segment (National Academies of Sciences, Engineering, and Medicine, 2022; Treiber & Kesting, 2025). The sandbox currently scales the free-flow speeds that Mejia and Sigua (2018) reported for rain conditions to fixed speed caps of 100, 100 and 96 km/h.**

5. **Kernel density estimation, network-constrained KDE and the Getis-Ord Gi∗ statistic** to test whether incident concentrations are statistically

significant (Berhanu et al., 2023; Alam & Tabassum, 2023). Segments are currently ranked by counts.

6. **Log-normal fitting of clearance times**, the **Volume Deviation Index with a two-sigma rule**, and the **volume-to-capacity ratio** (which requires section flows, passenger-car equivalents and segment capacities that are not in the record).

3.11.2 Predictive Analysis

The predictive tier forecasts traffic volume, congestion state, event uplift, incidents, clearance time, emissions and fleet mix. For each task, several candidate models are trained and compared against naive baselines on chronologically held-out data, and only a model that beats the baselines is published. The models are trained offline in Python. Their forecasts and held-out metrics are stored in gold tables and read by the dashboard and the mobile application.

3.11.2.1 Data Overview and Validation Design

Volume, CO_2 and fleet-mix models are trained on daily series from 1 January 2022 to 31 December 2025 (1,461 days). They are evaluated by rolling-origin (walk-forward) validation, in which the model is refitted at successive forecast origins and scored on the following 7 or 14 days, so that every test day lies after its training data. Two evaluation windows are reported: the most recent 20% of days (“80/20”) and the most recent 10% (“90/10”). Part of each nominal share is taken up by lag features at the start of the series, so slightly fewer days are scored. Each candidate is compared with two baselines: a seasonal-naive forecast that repeats the value of the same weekday one week earlier, and a climatology forecast (the average for that day of the year). The congestion classifier is trained on the live Waze feed and tested on the most recent days. The daily incident count models are scored on the final 20% of the record, like the volume and CO_2 models. The weather and speed models use a chronological holdout of the last 90 days with usable Waze speeds, and the severity and clearance-time models the last 20% of events.

3.11.2.2 Daily Corridor Volume Forecasting

Purpose: to forecast total daily entries on the corridor for staffing and resource planning. **Inputs:** the daily volume series and weather variables; for future days the weather is the training-period climatology, which prevents future observed weather from entering the forecast. **Candidate models:** Prophet, additive Holt-Winters exponential smoothing with a seven-day season, Holt's linear trend, SARIMAX(1,1,1)(1,1,1)$_7$ with weather regressors (Hyndman & Athanasopoulos, 2021), and a two-layer LSTM (64 and 32 units, 30-day input), each with and without weather. Prophet models the series as a sum of components:

$$y(t) = g(t) + s(t) + \beta^{\top} x_t + \varepsilon_t \quad (3.11.2.1)$$

where g(t) is a piecewise-linear trend with automatically detected changepoints, s(t) the weekly and yearly seasonality expressed as Fourier series, x_t the weather regressors and ε_t the error. No holiday or event regressor is used in the volume model. The additive Holt-Winters forecast is:

$$\hat{y}_{t+h} = \ell_t + h\, b_t + s_{t+h-m(k+1)} \quad (3.11.2.2)$$

where $\hat{y}_{t+h}$ is the forecast made at time t for h days ahead, with level ℓ_t, trend b_t, seasonal component s with period m = 7 and k the integer part of (h − 1)/m. **Selection: the candidate with the lowest WMAPE is published if it beats the best baseline on WMAPE and has a lower MASE than the seasonal-naive forecast scored on the same days (Section 3.11.3). Output:** the daily forecast with the evaluation metrics of every candidate. Forecasts are validated up to 14 days ahead; days beyond this are shown as unvalidated. **Feature:** Traffic – Predictive, booth staffing and the sandbox demand profile.

3.11.2.3 Congestion-State Classification

Purpose: to forecast whether each exit will be in a low, medium or high congestion state h hours ahead (h = 1 to 168). **Target:** the state at hour t + h, defined by the length-weighted Waze jam speed at the exit in that hour (20 km/h or more, or no jam reported, low; 10 to below 20 km/h medium; below 10 km/h high). The proponents chose these cut-offs from the corridor's own speed distribution; the dashboard labels the states Moving, Heavy and Severe. **Inputs:** exit, hour, weekday, lagged states and speeds at 1, 2, 3, 6, 24 and 48 hours, 6- and 24-hour rolling means, neighbouring-exit lags, the exit-by-hour historical profile computed from the training period only, and the horizon. **Candidates: an XGBoost classifier (Bentéjac et al., 2021) with 400 trees, depth 6 and learning rate 0.08, whose probabilities are calibrated by isotonic regression; a GRU network (48 units) over the last 24 hours; a single-layer LSTM, a stacked LSTM and a dual-path LSTM (Section 3.11.2.11) that read the last 72 hours of states, jam counts, jam speeds and neighbouring states, the last 28 days as daily summaries, and the known calendar of each forecast hour, and predict all 168 hours in one pass, with probabilities calibrated by isotonic regression; the dual-path LSTM with an XGBoost stage; and an MSTL decomposition of each exit's hourly jam speed into daily (24-hour) and weekly (168-hour) patterns with simple exponential smoothing of the remainder (Bandara et al., 2025), refitted daily. XGBoost minimizes a regularized loss:**

$$\mathcal{L} = \sum_i l(y_i, \hat{y}_i) + \sum_k \Omega(f_k), \qquad \Omega(f) = \gamma T + \frac{1}{2}\lambda \|w\|^2 \quad (3.11.2.3)$$

where l is the multi-class log loss, f_k the k-th tree, T its number of leaves and w its leaf weights. A random forest and SARIMAX are added in full runs. Persistence, the majority class and the exit-by-hour profile serve as baselines. Training rows whose target time falls after the training cut-off are dropped to prevent leakage. All candidates are scored on the same exit-hours of the 12 most recent days. A state is chosen from the probabilities by counting the high-state probability 1.5 times, so that missing a severe hour weighs as much as one and a half false alarms; the threshold for declaring a class should reflect the relative cost of misses and false alarms (Silva Filho et al., 2023). The same weighting is used in testing, in choosing the served model and in the stored state that the mobile application reads. Among the candidates that beat all three baselines, those whose accuracy is within 0.5 percentage points of the most accurate are treated as tied, and the one that catches the most severe hours is served, because accuracy differences that small are within the noise of a 12-day test. Output: the probability of each state by exit and hour. A cell is shown as congested when the combined probability of the medium and high states is at least 0.5, and as Severe rather than Heavy when the high-state probability is at least the medium-state probability. Only a model that beats all three baselines is served; otherwise the exit-by-hour profile is shown and labelled as such. XGBoost and the GRU are refitted every hour; the LSTM candidates are refitted in the weekly batch and rescored every hour. Feature: Live Map forecast, Traffic – Predictive, the fuzzy advisory and the mobile Corridor forecast.

3.11.2.4 Event-Day Uplift Estimation

Purpose: to estimate how much a Philippine Arena event raises traffic at each exit. **Inputs:** the event calendar, the holiday calendar and daily plaza volumes. **Procedure:** for every event day, the ratio of each exit's volume to its typical volume for the same weekday and month is computed; this forecasting baseline differs from the ±45-day control days used descriptively in Section 3.11.1.3. The median ratio across event days is the exit's uplift, $U_e = \text{median}_k\ (V_{e,k} / \tilde{V}_{e,k})$, where $\tilde{V}_{e,k}$ is the weekday-month median volume on days without events or holidays, and a percentile-bootstrap 95% confidence interval (2,000 resamples) is attached. Holidays are treated as a separate regime, and an uplift is considered material only when its interval lies above 1 and it rests on at least eight event days. Event days must come from the calendar. Selecting them by high volume would make the estimate circular. **Output:** uplift and expected additional vehicles by exit for upcoming events. **Feature:** Traffic – Predictive and the TOPSIS event ranking.

3.11.2.5 Incident Count Forecasting

Purpose: to forecast daily accident and breakdown counts. **Inputs:** lagged counts (1, 7, 14 and 364 days), rolling means, calendar and holiday flags, rainfall and log traffic volume. **Model:** a negative binomial generalized linear model, which allows the variance to exceed the mean (Lord et al., 2021):

$$\log \mu_t = \beta_0 + \beta^\top x_t, \qquad Var(y_t) = \mu_t + \alpha\mu_t^2 \quad (3.11.2.4)$$

Poisson regression, random forest, XGBoost, SARIMAX, LSTM and GRU models are trained as comparators. In the current code, the champion is the model with the highest R^2 among those with MASE at or below 1 against the seven-day naive forecast, a rule that differs from the WMAPE-based selection used for the volume and CO_2 models. In the holdout, same-day rainfall and volume are observed values, so the holdout error is optimistic relative to true forecasting. The models are scored on the final 20% of the record (Section 3.11.2.1), and holiday flags are taken from the Philippine holiday calendar (public.ph_holidays). The validation table on the dashboard also shows each model's training R^2 and the gap between training and validation R^2, so that over-fitting can be seen alongside the error metrics. **Output:** daily incident forecasts and per-exit and per-segment expected incidents, which weight the patrol positioning model. **Feature:** Incidents – Predictive and Prescriptive.

3.11.2.6 Incident Severity and Secondary-Incident Risk

Purpose: to estimate the likely severity of an accident and the risk of a secondary incident. **Model:** an ordinal logistic regression (Washington et al., 2020) for the ordered classes property damage only, injury and fatal:

$$logit\ P(Y \leq k|x) = \theta_k - \beta\text{^}\top x \quad (3.11.2.5)$$

and an XGBoost classifier as comparator. Both are compared with the majority-class baseline, and injury counts are excluded from the features to avoid target leakage. Secondary-incident risk is estimated by a ridge-penalized logistic regression on a label that marks accidents occurring within 2 km of a primary accident, in either direction, from its start until 30 minutes after it was cleared; logistic regression has been used to estimate the likelihood of freeway secondary crashes (Salek et al., 2021). **Feature:** Incidents – Predictive.

3.11.2.7 Clearance-Time Modelling

Purpose: to estimate how long an accident will take to clear. **Model:** a Cox proportional hazards model with a penalty of 0.1:

$$h(t|x) = h_0(t)\exp(\beta^{\top}x) \quad (3.11.2.6)$$

where $h_0(t)$ is the baseline hazard and $\exp(\beta)$ the hazard ratios of the covariates (Collett, 2023; Washington et al., 2020). Discrimination is measured by the concordance index. The dashboard's survival curves use the empirical Kaplan–Meier estimator (Equation 3.11.1.2), and the model currently treats all records as uncensored. **Feature:** Incidents – Predictive.

3.11.2.8 Spatial Incident Models

Purpose: to capture how the relationships between incidents and their predictors vary along the corridor. **Model:** a Poisson geographically weighted regression (Al-Hasani et al., 2021; Soroori et al., 2021) across the 20 exits with a fixed bandwidth, in which the coefficients vary with location:

$$\log \mu_i = \beta_0(u_i, v_i) + \sum_k \beta_k(u_i, v_i) x_{ik} \quad (3.11.2.7)$$

where (u_i, v_i) are the coordinates of exit i. A spatial LSTM that adds the lagged counts of the two nearest exits is trained for exit-day incident counts. Exit volume is taken from the hourly toll record for 19 of the 20 exits; the SCTEX interchange, which has no toll rows of its own, is assigned the volume of a typical exit. Both are evaluated against per-exit mean baselines. With only 20 locations, the GWR coefficients are interpreted descriptively. **Feature:** Incidents – Predictive (spatial risk).

3.11.2.9 Corridor CO_2 Forecasting

Purpose: to forecast daily corridor CO_2 for the next one to seven days. **Model:** gradient boosting regression (Bentéjac et al., 2021) with 300 trees, depth 3 and learning rate 0.05, trained on weekday, month, lags of up to 28 days, rolling means, rainfall and temperature climatology and holiday flags known in advance. The model builds an additive ensemble of trees:

$$F_m(x) = F_{m-1}(x) + \nu\, h_m(x) \quad (3.11.2.8)$$

where h_m is the m-th tree fitted to the residuals and v the learning rate. Polynomial ridge regression, an LSTM and a volume-derived forecast (forecast volume multiplied by CO_2 per vehicle) are comparators, scored by the same day-by-day projection used to serve the forecast. Because the CO_2 inventory is calculated from toll trips and their distances (Equation 3.9.3.5), this is a forecast of the inventory, not of measured emissions. **Feature:** Emissions – Predictive.

3.11.2.10 Fleet-Mix Forecasting

Purpose: to forecast the daily shares of Class 1, 2 and 3 vehicles, which must sum to one. **Model:** Dirichlet regression estimated by maximum likelihood, in which the shares follow a Dirichlet distribution whose parameters depend on the covariates:

$$(p_1, p_2, p_3) \sim Dir(\alpha_1, \alpha_2, \alpha_3), \log \alpha_c = x^{\top}\beta_c \quad (3.11.2.9)$$

XGBoost, Prophet and a multivariate LSTM in centred log-ratio space are comparators, and acceptance uses the Aitchison distance against the persistence and climatology baselines. **Feature:** Emissions – Predictive and the sandbox vehicle mix.

3.11.2.11 Hourly Volume Forecasting (48 Hours)

Purpose: to forecast hourly toll volume for the 48 hours after the latest upload, by plaza, direction and vehicle class, for shift-level planning. Series: 32 plaza series (exit, direction, entry or exit role and toll system), each with Class 1, 2 and 3 counts. Each trip is counted once, at the plaza where it paid, so the series add up exactly to the corridor total. Inputs: the last 72 hours of each series (class counts, observed corridor rainfall and calendar), the last 56 days as daily totals, and for each forecast hour its known calendar (hour, weekday, day of year, Philippine holiday from the holiday calendar, year-end week and an event day at the Philippine Arena, Philippine Sports Stadium or Clark) together with the same hour one and two weeks earlier. The rainfall of the forecast hours is never an input. Counts are transformed with log(1 + x) and standardized per series and class on the training hours.

Candidates: a single-layer LSTM (64 units); a stacked LSTM with two layers (64 and 32 units), in which the output sequence of the first layer is the input of the second (Cui et al., 2020); a dual-path LSTM; and the dual-path LSTM with an XGBoost correction fitted to its validation residuals. The dual-path LSTM reads the recent hours and the longer history in two branches: a short branch of two LSTM layers over the last 72 hours and a long branch of three LSTM layers over the last 56 days. A learned gate combines the two summaries, an approach related to multi-branch LSTM hybrids for traffic flow (Zheng et al., 2021):

$$h = g \odot h_{short} + (1 - g) \odot h_{long},$$

$$g = \sigma\left(W\left[h_{short}, h_{long}, e\right] + b\right) \quad (3.11.2.10)$$

where h_short and h_long are the outputs of the two branches, e the known calendar of the forecast hours, σ the logistic function and ⊙ element-wise multiplication. Each candidate predicts all 48 hours and the three classes in one pass and is trained with the Huber loss, early stopping and a learning-rate reduction. Baselines: the same hour one week earlier (seasonal naive) and the previous dashboard method, which spreads the same weekday's total from the previous week over that plaza's usual hourly shape.

Validation: the models are trained on forecast origins from 28 February 2022 to 30 September 2024, early-stopped on October to December 2024, and tested on every fifth hour of 2025 (1,709 forecast origins and 81,966 scored forecast hours). Selection: the candidate with the lowest WMAPE per plaza, direction, class and hour is chosen, a simpler candidate within 1% (relative) of the best is preferred, and a learned model is used only if it beats both baselines; otherwise the previous method is served (Makridakis et al., 2022). The chosen model is refitted on forecast origins up to 30 September 2025 and early-stopped on October to December 2025. A plaza series whose last 14 days or 56-day window has a missing day falls back to the previous method, and the run records the reason.

Uncertainty: an 80% band is set by split conformal prediction on the logarithmic residuals of the 2025 backtest (Angelopoulos & Bates, 2023), separately for each series, hour ahead and class, with a separate band for the corridor total:

$$\hat{y}^{lo} = (1 + \hat{y}) \cdot \exp(q_{0.10}) - 1,$$

$$\hat{y}^{hi} = (1 + \hat{y}) \cdot \exp(q_{0.90}) - 1 \quad (3.11.2.11)$$

where $q_{0.10}$ and $q_{0.90}$ are the 10th and 90th percentiles of the residuals ln(1 + y) − ln(1 + ŷ) in the calibration hours. The band was calibrated on January to June 2025 and checked on July to December 2025. Output: hourly forecasts for 48 hours with their band, stored in gold.ml_forecast_48h_runs and gold.ml_forecast_hourly_48h. Interpretation: the forecast starts from the end of the latest upload, so it describes the two days after the record until new data are loaded. Feature: Traffic – Predictive (a forecast line in the future zone of the volume chart and in its hourly view, and an evidence table of every candidate).

3.11.2.12 Methods Evaluated but Not Adopted, and Proposed Methods

Several methods in the design were either evaluated and rejected or not implemented. LSTM networks trained for daily volume and CO_2 did not beat the baselines and are retained only as comparators; for hourly volume and congestion state they are used (Sections 3.11.2.3 and 3.11.2.11). SARIMAX with weather was rejected for volume, and weather regressors added little accuracy once future weather was replaced by climatology. The following remain proposed: an XGBoost volume model with event and weather features; a true quantile regression forest for prediction intervals of the congestion forecast (the current random forest comparator is a standard classifier and is left out of the hourly refresh); GRU models trained separately on wet and dry speed sequences; a bidirectional LSTM with attention for travel time; multiple linear regression for emissions; and forecast combination (ensembles), which was not adopted because no combination was tested against the single champions. The dual-path LSTM with an XGBoost correction fits one model to another's residuals and is not a forecast combination.

3.11.3 Training and Testing of Models

Symbols in this chapter are defined locally for each group of equations. All models are trained and tested chronologically. Randomized k-fold cross-validation is not used because it would let models learn from observations that occur after the test days. Scalers and historical profiles are fitted on training data only, lag features never cross the training cut-off, and future exogenous values are replaced by information available at the forecast origin, except in the incident-count holdout, which uses observed same-day rainfall and volume and is therefore optimistic (Section 3.11.2.5). Forecast accuracy is measured with the metrics below, where y_t is the observed value, $\hat{y}_t$ the forecast and n the number of test points.

3.11.3.1 Mean Absolute Percentage Error (MAPE)

$$MAPE = 100\% \frac{}{n} \sum_{t=1}^{n} \left| \frac{y_t - \hat{y}_t}{y_t} \right| \quad (3.11.3.1)$$

MAPE is easy to communicate but undefined when $y_t = 0$ and unstable for small counts, so it is reported only for daily corridor totals.

3.11.3.2 Weighted Mean Absolute Percentage Error (WMAPE)

$$WMAPE = \frac{\sum_{t=1}^{n} |y_t - \hat{y}_t|}{\sum_{t=1}^{n} |y_t|} \times 100\% \quad (3.11.3.2)$$

WMAPE weights errors by actual volume, remains defined when individual values are zero, and is the primary ranking metric for regression tasks (Makridakis et al., 2022).

3.11.3.3 Root Mean Square Error (RMSE) and Mean Absolute Error (MAE)

$$RMSE = \sqrt{\frac{1}{n}\sum_{t=1}^{n}(y_t - \hat{y}_t)^2} \quad (3.11.3.3)$$

$$MAE = \frac{1}{n}\sum_{t=1}^{n}|y_t - \hat{y}_t| \quad (3.11.3.4)$$

Both are in the units of the target. RMSE penalizes large errors more heavily and equals the standard deviation of the errors only when their mean is zero (Hewamalage et al., 2021).

3.11.3.4 Mean Absolute Scaled Error (MASE)

$$MASE = \frac{\frac{1}{J}\sum_{j=1}^{J}|y_{T+j} - \hat{y}_{T+j}|}{\frac{1}{T-m}\sum_{t=m+1}^{T}|y_t - y_{t-m}|} \quad (3.11.3.5)$$

where J is the number of test points, T the number of training points and m the seasonal period (m = 7 for daily series). The denominator is the in-sample error of the seasonal-naive forecast, so MASE is scale-free and comparable across targets (Hyndman & Athanasopoulos, 2021). A value below 1 means the model's test error is smaller than the seasonal-naive error on the training data.

3.11.3.5 Classification and Survival Metrics

Classifiers are evaluated by accuracy, macro-averaged F1, the multi-class Brier score and the skill score against a reference model:

$$SS = \frac{acc - acc_{ref}}{1 - acc_{ref}} \quad (3.11.3.6)$$

where acc_ref is the accuracy of the best baseline (for congestion, the exit-by-hour profile), so SS > 0 means the model adds information beyond the historical pattern. Severity models are also compared with the majority-class rate. Survival models are evaluated by the concordance index, the probability that, of two incidents, the one predicted to clear sooner actually does. Information criteria (AIC and BIC) are recorded for the statistical volume models but are not used for selection, because they are not comparable across model families.

3.11.3.6 Model Selection, Accuracy Gate and Retraining

For each task, the published model is the candidate with the best held-out metric that also beats every baseline, with two stated exceptions. For congestion state, the served model is the best catcher of severe hours among the accepted candidates within 0.5 percentage points of the most accurate (Section 3.11.2.3). For the 48-hour volume forecast, a simpler model within 1% of the best WMAPE is preferred (Section 3.11.2.11). A weekly retraining batch refits all models on the latest data, keeps a backup of the active models and of their saved model files, and publishes a retrained model unless it stops beating a baseline that the previous model beat, its MASE rises from below 1 to 1 or more while it no longer beats its best baseline, an error metric rises by more than 25%, or a score (accuracy, AUC or R^2) falls by more than 0.05; otherwise the backup is restored. The batch runs every Sunday at 22:00 (Philippine time). The 25% and 0.05 tolerances were set by the proponents. The 48-hour model must also produce all 48 hours for every series, never a negative count, a corridor total equal to the sum of the plaza series, and a backtest WMAPE below that of the previous method. The XGBoost and GRU congestion candidates are refitted every hour, and the LSTM candidates in the weekly batch. Because champions are chosen on the same windows that are reported, the reported accuracy may be slightly optimistic; a separate validation block is planned. No model combination is used.

3.11.4 Prescriptive Analysis

The prescriptive tier turns forecasts into recommendations. Three components are mathematical optimization models that return an optimal solution for their stated objective and parameters. The remaining components are rule-based or scenario-based recommendations. All recommendations are computed by the API when the relevant page is viewed and are shown with their assumptions. Operators remain responsible for every decision.

3.11.4.1 Data Overview

The prescriptive models use the champion volume forecast (or the seasonal-naive baseline when no model passes the gate), the hourly plaza and hour shares of traffic, the number of toll booths per plaza (from OpenStreetMap), the predicted incidents by exit and segment, the Kaplan–Meier clearance-time curves, the congestion-state probabilities, the event-uplift estimates and the emission inventory. Economic parameters such as unit and delay costs are illustrative assumptions set in the API, not values estimated from NLEX data.

3.11.4.2 Toll-Booth Staffing Allocation

Decision variables: $x_{p,h}$, the booths staffed at plaza p in hour h, and $u_{p,h}$, the unserved demand. **Objective and constraints:**

$$\min \sum_{p,h} u_{p,h} \quad (3.11.4.1)$$

$$u_{p,h} \geq D_{p,h} - \kappa x_{p,h}, \qquad u_{p,h} \geq 0, \qquad x_{p,h} \in \left\{0,1,\ldots,\min\left(B_p, \left\lceil \frac{D_{p,h}}{\kappa} \right\rceil\right)\right\},$$

$$\sum_{p,h} x_{p,h} \leq P \quad (3.11.4.2)$$

where $D_{p,h}$ is the forecast demand (corridor forecast × plaza share × hour share), κ the throughput per booth-hour (350 vehicles by default, adjustable), B_p the booths at the plaza and P an optional staff pool. Because the objective is separable and each additional booth reduces a cell's unserved demand by a non-increasing amount, assigning booths one at a time to the cell with the largest reduction is optimal under the single pool constraint, so no solver is required. **Output:** an hourly staffing plan for the next seven days. Unlike queueing-based booth staffing (Bari et al., 2022; P. Wang et al., 2020), the allocation uses a fixed throughput per booth-hour and does not model queues. **Feature:** Traffic – Prescriptive (Booth Staffing Plan).

3.11.4.3 Patrol Positioning by Maximal Covering Location

Model: the maximal covering location problem (Baldomero-Naranjo et al., 2022), which positions K patrol units to cover the largest expected incident load within a response radius:

$$\max Z = \sum_{i \in I} w_i x_i \quad (3.11.4.3)$$

$$\sum_{j \in N_i} y_j \geq x_i \; \forall i \in I \quad (3.11.4.4)$$

$$\sum_{j \in J} y_j \leq K \quad (3.11.4.5)$$

$$x_i, y_j \in \{0,1\} \quad (3.11.4.6)$$

where w_i is the predicted incidents in zone i, y_j = 1 if a unit is stationed at site j, x_i = 1 if zone i is covered, and N_i the sites within the radius (8 km along the corridor by default) of zone i. The binary program is solved exactly with the YALPS branch-and-bound solver for the fleet size the operator selects. **Feature:** Incidents – Prescriptive.

3.11.4.4 Two-Stage Stochastic Clearance Protocol

Purpose: to decide how many response units to dispatch to an incident and whether to escalate when its clearance time is uncertain. **Model:** a two-stage stochastic mixed-integer program (Li & Grossmann, 2021). In the first stage, the number of units x is chosen. Clearance then follows one of four scenarios s (fast, typical, slow, extreme) defined by the Kaplan–Meier intervals up to the 50th percentile, 50th–80th, 80th–95th and above the 95th; each scenario's probability p_s is the fall in the Kaplan–Meier curve across its interval, which is close to 0.50, 0.30, 0.15 and 0.05 when the curve is estimated from many events; each scenario's duration is the conditional median of its interval. In the second stage, e additional units may be sent at the decision time:

$$\min \sum_{x} c_x z_x + \sum_{s \in S} p_s \sum_{x,e} q_{x,e,s} w_{x,e,s} \quad (3.11.4.7)$$

$$\sum_{x} z_x = 1, \qquad \sum_{e} w_{x,e,s} = z_x \; \forall x, s, \qquad z, w \in \{0,1\} \quad (3.11.4.8)$$

where c_x is the cost of dispatching x units, and $q_{x,e,s}$ the unit-hour cost plus the traffic-delay cost of scenario s. The delay is computed from a deterministic queue in which demand exceeds the reduced capacity during the blockage. The model reports the expected value of perfect information and the value of the stochastic solution. Unit cost (1,500 PHP per unit-hour) and delay cost (300 PHP per vehicle-hour) are illustrative values set in the API, while the effect of an extra unit (default 0.25) and the response-time saving (default 15 minutes) can be adjusted on the dashboard. **Feature:** Incidents – Prescriptive (Clearance Response Protocol).

3.11.4.5 Fuzzy Congestion-Response Advisory

Purpose: to grade the urgency of responding to a forecast congestion state. **Inputs: the forecast probability of high congestion p and the hours ahead h, for forecast hours 1 to 12 ahead.** Each input has three triangular membership functions (low, medium and high probability; soon, mid and far), and nine rules map their combinations to urgency values. The output is a weighted average, the zero-order (constant-consequent) form of Takagi–Sugeno inference:

$$U = \frac{\sum_{r=1}^{9} w_r z_r}{\sum_{r=1}^{9} w_r}, \qquad w_r = \min\left(\mu_{A_r}(p), \mu_{B_r}(h)\right) \quad (3.11.4.9)$$

where z_r is the urgency value of rule r. Scores of at least 0.7 are labelled Act, at least 0.4 Prepare, and otherwise Monitor, each with a fixed action text. The membership functions and rule values were set by the proponents and have not yet been validated with operators. **Feature:** Traffic – Prescriptive (Congestion Response Advisory).

3.11.4.6 TOPSIS Ranking of Interventions

The Technique for Order of Preference by Similarity to Ideal Solution (Chakraborty, 2022) ranks alternatives by their distance from an ideal and an anti-ideal solution. Criteria values are vector-normalized and weighted, and the closeness score is:

$$r_{ij} = \frac{x_{ij}}{\sqrt{\sum_i x_{ij}^2}}, \qquad v_{ij} = w_j r_{ij} \quad (3.11.4.10)$$

$$d_i^{\pm} = \sqrt{\sum_j \left(v_{ij} - v_j^{\pm}\right)^2}, \qquad C_i = \frac{d_i^-}{d_i^+ + d_i^-} \quad (3.11.4.11)$$

where the ideal point v_j^+ takes the best value of criterion j (the maximum for benefit criteria and the minimum for cost criteria), the anti-ideal v_j^- takes the worst, and d_i^+ and d_i^- are the distances of alternative i from them. Interval width and uncertainty width are cost criteria; the others are benefit criteria. Event interventions are ranked on extra vehicles, uplift, number of past events and interval width (weights 0.40, 0.25, 0.20 and 0.15). Emission strategies are ranked on expected avoided CO_2, the 5th-percentile avoided CO_2, whether the strategy is bounded by observed performance, and the width of its uncertainty range (weights 0.35, 0.25, 0.25 and 0.15). The weights were chosen by the proponents and have not been sensitivity-tested. **Feature:** Traffic and Emissions – Prescriptive.

3.11.4.7 Emission Strategy Estimation with Bootstrap Uncertainty

Three emission strategies are estimated. Under shorter lane blockages, each blockage is shortened to the 10th-percentile duration already achieved for its duration band and hour. Faster response in the busiest hours applies the same reduction only to the twelve busiest hours. A lower heavy-vehicle share is a what-if that shifts a share of vehicle-kilometres from Class 3 to Class 2; because the freight still moves, the saving per shifted vehicle-kilometre is the difference between the two classes' factors (798.2 − 689.4 = 108.8 g CO_2). The two incident-based strategies are given uncertainty ranges from 400 seeded bootstrap resamples (Efron & Hastie, 2021) of the observed incidents, reported as the 5th to 95th percentile. The idling CO_2 of each incident is computed with Equations 3.9.3.7 and 3.9.3.8. These ranges describe the likely spread of outcomes in the record, not a guaranteed reduction. The strategies are scenario estimates, not optimized policies. **Feature:** Emissions – Prescriptive.

3.11.4.8 Rule-Based Advisories

The system also issues rule-based advisories. A heavy-vehicle time window identifies the peak hours (within 10% of the busiest hour's vehicle-kilometres) and the quietest hours that could absorb the trucks. A peak toll-differential advisory lists the exits and hours with the most Class 3 CO_2; no price elasticity is estimated, so no saving is claimed. A patrol alert window marks the hours of each weekday whose mean incident count exceeds that day's average. These outputs follow fixed rules and are not optimized; their cut-offs are listed in Section 3.11.6. The system does not recommend driving speeds.

3.11.4.9 Proposed Prescriptive Methods

The design also proposed a weather-triggered speed advisory using PAGASA rainfall thresholds, a surge alert when forecast volume exceeds 80% of capacity, heuristic rerouting of emergency vehicles and commuters, variable speed limits from the fuzzy controller, and price-optimized dynamic tolling. These are not implemented. The weather-triggered advisory was withdrawn after the descriptive analysis showed little difference in high-incident-day probability between rainy and dry conditions, and the capacity-based alert requires segment capacities that are not in the record.

3.11.5 Agent-Based Simulation and Language-Model Features

3.11.5.1 Agent-Based Simulation Sandbox

The Scenario Sandbox is a microscopic, agent-based traffic simulation written in TypeScript that runs in the dashboard on an HTML5 Canvas. It is a testing environment in which operators compare interventions with a no-intervention baseline; it does not run the optimization models. Each vehicle is an agent with its own desired speed, time gap, maximum acceleration, politeness and reaction delay. Longitudinal motion follows the Intelligent Driver Model (Albeaik et al., 2022; Treiber & Kesting, 2025):

$$a = a_{\max}\left[1-\left(\frac{v}{v_0}\right)^4-\left(\frac{s^*(v,\Delta v)}{s}\right)^2\right] \quad (3.11.5.1)$$

$$s^*(v,\Delta v) = s_0 + vT + \frac{v\Delta v}{2\sqrt{a_{\max} b}} \quad (3.11.5.2)$$

where v is the speed, v_0 the desired speed, s the gap to the leading vehicle, Δv the approach rate, s_0 the minimum gap, T the time headway and b the comfortable deceleration. Lane changes follow the MOBIL model (Sunberg & Kochenderfer, 2022), which accepts a change when the driver's advantage, plus a politeness-weighted effect on neighbouring vehicles, exceeds a threshold and the new follower is not forced to brake harder than a safe limit:

$$\tilde{a}_c - a_c + \pi(\tilde{a}_n - a_n + \tilde{a}_o - a_o) > \Delta a_{th}, \qquad \tilde{a}_n \geq -b_{safe} \quad (3.11.5.3)$$

where a and ã are accelerations before and after the change for the changing vehicle (c), its new follower (n) and its old follower (o), π is the politeness factor, Δa_th the switching threshold and b_safe the maximum safe deceleration.

Agents and network: Class 1, 2 and 3 vehicles and motorcycles, plus wrecks and responders (ambulance, police, tow and work crews). The two carriageways of a chosen kilometre stretch are simulated with lane counts from OpenStreetMap, toll plazas, ramps and facilities. The stretch can be chosen by two exits or by typing a kilometre range, in which case the enclosing exits are selected automatically, and a shoulder breakdown can be placed only on the lane next to the shoulder. **Demand:** hourly entries by exit, direction and class from the warehouse, or the champion forecasts for a future date, with the forecast fleet mix. **Interventions:** lane closures over a stretch, speed limits and zones, counterflow lane reallocation, booth closures, and nine incident templates (including breakdown, collision, overturn, flood and road works) with durations drawn from calibrated NLEX clearance distributions. Interventions can also be entered as typed commands (Section 3.11.5.2). **Outputs:** average speed, throughput per minute, density, stopped vehicles, longest queue, travel time, unmet demand and CO_2 rate, compared with the baseline as the mean of seeded replications with a Student-t 95% confidence interval. **Verification:** an automated script checks the engine's calibration, scenario catalogue, closure geometry and assumptions (1,532 checks passed on 21 October 2026). **Limitations:** the two carriageways are simulated independently, lane data come from volunteer mapping, the CO_2 rate uses a speed-dependent multiplier that differs from the fixed-factor inventory, and the simulated queues have not yet been compared with observed incident queues.

3.11.5.2 Language-Model Features

SmartFlow NLEX uses large language models only as an interface layer. The models do not produce forecasts or solve optimization problems, and every number they present comes from the system's models or warehouse. Three features use them.

3.11.5.2.1 Sandbox Command Interpretation

In the Scenario Sandbox, an operator can type a command in English, Tagalog or a mixture, such as closing two lanes at a given kilometre during heavy rain. A GLM model from Z.ai, called through an OpenAI-compatible API and optionally routed through OpenRouter with zero data retention, converts the command into a structured plan of up to 21 action types. The API checks every field of the plan against the current scenario and the corridor limits, rejects unsupported or unsafe requests, and shows the plan to the operator. Nothing is applied until the operator confirms it. An offline check of the validation logic passed 65 of 65 cases on 21 October 2026.

3.11.5.2.2 Model Narratives

On the predictive and prescriptive pages, a GLM model writes short plain-language explanations of the displayed results. It receives only the values already shown on screen. Its answer is checked for percentages that do not appear in the input, and a fixed template is used if the model is unavailable or the check fails.

3.11.5.2.3 Mobile Assistant

The mobile Assistant uses a Qwen model through OpenRouter with tool calling (Schick et al., 2023). The model chooses among three tools that return the current corridor status, a corridor overview and the list of exits from the SmartFlow NLEX API, and it phrases the result for the commuter. A topic filter and the system instructions restrict it to the NLEX corridor. It does not access forecasts, give travel-time estimates or recommend departure times. These are proposed extensions. No retrieval-augmented generation index is used.

3.11.6 Decision Thresholds and Design Parameters

Several outputs of SmartFlow NLEX turn a continuous value into a label, colour, rank, alert or recommendation. Table 3.11.6.1 lists the decision rules that a user can see, its cut-off or value, where it appears and its basis. Where a published source defines the scale or the method, it is cited. Where the proponents chose the value, the basis is given as "Proponent". Proponent values were set from the corridor's own data or from the needs of the dashboard, have not been sensitivity-tested unless stated, and are reviewed with NLEX operators during user-acceptance testing (Section 3.13).

Of the KPIs in Table 3.6.5, the dashboard shows K1, K3, K4, K5, K7, K8, K9, K10, K15 and K16 and the accuracy component of K2. K6, K11 to K14 and the red-amber-green comparison of each KPI with its target are evaluated in Chapter 4.

Table 3.11.6.1 Decision Thresholds and Design Parameters

Rule	Cut-off or value	Shown in	Basis
Traffic and Live Map			
Waze jam level	0 free flow, 1 light, 2 moderate, 3 heavy, 4 severe, 5 blocked	Live Map, Overview	Waze (2023)
Live exit-direction status	Congested if the worst jam level is 3 or more or the slowest jam speed is below 10 km/h; slow for any other jam; clear when no jam was reported in the last 60 minutes. Shown green (clear), amber (slow) and red (congested)	Overview, Live Map	Proponent
Waze report filter	Hidden as disputed if confidence ≤ −1 or reliability < 5; drawn faintly as unconfirmed if reliability ≤ 5 and confidence ≤ 0	Live Map	Waze (2023) fields; cut-offs Proponent
Corridor match	Jams within 200 m of the corridor on an NLEX-named street; alerts within 3 km of an exit	Live Map, Overview	Proponent
Live feed freshness	Stale when the newest record is more than 30 minutes old; the Live Map refreshes every 15 seconds, the Waze panel every 30 seconds and the Overview every 60 seconds (the API caches live responses for 30 seconds); Redis snapshot expires after 10 minutes	Overview, Live Map	Proponent
Period comparison	Each total is compared with the preceding period of equal length (a rise is green for volume, red for incidents and CO_2)	Traffic, Incidents, Emissions	Proponent

Weekday peak hour	Hour with the highest mean Monday-to-Friday volume	Traffic – Descriptive	Proponent
Wet hour and wet day	Wet hour: mean corridor rainfall above 0.3 mm; wet day: mean of the day's hourly means above 0.3 mm	Weather filters, crash rate	Proponent
Hourly views	Offered only for date ranges of 14 days or less	Traffic, Emissions	Proponent
Holiday and event control days	Same weekday within ±45 days, excluding holidays and event days; at least four control days	Traffic – Descriptive	Giuliano and Lu (2021); window Proponent
Volume model acceptance	WMAPE below the better of seasonal naive and climatology, and MASE below the seasonal naive's own MASE on the same scored days; validated to 14 days and displayed to 90 days, with later days marked unvalidated	Traffic – Predictive	Hyndman and Athanasopoulos (2021); Makridakis et al. (2022)
Congestion states	Length-weighted exit-hour Waze speed: below 10 km/h High (Severe), 10 to below 20 km/h Medium (Heavy), 20 km/h or more or no jam Low (Moving)	Traffic – Predictive, Live Map forecast	Proponent (corridor speed distribution)
Displayed congestion state	Congested if p(Medium) + p(High) ≥ 0.5; Severe if p(High) ≥ p(Medium); probabilities above 0.95 or below 0.05 shown as ">95%" or "<5%"	Traffic – Predictive, Live Map forecast	Silva Filho et al. (2023); cut-off Proponent
Congestion model acceptance	Accuracy above the majority class, persistence and the exit-by-hour profile on the 12 most recent days; otherwise the profile is shown. Among accepted candidates within 0.5 percentage points of the most accurate, the one catching the most severe hours is served	Traffic – Predictive	Manibardo et al. (2022); tie margin Proponent
Severe-state weighting	A state is chosen after multiplying the high-state probability by 1.5 (one missed severe hour counts as 1.5 false alarms); used in testing, in	Traffic – Predictive (model table),	Silva Filho et al. (2023); weight Proponent

	choosing the served model and in the stored state	mobile Corridor forecast	
Congestion forecast freshness	"Refresh overdue" when the base time is more than 2 hours old; "expired" when no forecast hour remains; trailing hours with under 25% of their usual record count are dropped	Traffic – Predictive	Proponent
Event uplift	Material if the bootstrap 95% interval (2,000 resamples) lies above 1 and rests on at least eight event days; labelled Firm (interval width below 0.05), Fair (below 0.12) or Loose	Traffic – Predictive, TOPSIS ranking	Efron and Hastie (2021); cut-offs Proponent
TOPSIS event ranking	Weights 0.40, 0.25, 0.20 and 0.15; exits with extra vehicles above 0 or uplift above 1 are eligible; the top three are highlighted	Traffic – Prescriptive	Chakraborty (2022); weights Proponent
Congestion-response advisory	Act if urgency ≥ 0.7, Prepare if ≥ 0.4, otherwise Monitor; forecast hours 1 to 12 ahead; operator alert for the three most urgent Act segments	Traffic – Prescriptive	Varshney and Torra (2023); cut-offs Proponent
Booth staffing	350 vehicles per booth-hour by default (adjustable 150 to 800); at least one booth whenever demand is positive; capped at the booths on each carriageway; weekday and weekend profiles; plaza peak is the hour needing the most booths	Traffic – Prescriptive	Bari et al. (2022); values Proponent
Relative-volume words	Peak hour at 1.5 times the exit's median hour or more, busier than usual from 1.15, a usual hour from 0.85, quieter than usual from 0.5, otherwise a quiet hour; a jam below 0.7 times is flagged as a possible incident or roadworks	Live Map forecast	Proponent
Jam-profile details	Jams within 600 m of the exit; exit-hour basis needs at least 12 hours, direction at least 30 jam-hours, queue front at least 20 queues within	Live Map forecast	Proponent

	1,000 m; periods 05–09, 10–15, 16–20 and night		
48-hour model selection	Lowest WMAPE per plaza, direction, class and hour on the 2025 backtest; a simpler candidate within 1% (relative) of the best is preferred; a learned model must beat the same hour one week earlier and the previous profile method, otherwise the profile method is served	Traffic – Predictive	Makridakis et al. (2022); 1% Proponent
48-hour uncertainty band	80% split-conformal band from the 10th and 90th percentiles of the logarithmic residuals, per series, hour ahead and class; a separate band for the corridor total	Traffic – Predictive	Angelopoulos and Bates (2023); level Proponent
48-hour fallback and freshness	A series falls back to the profile method when a day of its last 14 days or of its 56-day window is missing; a run is stale 48 hours after the end of its history	Traffic – Predictive	Proponent
Rainy hour (48-hour evaluation)	Corridor mean rainfall of 2.5 mm/h or more in the forecast hour (moderate rain or heavier)	Traffic – Predictive (evidence table)	PAGASA (n.d.); Proponent
Incidents and Maintenance			
Severity class	Fatal if any fatality; otherwise Injury if anyone was injured; otherwise property damage only. The clearance protocol opens on Injury accidents	Incidents	Proponent rule on the record's fatality and injury counts; classes treated as ordered (Washington et al., 2020)
Hotspot segments	5-km bins of official kilometre posts from Km 12 (posts below grouped as "South of Km 12"), ranked by count; bars shaded by thirds of the top segment's count	Incidents – Descriptive	Bonera et al. (2024); bin Proponent
Wet-sample	Shown when a day-and-hour slot has	Incidents –	Proponent

warning	1 to 23 wet hours	Descriptive	
Rainfall labels	Light up to 2.5 mm/h, moderate above 2.5 and up to 7.5 mm/h, heavy above 7.5 mm/h, applied to the day's peak hourly rainfall	Incidents – Predictive	PAGASA (n.d.)
Clearance durations	Valid if above 0 and up to 1,440 minutes; longer durations censored at 1,440 minutes in survival curves; the protocol needs at least 30 events	Incidents	James et al. (2021); cut-offs Proponent
Clearance scenarios	Kaplan–Meier 50th, 80th and 95th percentiles split fast, typical, slow and extreme each weighted by the fall in the curve across its interval (about 0.50, 0.30, 0.15 and 0.05 on large samples); escalation decided at the median; 1 to 6 units, default 4	Incidents – Prescriptive	Li and Grossmann (2021); Proponent
Clearance targets	Projected mean clearance against the Kaplan–Meier restricted mean: accidents Green ≤ 30, Amber ≤ 45 minutes; breakdowns Green ≤ 10, Amber ≤ 18 minutes (indicative)	Incidents – Prescriptive	Federal Highway Administration (2021); provisional Proponent targets
Queue assumptions	2,000 vehicles per hour per lane, 3 lanes; a blockage halves capacity; demand at 0.5, 0.7 or 0.9 of capacity; unit cost 1,500 PHP per unit-hour; delay cost 300 PHP per vehicle-hour; extra-unit effect 0.25; response saving 15 minutes	Incidents – Prescriptive	National Academies of Sciences, Engineering, and Medicine (2022); values Proponent
Patrol positioning	Coverage radius 8 km (1 to 40); fleet 4 (1 to 12); demand is the corridor forecast shared by each exit's share of located accidents and breakdowns; "Reaches most risk" at 80% coverage or more	Incidents – Prescriptive	Baldomero-Naranjo et al. (2022); Proponent
High-incident day	Daily count at or above the 75th percentile of the preceding 365 days (the training 75th percentile when fewer than 180 days precede it); rain	Incidents – Predictive	Proponent

	effect significant at $p < 0.05$		
Patrol alert window	Hours whose mean incident count exceeds that weekday's mean; longest run shown; shading by percentile rank; uncertainty $\pm 1/\sqrt{n}$	Incidents – Prescriptive	Proponent
Predicted risk bands	Quartiles of the per-exit forecast: High at the 75th percentile or above, Elevated at the 50th, Moderate at the 25th, otherwise Low; "No data" for exits without history	Incidents – Predictive	Bonera et al. (2024); quartiles Proponent
Incident model acceptance	MASE ≤ 1 against the 7-day naive forecast, then highest R^2; a horizon is usable if WMAPE beats the 7-day naive forecast and MASE < 1; R^2 hidden below 30 scored days; underfit if training $R^2 < 0.25$, overfit if the training-validation gap exceeds 0.10 with training $R^2 \geq 0.30$; the training R^2 and the gap are shown for each model in the validation table	Incidents – Predictive	Hyndman and Athanasopoulos (2021); Proponent
Secondary incident	Another accident within 2 km in either direction, from the primary's start until 30 minutes after it was cleared	Incidents – Predictive (K16)	Salek et al. (2021); window Proponent
Breakdown response	3-hour blocks; groups under 20 flagged; "slowest" needs at least 5	Incidents – Predictive	Proponent
Hourly incident view	Daily forecast spread by the same weekday's hourly share over the previous 90 days	Incidents (hourly page)	Proponent
Maintenance status	Scheduled, In Progress, Completed, Cancelled; only In Progress works are drawn on the maps	Maintenance, maps	Proponent
Maintenance conflicts	Overlap in time, kilometre range and direction; Critical (same lane or a full closure) blocks saving; Nearby if on the same carriageway within 2 km; opposing carriageway noted; the Conflicts tile names up to two conflicting pairs, critical pairs first	Maintenance	Federal Highway Administration (n.d.-c); Proponent

Maintenance alerts and timing	Starting soon within 2 hours; start overdue; past end time; up to four alerts shown; a start within the day's six busiest hours is flagged; no quiet-hour suggestion for works over 12 hours; Km 12 to 87	Maintenance	Federal Highway Administration (n.d.-b); Proponent
Emissions			
Emission factors	CO_2 196.1, 689.4 and 798.2 g/km for Classes 1, 2 and 3 (low–high ranges 180.6–251.2, 492.9–836.3 and 667.8–944.5); NO_x 0.246, 5.626 and 6.513 g/km; PM2.5 0.00124, 0.119 and 0.138 g/km; fixed values with no speed term	Emissions	Ntziachristos and Samaras (2023); category mapping Proponent
Idling rate	0.3, 0.8 and 1.0 US gal of fuel per hour for Classes 1, 2 and 3, at 8.887 kg CO_2 per gallon of gasoline (Class 1) and 10.180 kg per gallon of diesel (Classes 2 and 3) (0.044, 0.136 and 0.170 kg CO_2 per vehicle-minute), weighted by the hour's class mix; accidents with no lane blockage or a blockage over 24 hours excluded	Emissions – Descriptive and Prescriptive	U.S. Department of Energy (2015) range; U.S. Environmental Protection Agency (2026); rates Proponent
Trip distance	Distance between the measured kilometre posts of the entry and exit plazas, counting only NLEX (NLEX Connector ends at the Harbor Link junction, SCTEX and SFEX ends at the SCTEX junction); open-section entries take the mean distance to the exits ahead up to Bocaue Barrier, bounded by the nearest and farthest; trips with no usable entry take the mean of known trips at the same plaza and class	Emissions	Lei et al. (2021) for entry-only records; open-section rule Proponent
CO_2 range	Low: nearest open-section exit with the lowest fitting factor per class; high: farthest exit with the highest	Emissions – Descriptive (Total CO_2 tile)	Ntziachristos and Samaras (2023); Proponent
Emission strategies	Blockage bands under 15, 15–60, 60–180 and over 180 minutes; floor	Emissions – Prescriptive	Proponent

	at the 10th percentile; 12 busiest hours; what-if shifting 1 percentage point of all vehicle-km from Class 3 to Class 2 (108.8 g CO_2 saved per shifted vehicle-km); the recommended strategy is the largest evidence-bounded saving; "long" blockages 60 minutes or more (180 or more on the delay tile); an incident giving 20% or more of a saving is flagged; top three 10-km stretches and top six hours		
Emission strategy ranking and range	TOPSIS weights 0.35 (expected avoided CO_2), 0.25 (5th-percentile avoided CO_2), 0.25 (bound achieved on NLEX) and 0.15 (uncertainty width, a cost criterion); range from 400 seeded bootstrap resamples, 5th to 95th percentile	Emissions – Prescriptive	Chakraborty (2022); Efron and Hastie (2021); weights Proponent
Heavy-vehicle advisories	Peak hours within 10% of the busiest hour's vehicle-km; quietest hours filled first; needs at least 12 hourly rows; the toll-differential advisory lists the top five exits by Class 3 CO_2	Emissions – Prescriptive	Kim et al. (2026); Proponent
Heavy vehicle	Classes 2 and 3, measured by vehicle-kilometres (the time-window advisory and the fleet what-if name Class 3 explicitly)	Emissions	Proponent
Air quality index	1 Good, 2 Fair, 3 Moderate, 4 Poor, 5 Very Poor, from OpenWeather's modelled estimate at each exit, not a roadside measurement; readings outside 1 to 5 discarded	Emissions – Descriptive	OpenWeather (n.d.)
CO_2 model acceptance	WMAPE below the best baseline and MASE < 1; validated to 7 days; longer views offered only where a horizon passes; co-champions within 0.15 percentage points of WMAPE	Emissions – Predictive	Hyndman and Athanasopoulos (2021); Proponent
Fleet-mix model	Aitchison distance below persistence and climatology, and skill ratio below 1; co-champions within 0.004; heavy-	Emissions – Predictive	Greenacre et al. (2023); Bayram et al.

	share surge above the training mean plus 1.5 standard deviations; "no material change" under 1 percentage point; trend steady within ±0.5 percentage points		(2022); Proponent
Data, Retraining, Simulation and Narratives			
Upload checks	Four gates (Section 3.9.2); toll files need every required column (eight, or seven for NLEX's hourly export), incident files at least two signature columns; NLEX's hourly export is processed with the client's rules for that export, and each load replaces the whole days it covers; a warning when plaza-days with at least 200 vehicles change by more than 25%; the ten most recent loads can be undone	Data Management	Proponent
Retraining gate	Weekly, Sunday 22:00; the new model is kept unless it loses a baseline the previous model beat, its MASE rises from below 1 to 1 or more while it no longer beats its best baseline, an error metric rises by more than 25%, or a score falls by more than 0.05	Data Management	Bayram et al. (2022); tolerances Proponent
Drift verdict	Drifted if the published forecast's WMAPE exceeds that of a same-weekday-last-week forecast over at least 7 scored days	Data Management	Bayram et al. (2022); Proponent
Audit-log severity	Critical for a failed restore; Warning for failed, refused, kept-previous, cannot-run, cancelled or deletion events; otherwise Info	Audit Log	Proponent
Sandbox replications	3 to 20 seeded runs (default 10); mean with a Student-t 95% confidence interval	Scenario Sandbox	Law (2023)
Sandbox demand and rain	Peak-hour inflow = daily volume × segment share ÷ 24 × 1.6 (about 6.7% of the daily volume, below the 8	Scenario Sandbox	Mejia and Sigua (2018) for rain caps;

	to 12% typical of urban roads), limited to 500–12,000 vehicles per hour; rain speed caps 100, 100 and 96 km/h		factor and limits Proponent
Sandbox advice	Poor outcome if an incident queue exceeds 120 m, unmet demand reaches the larger of 50 vehicles per hour and 3%, or a closed lane brings speed below 20 km/h or a drop above 30%; warning if more than 6 vehicles are stopped, a queue exceeds 80 m, or a closure uses 85% of 1,550 vehicles per hour per open lane; good if a speed-limit zone keeps average speed above 40 km/h	Scenario Sandbox	Thapa et al. (2021); Proponent
Sandbox calibration flags	Small-sample flag below 200 records; a 99th-percentile duration cap needs at least 1,000	Scenario Sandbox	Proponent
Model narratives	A model beats its benchmark if its MASE is more than 0.5% below the benchmark's (the seasonal naive's own MASE on the scored days where the page reports it, otherwise 1) and fails it if more than 0.5% above, otherwise it is level; accuracy is described against the baseline and the KPI design targets	Narratives	Proponent

3.12 Business Analytics Tools, Techniques, and Specific Applications

3.12.1 Descriptive Analytics

Tools: PostgreSQL 18 with PostGIS (SQL aggregation and spatial queries), Node.js/TypeScript API, Apache ECharts and Mapbox GL JS.

7. **Temporal aggregation:** hourly toll counts are summed to daily, weekly and monthly totals and averaged by hour of day and weekday to reveal recurring peaks.
8. **Local control-day comparison:** holiday and event days are compared with same-weekday days within ±45 days to measure their effect.
9. **Segment ranking:** incidents are counted in five-kilometre bins of official kilometre posts to rank high-incident segments.
10. **Time-normalized comparison:** crash rates per hour in wet and dry conditions are compared.
11. **Survival analysis:** Kaplan–Meier curves summarize clearance times.
12. **Emission accounting:** vehicle-kilometres are multiplied by per-class emission factors.
13. **Visualization:** time-series charts, hour-by-weekday heatmaps and a live corridor map.

3.12.2 Predictive Analytics

Tools: Python with pandas and NumPy, statsmodels, Prophet, scikit-learn, XGBoost, TensorFlow/Keras, PyTorch, lifelines, mgwr and SciPy; results stored in PostgreSQL and displayed with ECharts.

14. **Time-series forecasting: Prophet, Holt-Winters, Holt's linear trend, SARIMAX and LSTM for daily volume, with seasonal-naive and climatology baselines; single-layer, stacked and dual-path LSTMs, with an XGBoost residual correction, for 48-hour hourly volume, with same-hour-last-week and profile baselines; split conformal prediction for its uncertainty band.**

15. **Classification: XGBoost, GRU, single-layer, stacked and dual-path LSTMs, a dual-path LSTM with XGBoost, and MSTL decomposition with exponential smoothing for congestion state (a random forest and SARIMAX are added in full runs);** ordinal logistic regression and XGBoost for severity; logistic regression for secondary risk.

16. **Count, survival and compositional models:** negative binomial regression for incident counts, Cox proportional hazards for clearance time, Dirichlet regression for fleet mix, and geographically weighted regression for spatial variation.

17. **Machine-learning regression:** gradient boosting for corridor CO_2.

18. **Evaluation:** rolling-origin validation with WMAPE, MAPE, RMSE, MAE, MASE, accuracy, macro-F1, Brier score and concordance index, and a weekly retraining gate.

3.12.3 Prescriptive Analytics

Tools: TypeScript services, the YALPS linear and mixed-integer programming solver, and the TypeScript simulation engine on HTML5 Canvas.

19. **Optimization:** linear booth-staffing allocation, the maximal covering location model for patrols, and the two-stage stochastic mixed-integer clearance protocol.

20. **Rule-based inference:** a fuzzy congestion-response advisory and rule-based heavy-vehicle, tolling and patrol-window advisories.

21. **Multi-criteria ranking:** TOPSIS for event interventions and emission strategies.

22. **Uncertainty analysis:** seeded bootstrap ranges for emission strategies.

23. **Simulation:** the agent-based Scenario Sandbox (IDM and MOBIL) for what-if comparison of interventions.

24. **Language-model interface:** GLM (Z.ai) for sandbox commands and narratives; Qwen (OpenRouter) for the mobile assistant.

3.13 Software Quality Assurance Plans

This section defines how SmartFlow NLEX is verified and validated. It separates the verification already performed during development, with its evidence, from the tests planned for the evaluation phase. Tests that write data are run against a copy of the database, never the shared warehouse, because the shared tables are used by several developers and scheduled jobs.

3.13.1 Verification Performed During Development

Table 3.13.1.1 lists the verification carried out so far. Unit and integration tests are automated with the Node.js test runner for the API and the dashboard and with Python's unittest module for the analytics scripts. They are run together by a script that opens every database session in read-only mode. The tests are run manually before a merge, and no continuous-integration pipeline runs them. Verification also relies on self-checking scripts, model evaluation tables and manual reconciliation.

Table 3.13.1.1 Verification Performed During Development

Verification	Scope	Size / Result	Evidence
Sandbox engine verification	Calibration samplers, incident fallbacks, scenario catalogue, closure geometry, consistency of shared constants	1,532 checks, 0 failed (run 21 Oct 2026)	scenario-sandbox/scenarios/verify.ts
Sandbox command safety checks	Validation of language-model plans (sound, incomplete and hostile answers); kilometre-post conversion	65 checks, 0 failed (run 21 Oct 2026)	Back-End/scripts/sandbox-command-check.ts

Sandbox command evaluation	Live model interpretation of commands in English, Tagalog and Taglish	80 of 80 cases passed (7 Oct 2026; not re-run)	Back-End/scripts/sandbox-command-eval.ts
Clearance protocol verification	Kaplan–Meier, queue-delay formula, known optimum of the stochastic program, EVPI and VSS non-negativity	28 checks (requires database; to be re-run and recorded)	Back-End/scripts/verify-clearance-protocol.ts
Model validation	Rolling-origin holdout of every candidate against baselines	Stored per model and window	gold.ml_model_metrics, gold.ml_horizon_accuracy
Warehouse reconciliation	Layer row counts, duplicate keys and lineage across bronze, silver and gold	Manual run	Back-End/scripts/medallion/05-verify.sql
Automated unit and integration tests	Unit tests of access control, Overview ranking, queue length, period change, booth staffing, MASE, crash-rate ratio, incident KPI queries, patrol and clearance models, TOPSIS, fleet-mix shares, trip distance, the 48-hour forecast checks, maintenance, settings and sandbox rules; integration tests that compare API values with independent read-only database queries	24 unit and 23 integration test cases; results reported in Chapter 4	Back-End/tests, Front-End-Dashboard/tests, smartflow_scripts/tests, test-reports/run-all.sh

3.13.2 Unit Test

Unit tests check individual functions in isolation against expected values. The criteria in Table 3.13.2.1 apply to every unit test, and the scenarios in Table 3.13.2.2 cover the functions whose errors would most affect the reported results. The automated unit test cases (TC-UT-01 to TC-UT-24) apply these criteria to the functions behind the dashboard's displayed values, and their results are reported in Chapter 4. Where a function is written inline in a page component, the test extracts that exact code from the source file and runs it, so the application code is not changed for testing.

Table 3.13.2.1 Unit Test Criteria

Criterion ID	Criterion Name	Description	Pass/Fail Criteria
UTC-01	Computational Correctness	Deterministic calculations (emissions, ADT, deviation, metrics, optimization objective) match an independent hand or spreadsheet calculation.	Pass: difference ≤ 0.5% (or exact for integers). Fail: otherwise.
UTC-02	Response Time	Execution time of pure functions and of database-backed endpoints on the demonstration machine.	Pass: pure functions ≤ 50 ms; cached endpoints ≤ 500 ms (p95). Fail: otherwise.
UTC-03	Data Integrity	Loading the same file twice does not create duplicate records.	Pass: 0 duplicate natural keys after load. Fail: any duplicate.
UTC-04	Access Control	Requests without a valid session, without the second factor or with an unauthorized role are refused.	Pass: 401 for missing/invalid token or missing second factor; 403 for a role not permitted. Fail: any other response.
UTC-	Corridor	Records outside the corridor are	Pass: all boundary cases

05	Boundary Logic	rejected using plaza names and kilometre posts.	classified correctly. Fail: any misclassification.
UTC-06	Input Validation	Invalid inputs (missing fields, out-of-range values, unsupported requests) are rejected with a stated reason.	Pass: every invalid input rejected with the correct reason. Fail: any invalid input accepted.

Table 3.13.2.2 Unit Test Scenarios

Scenario ID	Scenario Name	Criterion ID	Test Input	Expected Output
UTS-01	Missing vehicle class	UTC-06	Toll record with a null vehicle class	Record rejected with reason “invalid vehicle class”
UTS-02	Duplicate upload	UTC-03	The same toll file uploaded twice	Second load inserts 0 records
UTS-03	Local-baseline deviation	UTC-01	A day 20% above the mean of its control days	Deviation = +20.0%
UTS-04	Emission calculation	UTC-01	100 Class 1 trips of 1 km	19.61 kg CO_2 (100 × 1 km × 196.1 g/km)
UTS-05	Forecast horizon bound	UTC-06	Request for 169 hours ahead	Request limited to the published 168-hour horizon
UTS-06	Command plan validation	UTC-06	Model plan closing a lane beyond the corridor	Plan rejected with an explanation
UTS-07	Unauthorized write	UTC-04	Maintenance write without a token; with an Incident Operator token	401 token_missing; 403 role_denied
UTS-08	Corridor boundary	UTC-05	Kilometre posts 11.9, 12.0, 87.0 and 89.1	Outside, inside, inside, outside

3.13.3 Integration Test

Integration tests confirm that data flow correctly between components. The planned scenarios are as follows. (1) An upload in check mode, then in load mode, appears in the silver and gold layers and in the corresponding dashboard chart. (2) A live Waze alert appears on the Live Map within the feed interval. (3) A maintenance schedule saved on the dashboard appears in the mobile application. (4) A section switched off on the Mobile App page disappears from the mobile application. (5) A weekly retraining run that fails the accuracy gate restores the previous model. Twenty-three integration test cases (TC-IT-01 to TC-IT-23) are automated. They compare the values returned by the API with independent queries on a read-only database connection, and cases whose steps write data are run only against a copy of the database. Their results are reported in Chapter 4.

3.13.4 Model and Data Validation

Model validation follows Section 3.11.3: chronological rolling-origin evaluation, comparison with seasonal-naive and climatology baselines, and the weekly accuracy gate. Results are reported per task with their grain, horizon and evaluation window. Data validation checks completeness (missing days and timestamp coverage), uniqueness of natural keys, valid ranges (kilometre posts, durations, AQI), consistent exit names and the freshness of live feeds, and records known gaps in a data-gap register.

3.13.5 Security Test

Security testing covers a role-by-endpoint access matrix, attempts to bypass the second factor, tampered tokens, attempts to change one's own role, injection inputs on validated endpoints, rate limiting, a dependency vulnerability audit, a secrets scan of the repositories, and verification that the database connection is encrypted. Findings are rated by severity and must be resolved or documented before user-acceptance testing.

3.13.6 Stress Test

Stress tests measure performance under load with a scripted load-testing tool such as k6, against a separate copy of the API and database. The load model is derived from the expected users: up to 50 concurrent operators and several hundred mobile users reading forecasts.

Table 3.13.6.1 Stress Test Criteria

Criterion ID	Criterion Name	Description	Pass/Fail Criteria
STC-01	Latency (p95)	Response time of analytics and forecast endpoints under load	Pass: p95 < 3 s for cached reads and < 10 s for cold reads at 50 concurrent users
STC-02	Error Rate	Share of failed requests (HTTP 5xx or time-outs)	Pass: < 1%
STC-03	Recovery	Availability after the load ends	Pass: service responds normally within 60 s without a manual restart
STC-04	Resource Ceiling	CPU and memory of the API host and database connections in use	Pass: CPU and memory ≤ 80%; database connections below the

			configured maximum

Table 3.13.6.2 Stress Test Scenarios

Scenario ID	Scenario Name	Criterion ID	Test Input	Expected Output
STS-01	Operator ramp-up	STC-01, STC-02	0 to 50 virtual users over 10 minutes on dashboard endpoints	Criteria met at every step
STS-02	Mobile forecast load	STC-01, STC-02	Up to 500 virtual users requesting corridor status and forecasts	Cached responses served; error rate < 1%
STS-03	Large upload	STC-03, STC-04	A full-year toll export uploaded in check mode	Completes without exhausting connections; service remains responsive
STS-04	Run to failure	STC-03	Load increased until the error rate exceeds 1%	Degradation point recorded; service recovers

3.13.7 User-Acceptance Test

User-acceptance testing (UAT) determines whether the dashboard and the mobile application meet their users' needs. Participants are selected by purposive sampling: 10 NLEX operators (5 incident operators and 5 TCC operators), 5 NLEX data analysts and 15 commuters and drivers who use the corridor. Under the problem-discovery model summarized by Budiu (2021), the share of usability problems found grows as $1 - (1 - L)^n$, where n is the number of users and L the probability that one user finds a given problem. With the average L of about 0.31, five users reveal about 85% of problems (Budiu, 2021). Each profile therefore has at least five participants, and testing is run in two rounds. Each participant completes the scenarios for their role. The proponents record task success, time on task and the participant's decision on each recommendation (accept, modify or reject), and participants complete the 10-item System Usability Scale (SUS), whose mean score on the 0–100 scale is compared with the commonly used benchmark of 68 (Hyzy et al., 2022). Participation is voluntary, informed consent is obtained, and no personal data beyond role and responses are collected, in accordance with the Data Privacy Act of 2012.

Table 3.13.7.1 User-Acceptance Test Scenarios for Incident Operators

Scenario ID	Scenario Name	Description	Test Input	Expected Output
UAT-IO-1	Locate a live incident	Find a live Waze alert on the Live Map and identify its exit and kilometre post	Current live feed	Alert located with correct exit and km post within 60 s
UAT-	Review patrol	Run patrol positioning for	Fleet size 4,	Participant

IO-2	positioning	four units and interpret the covered zones	radius 8 km	explains the recommended positions; decision recorded
UAT-IO-3	Review the clearance protocol	Read the recommended dispatch and escalation for a typical accident	Accident, default costs	Participant states the units to dispatch and when to escalate; decision recorded

Table 3.13.7.2 User-Acceptance Test Scenarios for Data Analysts

Scenario ID	Scenario Name	Description	Test Input	Expected Output
UAT-DA-1	Verify an emission total	Compare one month of corridor CO_2 on the Emissions page with an independent calculation	One month of toll data and the emission factors	Values agree within 0.5%
UAT-DA-2	Upload in check mode	Upload a toll export in check-only mode and interpret the rejection report	Sample export with deliberate errors	All deliberate errors reported with correct reasons; nothing saved
UAT-DA-3	Review event impact	Identify the largest event-day deviations for the past year	Event panel, last 12 months	Deviations and their baselines correctly interpreted

Table 3.13.7.3 User-Acceptance Test Scenarios for TCC Operators

Scenario ID	Scenario Name	Description	Test Input	Expected Output
UAT-TC-1	Plan for a forecast peak	Select the next day's peak hour on the congestion forecast and read the exits at risk	Forecast picker	Forecast rendered within 5 s; exits at risk identified
UAT-TC-2	Review booth staffing	Read the staffing plan for the busiest plaza and judge its acceptability	Default throughput 350 veh/booth-h	Decision (accept, modify or reject) and reasons recorded

UAT-TC-3	Corridor overview	View the status of the whole corridor at once	Live Map, full extent	All exits from Km 12 to Km 87 visible with their status
UAT-TC-4	What-if lane closure	Simulate a two-lane closure at Bocaue and compare it with the baseline	Two lanes closed, 1 hour, weekday peak	Change in queue length, travel time and throughput reported with confidence intervals

Table 3.13.7.4 User-Acceptance Test Scenarios for Commuters and Drivers

Scenario ID	Scenario Name	Description	Test Input	Expected Output
UAT-CM-1	Check the corridor	Find the current condition at a chosen exit	Dashboard and Corridor tabs	Condition found within 60 s
UAT-CM-2	Plan with the forecast	Choose a time tomorrow and read the expected condition at the destination exit	Corridor forecast view	Expected condition correctly read
UAT-CM-3	Post a report	Post a road report with location and photo	Community tab	Report appears in the feed
UAT-CM-4	Ask the assistant	Ask about current conditions in English or Tagalog, and ask an off-topic question	Assistant tab	Correct live status returned; off-topic question declined

3.14 Deployment Plans

This section describes how SmartFlow NLEX is deployed for the capstone demonstration and how it would be deployed in production for NLEX. The two are kept separate: the first describes what is running, and the second is a proposal.

3.14.1 Current Deployment (Capstone Demonstration)

Table 3.14.1.1 Components and Hosts of the Current Deployment

Component	Technology	Host
Operator dashboard	Next.js 15 static export, React 19, TypeScript, packaged with Electron	Operator workstation (Windows desktop application, built with electron-builder); during development a launcher script starts the API and the development server and opens the same Electron window on it
API	Node.js, Express, TypeScript	Developer workstation; a demonstration instance on Render (free tier) used by the mobile application
Data warehouse	PostgreSQL 18 with PostGIS 3.6	AWS RDS, Asia Pacific (Singapore) region; one shared instance
Authentication	Supabase Auth with TOTP multi-factor authentication	Supabase cloud (separate projects for the dashboard and the mobile application)
Model training and refresh	Python scripts	GitHub Actions hosted runners: hourly congestion refresh and weekly retraining batch (three jobs); model files kept in the Actions cache
Live Waze ingestion	Polling process	Separate ingestion host (about every 2 minutes)
Language models	GLM (Z.ai); Qwen (OpenRouter)	Provider APIs
Mobile application	Expo and React Native	Development builds on Android and iOS devices

3.14.2 Release Procedure

Each member develops on a feature branch. Before integrating, the integrator creates a backup branch of the shared Main-Dashboard branch, merges the feature branch, runs the verification scripts and a local smoke test of every page, and then pushes the merged branch to the team repository. Database changes are applied with versioned SQL scripts. The desktop installer is built with electron-builder once the build has passed these checks.

3.14.3 Rollback and Maintenance

If a release causes a critical defect, the code is rolled back by restoring the backup branch created before the merge. Model outputs are rolled back by the weekly retraining batch, which restores the previous models whenever a retrained model fails the accuracy gate. Maintenance consists of monitoring the retraining log, the audit log, the freshness of the live feed and the served-forecast log, updating the event calendar, and loading new NLEX exports through the Data Management page. Automated database snapshots and alerting are not yet configured.

3.14.4 Proposed Production Deployment for NLEX

For production use, the proponents propose the following measures:

25. Host the database on a dedicated managed PostgreSQL instance with automated backups, point-in-time recovery, certificate-verified TLS, least-privilege roles for the API and the jobs, and private network access.

26. Run the API on a managed container service behind HTTPS, with authentication on every write endpoint, security headers and correct handling of proxy addresses for rate limiting.

27. Add failure alerts to the hosted scheduled jobs, and give each job its own output tables.

28. Add a staging environment and automated build, type-check and test runs on every change.

29. Store roles in server-controlled metadata, disable public sign-up, and mask personal fields in the analytics layers.

30. Distribute a code-signed desktop installer with automatic updates, publish the mobile application to the app stores with server push notifications, and monitor uptime and performance.

References

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Appendix

A.1 Glossary

Term	Meaning	Computation	Where it appears
Local control-day comparison	Measures how much a holiday or event changes traffic.	Percentage deviation of the day's volume from the mean of the same weekday within ±45 days, excluding holidays and events (Eq. 3.9.3.4).	Descriptive analytics (3.11.1.3)
Average daily traffic (ADT)	Mean daily toll-plaza entries.	Sum of daily entries over valid days divided by the number of valid days (Eq. 3.9.3.1).	Data gathering (3.9.3)
Exposure-normalized crash rate ratio	Compares crash risk in wet and dry hours.	Crashes per wet hour divided by crashes per dry hour (Eq. 3.11.1.1).	Descriptive analytics (3.11.1.5)
Mean Time to Clear (MTTC)	Average time an incident occupies the carriageway.	Mean of site-cleared time minus event start for incidents with valid timestamps, reported with median, 90th percentile and coverage.	KPI K9; descriptive analytics (3.11.1.6)
Kaplan–Meier estimator	Empirical probability that an incident is still uncleared after t minutes.	Product over clearance times of (1 − cleared / still uncleared) (Eq. 3.11.1.2).	Descriptive and prescriptive analytics
Baseline emissions	Estimated CO_2 from vehicle travel on NLEX.	Trips × each trip's NLEX distance between its entry and exit plazas × fixed per-class factor (196.1/689.4/798.2 g CO_2/km), with a low–high range (Eqs. 3.9.3.5, 3.9.3.6 and 3.9.3.9).	Emission features (3.9.3); Emissions page
Vehicle-kilometres (trip-based)	Distance driven on NLEX by toll trips.	Sum over trips of the distance between the entry and exit plazas; open-section entries use the mean	Emission features (3.9.3); Emissions

		distance to the exits ahead (Eq. 3.9.3.6).	page
Delay-induced CO_2	Additional CO_2 from vehicles idling in the queue behind a lane-blocking incident.	$\varphi_h \times (\text{flow}/60) \times T^2/2$ per incident, with T the lane-blockage minutes and φ_h the idling CO_2 per vehicle-minute at that hour's class mix (Eqs. 3.9.3.7 and 3.9.3.8).	KPI K8; Emissions page
Prophet	Decomposable time-series model with trend, seasonality and regressors.	$y(t) = g(t) + s(t) + \beta^T x_t + \varepsilon_t$ (Eq. 3.11.2.1).	Champion daily volume model (3.11.2.2)
Holt-Winters exponential smoothing	Forecasting method with level, trend and seasonal components.	Additive form with a seven-day season (Eq. 3.11.2.2).	Volume benchmark (3.11.2.2)
SARIMAX	Seasonal ARIMA with external regressors.	SARIMAX$(1,1,1)(1,1,1)_7$ with weather regressors.	Volume comparator (rejected)
LSTM / GRU	Recurrent neural networks for sequences.	Trained on lagged series; compared with baselines.	Daily volume and CO_2 comparators; 48-hour hourly volume model (stacked LSTM); congestion-state candidates (3.11.2.3, 3.11.2.11)
Dual-path LSTM	Recurrent network with a short-term and a long-term LSTM branch joined by a learned gate.	$h = g \cdot h_short + (1 - g) \cdot h_long$, $g = \sigma(W[h_short, h_long, e] + b)$ (Eq. 3.11.2.10).	48-hour volume and congestion candidates (3.11.2.3, 3.11.2.11)
XGBoost	Gradient-boosted decision trees with regularization.	Minimizes a regularized loss over trees (Eq. 3.11.2.3).	Congestion-state candidate (3.11.2.3); residual correction in

			the hybrid 48-hour candidate (3.11.2.11)
MSTL	Decomposition of a series into trend, several seasonal patterns and remainder.	Daily (24 h) and weekly (168 h) seasons; remainder smoothed exponentially.	Congestion-state candidate (3.11.2.3)
Negative binomial regression	Count model for overdispersed data.	$\log \mu = \beta_0 + \beta^T x$; $Var = \mu + \alpha\mu^2$ (Eq. 3.11.2.4).	Incident count forecast (3.11.2.5)
Ordinal logistic regression	Model for ordered categories.	$\text{logit } P(Y \le k) = \theta_k - \beta^T x$ (Eq. 3.11.2.5).	Severity classification (3.11.2.6)
Cox proportional hazards model	Survival model for time to clearance.	$h(t\|x) = h_0(t) \exp(\beta^T x)$ (Eq. 3.11.2.6); evaluated by the concordance index.	Clearance-time model (3.11.2.7)
Geographically weighted regression (GWR)	Regression with coefficients that vary by location.	Local Poisson regression across the 20 exits (Eq. 3.11.2.7).	Spatial incident model (3.11.2.8)
Gradient boosting regression	Additive ensemble of regression trees.	$F_m = F_{m-1} + vh_m$ (Eq. 3.11.2.8).	Champion CO_2 forecast (3.11.2.9)
Dirichlet regression	Model for proportions that sum to one.	Shares ~ Dirichlet(α), log α_c = x^Tβ_c (Eq. 3.11.2.9).	Fleet-mix forecast (3.11.2.10)
WMAPE	Weighted mean absolute percentage error.	$\Sigma\|y - \hat{y}\| / \Sigma\|y\| \times 100\%$ (Eq. 3.11.3.2).	Primary ranking metric (3.11.3)
MAPE / RMSE / MAE	Standard forecast error metrics.	Equations 3.11.3.1, 3.11.3.3 and 3.11.3.4.	Model evaluation (3.11.3)

MASE	Scale-free error relative to the seasonal-naive forecast.	Test MAE divided by in-sample seasonal-naive MAE with m = 7 (Eq. 3.11.3.5).	Model evaluation and accuracy gate
Skill score	Improvement of a classifier over a reference.	(accuracy − reference accuracy) / (1 − reference accuracy) (Eq. 3.11.3.6).	Congestion classifier evaluation (KPI K2)
Seasonal-naive and climatology baselines	Simple forecasts every model must beat.	Value one week earlier; average for the same day of the year.	Model selection (3.11.2.1)
Split conformal prediction	Interval method that sets the band from held-out residual quantiles.	Band = (1 + forecast) × exp(10th and 90th percentiles of log residuals) − 1 (Eq. 3.11.2.11).	48-hour forecast band (3.11.2.11)
Rolling-origin evaluation	Chronological testing at successive forecast origins.	Refit at each origin and score the next 7–14 days.	Model evaluation (3.11.2.1)
Accuracy gate	Rule that decides whether a retrained model is published.	Must beat the baselines, not lose its MASE advantage over its best baseline, and be no more than 25% worse than the previous run.	Weekly retraining (3.11.3.6)
Booth staffing allocation	Linear allocation of toll booths to demand.	Minimize unserved demand subject to booth and staff limits (Eqs. 3.11.4.1–3.11.4.2).	Traffic – Prescriptive
Maximal covering location problem (MCLP)	Places patrol units to cover the most expected incidents.	Binary integer program (Eqs. 3.11.4.3–3.11.4.6).	Incidents – Prescriptive
Two-stage stochastic program	Optimization under uncertain clearance times.	Dispatch decision, then recourse escalation over four scenarios (Eqs. 3.11.4.7–3.11.4.8).	Clearance Response Protocol
Fuzzy inference	Grades the urgency of a congestion response.	Weighted average of nine rule outputs (Eq. 3.11.4.9).	Congestion Response Advisory

advisory			
TOPSIS	Ranks alternatives by closeness to an ideal solution.	$C = d^- / (d^+ + d^-)$ (Eqs. 3.11.4.10–3.11.4.11).	Event and emission strategy ranking
Bootstrap uncertainty range	Sampling uncertainty of a strategy estimate.	400 seeded resamples of observed incidents; 5th–95th percentile.	Emission strategies (3.11.4.7)
Intelligent Driver Model (IDM) and MOBIL	Car-following and lane-changing rules of the simulated vehicles.	Equations 3.11.5.1–3.11.5.3.	Scenario Sandbox (3.11.5.1)
Language-model interface	Interprets commands or explains results; does not forecast or optimize.	GLM (sandbox, narratives); Qwen with tool calling (mobile assistant); outputs validated by the system.	Section 3.11.5.2
Medallion architecture	Layered warehouse design.	Bronze (raw), silver (cleaned), gold (analysis-ready and model outputs).	Sections 3.9–3.10